Foundations of Risk Management 目录

Quantitative Analysis 目录

Financial Markets and Products 目录

Valuation and Risk Models 目录

1. Case studies that feature financial engineering by way of complex derivatives include Bankers Trust, the Orange County case, and Sachsen Landesbank. In regard to these financial engineering cases, each of the following statements is true EXCEPT which is false?
通过复杂衍生品进行金融工程的案例研究包括 Bankers Trust、橙县案例以及萨克森州银行（Sachsen Landesbank）。关于这些金融工程案例，下列各项陈述均为真，除了哪一项为假?

A. Bankers Trust (BT) proposed an overly complex swap to their clients (P&G and Gibson Greetings) but the swaps experienced colossal losses; the clients sued BT, who never recovered from the ensuing reputational damage
Bankers Trust（BT）向其客户（宝洁 P&G 和 Gibson Greetings）提出了过于复杂的互换方案，但这些互换产生了巨额亏损；客户起诉了 BT，而 BT 也从未从随后造成的声誉损害中恢复

B. Orange County's treasurer (Robert Citron) borrowed through the repo market to purchase inverse floating-rate notes – positions that Citron later said he did not understand – but the combination of excessive leverage and embedded interest-rate risk generated losses that ultimately forced Orange County to file for bankruptcy
橙县财政官（Robert Citron）通过回购市场（repo market）借款，用于购买反向浮动利率票据——Citron 事后表示自己并不了解这些头寸——但过度杠杆与内嵌利率风险的组合最终导致巨额亏损，迫使橙县不得不申请破产

C. Sachsen Landesbank set up off-balance-sheet vehicles (guaranteed by Sachsen) that held highly-rated U.S. mortgage-backed securities; the operation was highly profitable, but the 2007-08 subprime crisis wiped out Sachsen's capital
萨克森州银行（Sachsen Landesbank）设立了由萨克森州银行提供担保的表外工具，这些工具持有高评级的美国抵押贷款支持证券；该业务在一段时间内利润颇丰，但 2007–08 年的次贷危机却抹去了萨克森州银行的全部资本

D. If the purpose of the position is designated as hedging (rather than speculation) and if the hedge consists only of some combination(s) of forwards, swaps and/or options–which are the primary building blocks–then the firm can avoid problems suffered by the financial engineering case studies because the firm avoids undue sophistication
如果某一头寸的目的被指定为套期保值（而非投机），且该套保仅由远期、互换和/或期权——这些主要构件——的某种组合构成，那么公司就可以避免这些金融工程案例中遭遇的问题，因为公司避免了过度复杂化

答案:D
解析:
D 选项错误。

- 对冲并不能避免那些可能未被理解的嵌入风险；虽然对冲/投机的意图很重要，但这并不能完全保证头寸不存在风险。更重要的是，复杂头寸本身就是由这些基本构件组合而成，因此即便是复杂的对冲头寸，也无法避免承担重大风险。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Valuation and Risk Models

2. A very small sample of 10 public companies has one target variable (i.e., Dividend) and four features (i.e., Earnings, LargeCap, Retail, and Tech). Both Dividend and Earnings are binary variables where respectively, one (1) indicates the company recently paid a dividend, and its earnings dropped in the previous year. The data frame is displayed below.

Dividend	Earnings	LargeCap	Retail (%)	Tech
1	0	0	80	1
1	0	1	40	1
1	0	1	20	0
1	0	1	60	0
0	1	0	30	1
0	1	0	40	0
0	1	1	40	0
1	1	1	30	0
1	1	1	20	0
1	1	1	20	1

Based on Dividend as the output (aka, target) variable, the base Gini coefficient is 0.420. That's because $1-[(7/10)^2+(3/10)^2]$. What is the information gain if we split (as the root node's feature) the Earnings feature?
一组仅包含 10 家上市公司的极小样本具有一个目标变量（即 Dividend）和四个特征（即 Earnings、LargeCap、Retail 和 Tech）。

Dividend 和 Earnings 均为二元变量，其中取值为 1 表示公司最近支付了股利，且其上一年的盈利出现了下滑。数据框如下所示。

Dividend	Earnings	LargeCap	Retail (%)	Tech
1	0	0	80	1
1	0	1	40	1
1	0	1	20	0
1	0	1	60	0
0	1	0	30	1
0	1	0	40	0
0	1	1	40	0
1	1	1	30	0
1	1	1	20	0
1	1	1	20	1

以 Dividend 作为输出（即目标）变量时，其基础 Gini 系数为 0.420。这是因为 $1-[(7/10)^2+(3/10)^2]$。如果我们以 Earnings 作为根节点的特征进行划分，其信息增益是多少?

A. Zero

B. 0.120

C. 0.348

D. 0.875

答案:B

解析:

B 选项正确。

- 我们需要计算以 Earnings 作为特征进行划分时的加权 Gini 系数。对于 Earnings=0 的四个分支，所有公司都支付股利，因此该分支是纯的，Gini 系数为零！对于 Earnings = 1 的六个分支，其中 3 家公司支付股利，3 家公司不支付股利，其 Gini 系数为 $1–(3/6)^2–(3/6)^2 = 0.50$。因此，加权 Gini 系数为 $0*4/10 + 0.50*6/10 = 0.30$。因此，信息增益为 0.120，因为以 Earnings 作为特征进行划分时，从基础 Gini 系数 0.420 下降到 0.300。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]

科目: Financial Markets and Products

3. About swaps, each of the following is true EXCEPT which is false?

关于互换（swaps），下列各项陈述均为真，除了哪一项为假?

A. At any given time during the life of a bilateral OTC swap, at least one of the counterparties incurs both market and credit risk
在一笔双边场外（OTC）互换的存续期内的任意时点，至少有一方交易对手承担市场风险和信用风险

B. A difference between the typical interest rate swap and currency swap is whether the principal (aka, notional principal) is exchanged
典型利率互换与货币互换之间的一个区别在于是否交换本金（即名义本金）

C. Both overnight indexed swaps (OIS) and volatility swaps need to determine a realized (aka, actual) financial variable at the end of a period or tenor
隔夜指数互换（OIS）和波动率互换都需要在某一期间或期限结束时确定一个已实现（即实际的）金融变量

D. A commercial bank funded by short-term floating-rate deposits that extends longterm fixed-rate loans can hedge market risk by entering a swap where it is the floating-rate payer and fixed-rate receiver
一家由短期浮动利率存款融资、却发放长期固定利率贷款的商业银行，可以通过进入一笔自身支付浮动利率、收取固定利率的互换来对冲市场风险

答案:D

解析:

D 选项正确。

- 银行通过进入一笔自身支付浮动利率、收取固定利率的互换来对冲市场风险，这是相反的套期保值！在 A、B 和 C 中，每个陈述都是正确的

考察： 偿付能力 II 对保险公司的资本要求　　**重要程度：** [■■□]
科目： Financial Markets and Products

4. A large international bank uses EWMA for their risk system, with a decay factor of 0.92. On day T the forecast volatility for two variables, X&Y, are identical at 1.25 and the current correlation forecast is 15.0. If on day T the realized returns for X & Y are 0.50 and -0.50 respectively, which of the following is closest to the new correlation?
一家大型国际银行使用 EWMA 模型进行风险系统管理，衰减因子为 0.92。在第 T 天，两个变量 X 和 Y 的预测波动率均为 1.25，当前的预测相关系数为 15.0。如果第 T 天的实际回报分别为 0.50 和 -0.50，则以下哪个选项最接近新的相关系数?

A. 12.5%

B. 13.4%

C. 15.0%

D. 16.2%

答案:B
解析:
A 选项错误。

- 12.5% 是如果使用昨天的波动率计算相关系数的结果

B 选项错误。

- 回想 EWMA 方差和协方差的公式：

$$\sigma_{T+1}^2 = \lambda\sigma_T^2 + (1-\lambda)r_T^2$$
$$cov_{T+1}^{\square} = \lambda cov_T^{\square} + (1-\lambda)r_{X,T}^{\square}r_{Y,T}^{\square}$$

  因此，第 T 天的协方差预测为 $0.15 \times 0.0125 \times 0.0125 = 2.34 \times 10^{-5}$.

  我们需要更新 X、Y 和 cov(X,Y) 的方差估计：

  $\text{Variance}(X) = 0.92 \times (0.0125^2) + (0.08) \times (0.005^2) = 0.00014575$, $\text{Std}(X) = 1.21\%$。

  $\text{Variance}(Y) = 0.92 \times (0.0125^2) + (0.08) \times (-0.005^2) = 0.00014575$, $\text{Std}(Y) = 1.21\%$。

  $\text{Cov}(X,Y) = 0.92 \times (2.34 \times 10^{-5}) + 0.08 \times (0.005) \times (-0.005) = 1.956 \times 10^{-5}$。

  $\text{Corr}(X,Y) = \text{Cov}(X,Y)/(\text{Std}(X) \times \text{Std}(Y)) = 13.4\%$。

C 选项错误。

- 15.0% 是当前的相关系数

D 选项正确。

- 16.2% 是如果使用 0.5% 的回报率而不是 0.05% 的回报率计算得到的结果

考察： 偿付能力 II 对保险公司的资本要求　　**重要程度：** [■■□]
科目： Financial Markets and Products

5. According to GARP, each of the following was a causal factor in the 2007-2009 global financial crisis (GFC) EXCEPT which is not a causal factor? 根据 GARP，下列各项被认为是 2007-2009 年全球金融危机（GFC）的原因，除了哪一项不是原因?

A. Low interest rates
低利率

B. The originate-to-distribute (OTD) business model and securitization, especially CDOs
发起-分销（OTD）业务模式和证券化，特别是 CDO

C. An unexpected spike in prepayments due to an acceleration in repeat refinancing
由于重复再融资的加速，预付意外飙升

D. Dubious lending practices and risky mortgage loan products (e.g., NINJA) and loan features (e.g., teaser rates)//可疑的贷款实践和风险抵押贷款产品（例如 NINJA）和贷款特征（例如 teaser rates）

答案:C
解析:
A 选项错误。

- 低利率（请注意其在住房需求和投资者供给中的双重作用）:“住房需求的增长及随之而来的抵押贷款融资，部分是由 2000 年代初期存在的低利率环境推动的。这种需求助推了房价的显著上涨。低利率环境也促使投资者，包括机构投资者，寻找能提高收益的投资。他们在次级抵押贷款中找到了这种收益，次级贷款的利率通常比优质贷款高出最多 300 个基点。”——引自 2020 年 FRM 一级《风险管理基础》第 10 版

B 选项错误。

- 银行向“发起-分销”（OTD）业务模式的转变助长了道德风险，因为这种模式降低了对借款人信用状况的严格审查。

C 选项正确。

- 再融资使借款人能够在最初的诱导利率期结束后，避免因利率重置而带来的更高利率；但问题在于，住房价格的下跌终止了这种曾推动许多次级抵押贷款的再融资行为。因此，提前还款实际上并不是危机的真正原因。就（A）、（B）和（D）而言，这些选项都是真实的。虽然导致危机的原因有很多（部分有争议），但根据 GARP 第 10 章，成因包括:

D 选项错误。

- 风险抵押贷款产品，特别是诱导利率；例如，2/28 可调利率抵押贷款。（但也有只付利息的诱导利率期和其他“创新”，如 NINJA 贷款）

考察: 偿付能力 II 对保险公司的资本要求　**重要程度:** [■■□]
科目: Financial Markets and Products

6. A hedge fund's big data algorithm can predict the market's direction on five out of eight days (62.5%). Each day's prediction is either a success (e.g., market goes up and algo predicts up) or a failure (e.g., market goes down but algo predicts up). If we dubiously assume the predictions are independent, the binomial distribution fits a series of daily predictions over, say, a week or a month. Over two months, the probability of each day's predication being successful, p, equals 5/8 or 62.5% and the number of days, n, equals 60. We observe that n*p = 60*62.5% = 37.5 and n*(1-p) = 60*37.5% = 22.5, and both of these values (i.e., 37.5 and 22.5) are greater than 10; this satisfies a conventional test that says we can use the normal to approximate the binomial. For example, if p were only 1.0%, then n*p = 6, but 6 is less than 10, and such a binomial is deemed to be too skewed to be approximated by the normal distribution. But ours passes the test so we will approximate with the normal distribution. If we do rely on the normal distribution to approximate this binomial where p = 5/8 and n = 60, what is the probability that the algo makes a correct prediction on only half the days or worse; i.e., where X is the number of successful predictions and we approximate with the normal distribution, what is the Pr(X <= 30)?
一个对冲基金的大数据算法可以预测市场的方向，在八天中有五天（62.5%）成功。每天的预测要么成功（例如，市场上涨，算法预测上涨）要么失败（例如，市场下跌，算法预测上涨）。如果我们假设预测是独立的，二项分布适合一系列每日预测，例如一周或一个月。在两个月内，每天预测成功的概率 p 等于 5/8 或 62.5%，天数 n 等于 60。我们观察到 n*p = 60*62.5% = 37.5 和 n*(1-p) = 60*37.5% = 22.5，这两个值（即 37.5 和 22.5）都大于 10；这满足一个常规测试，即我们可以使用正态分布来近似二项分布。例如，如果 p 只有 1.0%，那么 n*p = 6，但 6 小于 10，这样的二项分布被认为过于偏斜，无法用正态分布近似。但我们的通过了测试，所以我们使用正态分布近似。如果我们依赖正态分布来近似这个二项分布，其中 p = 5/8 和 n = 60，那么算法在只有一半天数或更少天数正确预测的概率是多少；即当 X 是成功预测的数量时，我们使用正态分布近似，Pr(X <= 30) 是多少?

A. 2.2750%

B. 8.3500%

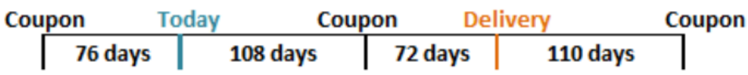
C. 11.090%

D. 14.6667%

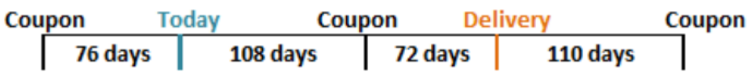
答案:A
解析:
A 选项正确。

- 方差 $= n \times p \times (1-p)$ 和标准差 $= \sqrt{n \times p \times (1-p)}$; 在这种情况下，标准差 $= \sqrt{60 \times (5/8) \times (3/8)} = 3.750$。其中 $Z = (30 - 60 \times 5/8)/3.750 = -2.0$，所以 $\Pr(Z < -2.0) = 2.2750\%$。

7. Suppose that it is known that a bond will be delivered in 180 days under the terms of a futures contract. The timing of coupon payments is shown below. The last coupon on the bond was paid 76 days ago, and the next coupon will be paid in 108 days. The coupon after the next one will be paid in 290 days (i.e., 110 days after delivery). Here is the timeline: next one will be paid in 290 days (i.e., 110 days after delivery). Here is the timeline:



Assume that the risk-free interest rate for all maturities is 5.0% with continuous compounding and that the delivered bond pays a coupon of 9.0% semi-annually with a current quoted (clean) price of USD 124.80 and a conversion factor of 1.300. Which of the following is nearest to the quoted futures price?
假设根据期货合约条款，已知一只债券将在 180 天后交割。息票支付的时间安排如下所示。该债券的上一次息票支付是在 76 天前，下一次息票将在 108 天后支付。再下一次息票将在 290 天后（即交割后 110 天）支付。时间轴如下所示:



假设所有期限的无风险利率为 5.0%（连续复利），这只被交割债券的票息为 9.0%，每半年付息一次，目前的报价（净价）为 124.80 美元，转换因子为 1.300。下列哪个最接近该期货合约的报价期货价格?

A. $95.00

B. $98.22

C. $123.50

D. $125.27

答案:A
解析:
A 选项正确。

Assumptions	
Face	$100.00
Current Quoted (Clean) Price	$124.80
Coupon	9.0%
Interest rate	5.0%
Conversion Factor	1.300
Delivery (days)	180
Last Coupon (-days)	76
Next Coupon (+ days)	108
Coupon thereafter	182

Timeline	
7/1/2023	Coupon payment
-76 days	
9/15/2023	Current time
108 days	
1/1/2024	(Next) coupon payment
72 days	
3/13/2024	Maturity of futures
110 days	
7/1/2024	Coupon payment

Solution (Hull's Example 6-2)	
Accrued Interest	$1.859
Cash (Dirty, Full) Price	$126.6587
PV of next coupon	$4.434
Cash Futures Price	$125.2760
Days Accrued @ delivery	72
Days Remain @ delivery	110
Futures price, Quoted, 12% bond	$123.4958
Futures price, Quoted, CTD	$94.9968

COC: $F_0=(S_0-I)*EXP(rT)$	
1. COC to calculate Cash forward	
Spot:	$126.659
Income:	$4.434
Forward cash price:	$125.276
2. Then dirty --> clean & standardize	
Subtract AI	$1.780
Quoted Price, 12% bond =	$123.496
Divide by CF of 1.30	$94.997

8. A risk analyst is using the EWMA model to build a daily update of correlation and covariance rates. The two variables are random and are called X and Y. The most recent covariance weight (aka, lambda in the EWMA model) from day n-1 is 0.7. The correlation between X and Y on day N-1 is estimated to be 0.55. X and Y had estimated standard deviations of 0.015 and 0.017 with percentage changes of 3% and 4% respectively on day N-1. The updated covariance between X and Y on day n is nearest to which of the following?
一位风险分析师使用 EWMA 模型构建每日更新的相关性和协方差率。这两个变量是随机的，称为 X 和 Y。从第 n-1 天获取的最新协方差权重（即 EWMA 模型中的 lambda）为 0.7。第 N-1 天 X 和 Y 之间的相关性估计为 0.55。X 和 Y 在第 N-1 天的估计标准差分别为 0.015 和 0.017，百分比变化分别为 3% 和 4%。第 n 天 X 和 Y 之间的更新协方差最接近以下哪个选项?

A. 1.48×10^{-4}

B. 4.58×10^{-4}

C. 1.18437

D. 0.04581

答案:B
解析:
B 选项正确。

- 我们可以使用如下公式计算 EWMA（指数加权移动平均）:

  $Cov_N = \lambda \times Cov_{N-1} + (1-\lambda) \times X_{n-1} \times Y_{n-1}$

  其中，λ 表示第 $n-1$ 天最近协方差的权重；

  X 表示第 $n-1$ 天变量 X 的百分比变化；

  Y 表示第 $n-1$ 天变量 Y 的百分比变化。

  我们知道 $Cov_{xy} = \rho_{xy} \times \sigma_x \times \sigma_y$。

  $Cov(n-1) = 0.55 \times 0.015 \times 0.017 = 0.00014025$。

  $Cov_N = 0.7 \times 0.00014025 + (1-0.7) \times 3\% \times 4\% = 0.000458175 = 4.58 \times 10^{-4}$

9. Which of the following is TRUE about the SWIFT (Society for Worldwide Interbank Financial Telecommunication) case study? 以下关于 SWIFT（环球银行金融电信协会）案例研究，哪一项说法是正确的?

A. The 2015-16 cyber-attack (aka, hack) demonstrated that the SWIFT network was unreliable, and it was subsequently phased out
2015-16 年网络攻击（又称“黑客”）表明 SWIFT 网络不可靠，随后被逐步淘汰

B. The 2015-16 cyber-attack (aka, hack) successfully exploited vulnerabilities to achieve the theft of about USD 81.0 million
2015-16 年网络攻击（又称“黑客”）成功利用漏洞，成功窃取了约 8100 万美元

C. The 2015-16 cyber-attack (aka, hack) was an unsuccessful attempt to steal money, and it demonstrated the SWIFT network is essentially impervious to attacks
2015-16 年网络攻击（又称“黑客”）是一次失败的尝试窃取资金，并证明了 SWIFT 网络基本上对攻击免疫

D. The 2015-16 cyber-attack (aka, hack) was a fictitious news account but the negative press nonetheless shook confidence sufficiently in the network that transactions ground to a halt for several weeks
2015-16 年网络攻击（又称“黑客”）是一个虚假的新闻报道，但负面新闻仍然足以动摇网络的信心，导致交易暂停数周

答案:B
解析:
B 选项正确。

- 2015-16 年网络攻击（又称“黑客”）成功利用漏洞，窃取了约 8100 万美元。关于（A）、（C）、（D），每一项都是错误的。

10. A credit portfolio contains some number of independent credit-sensitive assets with identical default probabilities; as the defaults are i.i.d., we can use the binomial distribution to characterize the number of defaults. We are told the expected number of defaults is 4.0 with a variance of 3.80. Which is nearest to the probability of exactly four defaults; Pr(X = 4 | binomial with mean of 4.0 and variance of 3.80)?
一个信用组合包含一些具有相同违约概率的独立信用敏感资产；由于违约是独立同分布的，我们可以使用二项分布来描述违约的数量。我们被告知期望违约数为 4.0，方差为 3.80。哪一个最接近恰好四个违约的概率；Pr(X = 4 | 二项分布，均值为 4.0，方差为 3.80)?

A. Less than 0.01%

B. 10.0%

C. 20.0%

D. 33.3%

答案:C
解析:
C 选项正确。

- 二项分布的均值为 $p \times n$，方差为 $p \times (1-p) \times n$。已知 $p \times n = 4.0$，$p \times (1-p) \times n = 3.80$。我们可以将第一个公式代入第二个公式，据此求得投资组合中的资产数量以及违约概率：若 $p \times (1-p) \times n = [p \times n] \times (1-p) = 3.80$，又已知 $p \times n = 4.0$，则有 $4.0 \times (1-p) = 3.80$，解得 $p = 1 - \frac{3.80}{4.0} = 5.0\%$。因为 $n \times p = 4.0$，所以 $n = \frac{4}{0.050} = 80$。所以，$\Pr(X=4) = \text{BINOM.DIST}(4, 80\text{次实验}, 0.050, \text{false}) = 20.043\%$。

11. At the request of his boss at a large commercial bank, Jason is using the bank's internal model to perform a valuation of a freshly constructed pass-through mortgage-backed security (MBS) pool. The pool contains about 500 mortgages with a principal value of $400 billion; all of the mortgages originated within the last 30 days. The pool's weighted average coupon (WAC) is 7.50%. Jason is going to assume a PSA rate of 140%. This multiplies by 1.40 the baseline 100% PSA rate that starts at 0.20% CPR each month for 30 months, then levels at 6.0% CPR. Each of the following is true EXCEPT which is false?
Jason 是一家大型商业银行的老板，请求他使用银行的内部模型对一个新构建的转按揭抵押贷款支持证券（MBS）池进行估值。该池包含约 500 笔抵押贷款，本金总额为 4000 亿美元；所有抵押贷款均在最近 30 天内发放。该池的加权平均利率（WAC）为 7.50%。Jason 将假设 PSA 率为 140%。这相当于将 100% PSA 基准利率（每月 0.20% CPR，持续 30 个月，然后稳定在 6.0% CPR）乘以 1.40。以下哪一项陈述是正确的，除了哪一项是错误的?

A. The pass-through security will pay investors a coupon rate less than 7.50%
转按揭抵押贷款支持证券将支付投资者低于 7.50% 的利率

B. The pool factor declines over time due to both scheduled and unscheduled prepayments
由于计划和非计划提前还款，池因子随时间推移而下降

C. An increase in the assumption for the PSA rate (e.g., 140% to 170%) implies an increase in present value of this pool
假设 PSA 率增加（例如，从 140% 到 170%）意味着该池的现值增加

D. The single month mortality (SMM) rate excludes scheduled prepayments, and after three years, it will be about 0.7285%
单月死亡率（SMM）率不包括计划提前还款，三年后约为 0.7285%

答案:C
解析:
C 选项错误。
相反，提高 PSA 率实际上反映了未计划提前还款的增加，这将降低 MBS 池的价值。
关于 A、B 和 D，每一项都是正确的。具体来说：

- 转按揭抵押贷款支持证券支付给投资者的利率将低于 7.50%
- 由于计划和非计划提前还款，池因子（Pool Factor）会随时间逐渐下降
- 单月死亡率（SMM）不包括计划提前还款，三年后大约为 0.7285%

12. Scott, a risk manager at a small deep-value strategy, is trying to ensure his company is using a coherent risk benchmark. The deep value strategy can be volatile, and thus, Scott wants to see how his capital at risk would adjust if the portfolio had the same holdings but in a smaller proportion to increase its allocation to the risk-free asset. Which principal of a coherent benchmark is reflected in the below scenario?

Portfolio	**Scenario 1**	**Scenario 2**
Portfolio	$20,000,000	$20,000,000
Risk-free Allocation	1.0%	11.0%
Risky Allocation	$19,800,000	$17,800,000
Volatility	24.49%	24.49%
Capital at Risk	$4,850,000	$4,360,101
Risk free Capital	$200,000	$2,200,000
Net Capital at Risk	$4,650,000	$2,160,101

Scott 是一家小型深度价值策略公司的风险经理，他正在努力确保公司采用连贯的风险基准。深度价值策略可能会有较大波动，因此 Scott 希望了解，如果投资组合持有相同标的资产，但通过减少比例、增加无风险资产配置后，其风险资本会如何变化。下表所反映的情景体现了哪一项连贯风险基准的原则？

投资组合	**情景 1**	**情景 2**
投资组合总额	$20,000,000	$20,000,000
无风险资产配置	1.0%	11.0%
风险资产配置	$19,800,000	$17,800,000
波动率	24.49%	24.49%
风险资本	$4,850,000	$4,360,101
无风险资本	$200,000	$2,200,000
净风险资本	$4,650,000	$2,160,101

A. Translation Invariance
平移不变性

B. Homogeneity
齐次性

C. Subadditivity
次可加性

D. Monotonicity
单调性

答案:A

解析:

A 选项正确。

- 平移不变性表示添加（n）会减少所需的额外现金（n）。简单来说，添加现金会降低风险。对于上面的例子，风险无分配的增加减少了风险分配。风险资本随后被风险无分配减少。

B 选项错误。

- 齐次性表示将投资组合规模乘以常数会导致风险度量按比例变化。

C 选项错误

- 次可加性表示两个或更多投资组合的风险之和不超过各个组合风险的和。添加更多资产或资产类别（如商品）到投资组合中应该导致总体风险度量不超过各个单独风险度量的和。

D 选项错误

- 单调性意味着如果一个投资组合的风险大于或等于另一个投资组合，那么它也应该有大于或等于的风险度量。它确保添加更多资产或增加现有资产的分配不会增加风险度量。

13. Barbara is a certified FRM who previously generated income statement and profit projections over a five-year horizon in response to her client's request. Subsequent to the coronavirus outbreak, her client asks Barbara to generate revised financial projections (income statement and balance sheet) and provide the best single point estimate of future revenue, profit, and equity. Barbara utilizes Monte Carlo simulation. Although her client has asked for a single point estimate of these future financial metrics, Barbara perceives that the virus (and the consequential responses) render economic predictions extremely difficult and necessarily laced with great uncertainty. Which of the following is probably her BEST approach to the problem?
Barbara 是一位认证的 FRM，她之前根据客户要求，生成了一份五年的收入报表和利润预测。随后新冠病毒爆发，客户要求她重新生成修订后的财务预测（收入报表和资产负债表），并提供未来收入、利润和股权的最佳单一估计值。Barbara 使用蒙特卡罗模拟。尽管客户要求她提供这些未来财务指标的单一估计值，但 Barbara 认为病毒（及其后续反应）使得经济预测极其困难，并且必然带有极大的不确定性。以下哪一项可能是她解决这个问题的最佳方法?

A. She can comply by simply selecting the most probable future outcome; the Code has generally nothing to say about this technical task
她可以简单地选择最可能的未来结果；Code 通常对此技术任务没有具体规定

B. She can comply but she should qualify her findings to avoid overstating the accuracy of her projections; for example, she can plot a future distribution
她可以遵守，但应该对其发现进行限定，以避免高估预测的准确性；例如，她可以绘制未来分布

C. She should work hard to generate the most precise estimate possible; for example, "sales will go down by 10.85%" is more useful than "sales will go down by 10%"
她应该努力生成最精确的估计；例如，" 销售将下降 10.85%" 比" 销售将下降 10%" 更有用

D. She should avoid the temptation to offer any revised future estimates in the face of several "unknown unknowns" because some future values cannot be estimated with sufficient accuracy
她应该避免在面对多个" 未知未知" 时提供任何修订后的未来估计，因为一些未来值无法以足够的准确度估计

答案:B

解析:

B 选项正确。她可以遵守，但应该对其发现进行限定，以避免高估预测的准确性；例如，她可以绘制未来分布。
根据（4.）基本责任，GARP 成员：

4.1 应遵守所有适用于 GARP 成员专业活动的适用法律、规则和法规（包括本 Code），并不得故意参与或协助任何违反此类法律、规则或法规的行为。

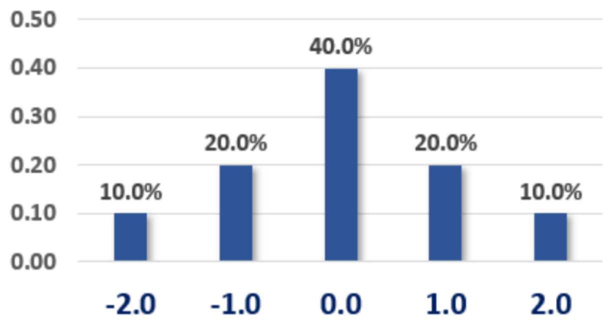
4.2 应承担道德责任，不能外包或委托这些责任给他人。

4.3 应理解其雇主或客户的需求和复杂性，并应提供适当和合适的风险管理服务和建议。

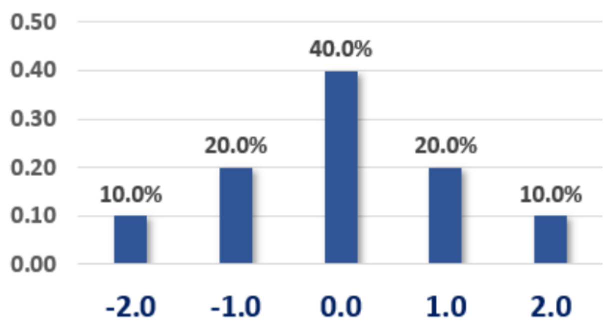
4.4 应勤勉地不夸大结果或结论的准确性或确定性。

4.5 应明确披露其特定知识、行业实践和适用法律和法规在风险评估方面的相关限制。

14. Consider the probability mass function (pmf) below. For example, Pr(X = -1.0) = 20.0%.



As we can see, this distribution is symmetrical, so we know that its skewness is zero without performing any calculations. We are told the variance is 1.20 (although we can calculate the variance). What is this distribution's kurtosis?
请考虑下方的概率质量函数（pmf）。例如，Pr(X = -1.0) = 20.0%。



如图可见，该分布是对称的，因此即使不计算也可以知道其偏度为零。已知方差为 1.20（虽然我们也可以计算方差）。那么该分布的峰度是多少?

A. Zero

B. 2.50

C. 3.60

D. 4.40

答案:B
解析:
B 选项正确。
$(-2)^4 \times 10.0\% + (-1)^4 \times 20.0\% + 0^4 \times 40.0\% + 1^4 \times 20.0\% + 2^4 \times 10.0\% = 3.60$，但我们还需要标准化，因此峰度 $= 3.60/1.20^2 = 2.50$。
峰度是第四中心距（即关于均值的四阶矩），再对标准差的四次方（即二阶矩的平方）进行标准化。因此，分子为 $E[X-\mu(x)]^4$，就是第四中心距；分母为 $\sigma(X)^4$，实现对该中心矩的标准化。参见 `https://en.wikipedia.org/wiki/Standardized_moment`

15. A U.S. corporate bond that matures on April 1st, 2026, pays a semi-annual coupon of 9.0% per year. The coupon payment dates are Jan 1st and July 1st. Settlement is on April 7th, 2024. The year 2024 is a leap year. The bond's yield happens to be about 7.00%. If the cash (aka, dirty) price is \$1,064.50, then which is nearest to the quoted (aka, clean) price? (Inspired by GARP EOC PQ 17.13)

一只美国企业债券将于 2026 年 4 月 1 日到期，每年支付两次 9.0% 的息票。息票支付日期为 1 月 1 日和 7 月 1 日。结算日期为 2024 年 4 月 7 日。2024 年是闰年。该债券的收益率约为 7.00%。如果现金价格（aka，脏价）为 \$1,064.50，那么哪个最接近报价（aka，清洁价）?（inspired by GARP EOC PQ 17.13）

A. \$960.14

B. \$974.50

C. \$1,040.50

D. \$1,088.50

答案:C

解析:

C 选项正确。 True: \$1,040.50

按 30/360 天计法，共有 $30 \times 3 + 6 = 96$ 天。应计利息（AI）为 $\$1,000 \times 9.0\%/2 \times 96/180 = \$24.00$。

因此，债券的报价（净价）为 $\$1,064.50 - 24.00 = \$1,040.50$。

请注意，我们也可以从上一个付息日的债券价格推出现金价（题目已给出）：$\$1,045.151$ 是该债券在 2024 年 1 月 1 日的价格，参数为 $N = 5$，$I/Y = 3.5\%$，$PMT = 45$，$FV = 1,000$。然后将其复利到结算日：$\$1,045.151 \times (1 + 7.0\%/2)^{96/180} = \$1064.50$。

16. When is the soonest that an obligation rated AAA (the highest rating) can default?

	AAA	AA	A	BBB	BB	B	CCC to C	Default
AAA	91.00	9.00	-	-	-	-	-	-
AA	1.00	84.00	7.00	5.00	3.00	-	-	-
A	0.40	2.00	91.60	3.00	2.00	1.00	-	-
BBB	-	0.60	2.00	88.50	5.00	3.00	0.90	-
BB	-	0.40	1.00	3.00	79.60	10.00	4.00	2.00
B	-	-	0.80	3.00	9.00	67.20	9.00	11.00
CCC - C	-	-	-	2.00	10.00	15.00	58.00	15.00

什么时候违约概率最高的债券（AAA 评级）最早可能违约?

	AAA	AA	A	BBB	BB	B	CCC to C	Default
AAA	91.00	9.00	-	-	-	-	-	-
AA	1.00	84.00	7.00	5.00	3.00	-	-	-
A	0.40	2.00	91.60	3.00	2.00	1.00	-	-
BBB	-	0.60	2.00	88.50	5.00	3.00	0.90	-
BB	-	0.40	1.00	3.00	79.60	10.00	4.00	2.00
B	-	-	0.80	3.00	9.00	67.20	9.00	11.00
CCC - C	-	-	-	2.00	10.00	15.00	58.00	15.00

A. During the first year (i.e., by the end of the first year) with a margin probability of 2.0%
在第一年（即第一年末），违约概率为 2.0%

B. During the second year (i.e., between the end of the first and second year) with a joint probability of 0.1879% (or 0.001879 or 1.879×10^{-3})
在第二年（即第一年末和第二年末之间），联合概率为 0.1879%（或 0.001879 或 1.879×10^{-3}）

C. During the third year (i.e., between the end of the second and third year) with an unconditional probability of 0.00540% (or 0.0000540 or 5.40×10^{-5})
在第三年（即第二年末和第三年末之间），无条件概率为 0.00540%（或 0.0000540 或 5.40×10^{-5}）

D. During the fourth year (i.e., between the end of the third and fourth year) with a conditional probability of 2.00%
在第四年（即第三年末和第四年末之间），条件概率为 2.00%

答案:C

解析:

C 选项正确。正确答案：在第三年内（即第二年末与第三年末之间），无条件违约概率为 0.00540%（或 0.0000540 或 5.40×10^{-5}）。

在第一年末，AAA 评级的最严重降级仅为 AA，概率为 9.0%。若第二年初为 AA 评级，则最严重可降级为 BB，条件概率为 3.0%（如果降级为 BBB 或更高评级，该债务在第三年末之前无法违约）。若第三年初为 BB 评级，则该债务在第三年内违约的条件概率为 2.0%。由此，累计概率为 $9.0\% \times 3.0\% \times 2.0\% = 0.005400\%$。

17. Arguably the U.S. savings and loan (S&L) crisis in the 1980s had multiple causes, but among the following which is the BEST summary explanation of the S&L crisis?
可以说 1980 年代的美国储蓄和贷款危机有多重原因，但以下哪一项是对 S&L 危机的最佳总结解释?

A. S&Ls engaged in a practice called "riding the yield curve" but this was one of the central bank's monetary policy tools and was not meant to be performed by private banks
S&Ls 参与了一种被称为"骑乘收益率曲线"的实践，但这只是中央银行货币政策工具之一，并非私人银行应执行的操作

B. The Fed's restrictive monetary policy led to a dramatic increase in short-term interest rates which wiped out bank's interest rate spread due to their asset-liability mismatch
美联储的紧缩货币政策导致短期利率大幅上升，由于资产负债不匹配，银行利率差额被抹平

C. The Fed's expansionary monetary policy led to a dramatic decrease in short-term interest rates which wiped out bank's interest rate spread due to their asset-liability mismatch
美联储的扩张性货币政策导致短期利率大幅下降，由于资产负债不匹配，银行利率差额被抹平

D. New regulations introduced in the early 1980s hindered the ability of the savings and loan (S&L) industry to lend and/offer new loan products that could help it grow out of its problems
1980 年代初引入的新法规阻碍了储蓄和贷款（S&L）行业的放贷能力和/或提供新的贷款产品，这些产品可能有助于其摆脱困境

答案:B
解析:
B 选项正确。美联储的紧缩货币政策导致短期利率大幅上升，由于资产负债不匹配，银行利率差额被抹平。
正如 GARP 解释的那样，"然而，1970 年代末的通货膨胀促使美联储实施紧缩货币政策，导致短期利率大幅上升。短期利率上升推高了储蓄和贷款（S&L）的融资成本，抹平了它们依赖的利率差额，导致它们的利润率下降。短期融资成本的飙升（需要为长期固定利率抵押贷款提供资金）意味着 S&L 在许多长期住宅抵押贷款组合上产生了负净利息差。"

18. A discrete random variable is characterized by the probability mass function (pmf) as given by $f(x) = x \cdot a$, and its domain is the set of integers $\{6, 7, 8, 9, 10\}$. What is the variable's expected value?
一个离散随机变量由概率质量函数（pmf）给出，形式为 $f(x) = x \cdot a$，其定义域为整数集合 $\{6, 7, 8, 9, 10\}$。该变量的期望值是多少?

A. 6.67

B. 8.00

C. 8.25

D. 9.33

答案:C
解析:
C 选项正确。 True: 8.25
我们知道 $6a + 7a + 8a + 9a + 10a = 1.0$，因此 $a = 1/40 = 0.0250$，概率质量函数为 $f(x) = x/40 = 0.0250 \times x$。
期望值等于 $1/40 \times (6^2 + 7^2 + 8^2 + 9^2 + 10^2) = 8.25$。

19. Assume the following six-month forward rates that are quoted per annum with semi-annual compounding:

- $F(0, 0.5) = S(0.5) = 2.0\%$; i.e., the six-month spot rate

- $F(0.5, 1.0) = 4.0\%$
- $F(1.0, 1.5) = 6.0\%$

If the price of a 7.0% semi-annual coupon bond is \$103.20, which is nearest to the 2-year spot rate?
假设以下六个月远期利率以年化利率报价，半年付息一次：

- $F(0, 0.5) = S(0.5) = 2.0\%$; i.e., the six-month spot rate
- $F(0.5, 1.0) = 4.0\%$
- $F(1.0, 1.5) = 6.0\%$

如果 7.0% 半年付息债券的价格为 \$103.20，哪个最接近 2 年期的即期利率？

A. 4.30%

B. 5.40%

C. 6.60%

D. 7.90%

答案:B
解析:
B 选项正确。 True: 5.40%

20. In regard to through-the-cycle (TTC) versus at-the-point (aka, point in time, PIT) approaches to credit ratings, each of the following statements is true EXCEPT which is false?
关于信用评级中的通过周期（TTC）与时点（aka，时点，PIT）方法，以下哪一项陈述是正确的，除了哪一项是错误的？

A. Agency (i.e., external) credit ratings tend to be through-the-cycle (TTC)
机构（即外部）信用评级倾向于通过周期（TTC）

B. Through-the-cycle (TTC) is conditional, while at-the-point (PIT) is unconditional
通过周期（TTC）是条件性的，而时点（PIT）是无条件的

C. Credit spreads are at-the-point (PIT) and PIT measures incorporate more information
信用利差是时点（PIT）的，PIT 度量包含更多信息

D. During crisis periods, PIT approaches imply higher expected and unexpected loss (EL & UL) such that PIT tends to be pro-cyclical
在危机期间，PIT 方法意味着更高的预期和意外损失（EL & UL），因此 PIT 倾向于顺周期

答案:B
解析:
B 选项错误。正确的说法是：通过周期（TTC）是无条件的，而时点（PIT）是有条件的。
就（A）、（C）和（D）而言，均为**正确**说法。
关于（A）选项：外部评级机构的信用评级倾向于通过周期（TTC）。
关于（C）选项：信用利差属于时点（PIT），而 PIT 度量理论上包含更多信息。但请注意，这一点是有争议的，一些人认为某些 PIT 方法（如结构型方法）中很多所谓的信息其实是噪音伪装的。
关于（D）选项：在危机期间，PIT 方法意味着更高的预期和意外损失（EL & UL），因此 PIT 方法具有顺周期性。

21. Alice, Bert, Chris, Don, Eva, and Fred are individual investors. Each of them exhibits a particular behavioral bias.

- Alice (A) invested in the stock Cloudera (CLDR) and bad news (aka, new information) renders her original thesis obsolete but she is reluctant to sell today because an exit implies the realization of a -30.0% loss on the position and she much prefers to sell after her position experiences a double-digit gain

- Bert (B) can buy a new smartphone for $79.00 but he cannot resist a sale and prefers to pay $100.00 because it represents a 50.0% discount from the retail (MSRP) price
- Chris (C) attends an investment conference but he avoids the seminar focusing on recession risks because he is overweight homebuilders and he worries the topic will make him anxious with worry
- Don (D), who enjoys food shopping, tends to be price-conscious (e.g., he seeks bargains) when he pays cash at Sprouts or Trader Joes, but when he uses his Amazon credit card at Whole Foods he doesn't worry about the cost because he doesn't look at the statement for several days or weeks
- Eva (E) purchased Facebook (FB) at $160.00 because his firm's price target was $200.00, and he decides to ignore new information until the price reaches this level
- Fred (F) was previously a patient buy-and-hold investor who purchased high-conviction stocks and only checked his portfolio once a week, but last year he signed up for a subscription to Seeking Alpha and since that time his buy/sell transactions have quintupled because he's reading news about his portfolio holdings every day

Which of the following correctly matches the individual investor to his or her behavioral bias?
Alice、Bert、Chris、Don、Eva 和 Fred 是单独的投资者。他们每个人都有特定的行为偏差。

- Alice（A）投资了 Cloudera（CLDR）股票，坏消息（aka，新信息）使得她的原始观点过时，但她今天不愿意卖出，因为退出意味着 30.0% 的损失，她更喜欢在她的头寸经历双位数收益后再卖出
- Bert（B）可以以 $79.00 购买一部新智能手机，但他无法抗拒销售，更喜欢支付 $100.00，因为它代表了 50.0% 的折扣零售价（MSRP）
- Chris（C）参加投资会议，但避免参加关于经济衰退风险的研讨会，因为他超配了住房建筑商，担心这个话题会让他焦虑
- Don（D）喜欢食品购物，在 Sprouts 或 Trader Joes 使用现金时倾向于价格敏感（例如，他寻求折扣），但在 Whole Foods 使用亚马逊信用卡时他不担心成本，因为他几天或几周不看账单
- Eva（E）以 $160.00 购买了 Facebook（FB）股票，因为他的公司的目标价格是 $200.00，他决定忽略新信息，直到价格达到这个水平
- Fred（F）以前是一位持有并等待的投资者，购买了高信心股票，每周只检查一次投资组合，但去年他注册了 Seeking Alpha 订阅，从那时起他的买卖交易量增加了五倍，因为他每天阅读关于他的投资组合持仓的新闻

哪一项正确地将单独的投资者与其行为偏差相匹配?

A. A = Home bias, B = Mental accounting, C = Ostrich effect, D = Herding, E = Groupthink, F = Framing
A = 本土偏好，B = 心理会计，C = 鸵鸟效应，D = 羊群效应，E = 群体思维，F = 框架效应

B. A = Loss aversion, B = Framing, C = Ostrich effect, D = Mental accounting, E = Anchoring, F = Feedback effects
A = 损失规避，B = 框架效应，C = 鸵鸟效应，D = 心理会计，E = 锚定，F = 反馈效应

C. A = Groupthink, B = Herding, C = Home bias, D = Framing, E = Loss aversion, F = Ostrich effect
A = 群体思维，B = 羊群效应，C = 本土偏好，D = 框架效应，E = 损失规避，F = 鸵鸟效应

D. A = Mental accounting, B = Home bias, C = Loss aversion, D = Groupthink, E = Feedback effects, F = Herding
A = 心理会计，B = 本土偏好，C = 损失规避，D = 群体思维，E = 反馈效应，F = 羊群效应

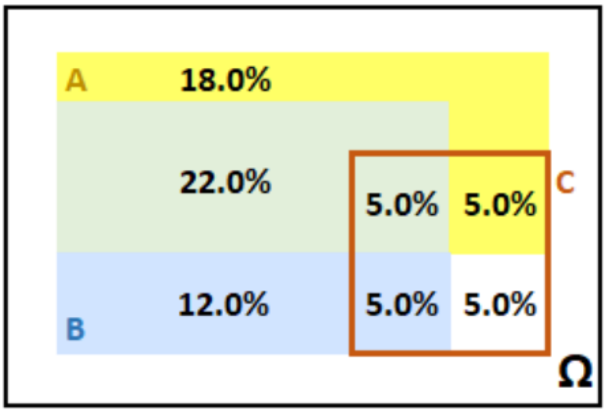
答案:B
解析:
B 选项正确。正确匹配：A = 损失规避，B = 框架效应，C = 鸵鸟效应，D = 心理会计，E = 锚定，F = 反馈效应
根据 GARP 教材 Box 8.3 的定义：[1]

- **锚定与参考（Anchoring and referencing）**：指用心理参照点为决策提供背景（如用已有价格点对比新价格是否具有吸引力）。锚点可能以非理性的方式影响决策过程。此外，一组相关决策中的多个参考点可能缺乏连贯性。
- **反馈效应（Feedback effects）**：正面反馈的频率与否会不理性地影响决策者坚持决策的能力。

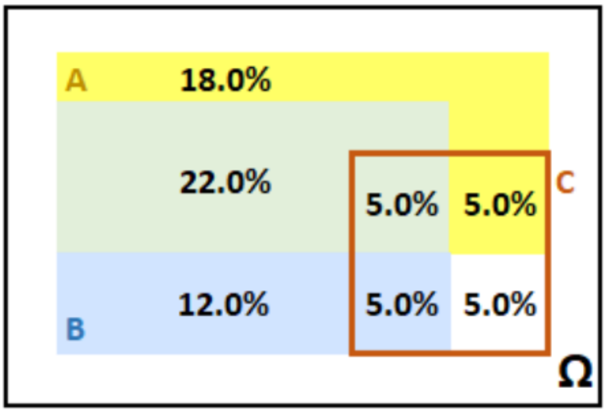
- **框架效应（Framing）**：选择被如何表述会推动决策者倾向于某种结果。例如，消费者可能为了省下 10 美元愿意为手机壳寻求 50% 优惠，但买 200 美元手机时却不愿意为这同样的 10 美元付出努力（因为在大额消费中，这只占较小比例）。
- **群体思维（Groupthink）**：指群体中的个体倾向于克服对高风险决策的质疑（或保持沉默）以维持群体共识。该共识本身可能受主导个体、有缺陷的目标设定或对模糊证据的选择性解读所影响。
- **羊群效应（Herding）**：投资者在投资或在动荡市场减少损失时，倾向于模仿他人行为。在风险管理中，羊群效应可能导致太多投资者使用同一风险指标或设置相同止损，进而引发剧烈的市场抛售。
- **本土偏好（Home bias）**：投资者偏向投资本国资产，而不是建立全球多元化投资组合，这可能是由于对海外市场的不确定性感到不安。
- 损失规避（Loss aversion）：实验证明，大多数人对潜在损失的敏感程度高于同等规模的潜在收益。这导致决策者倾向选择确定结果，而放弃可能获更大利润但存在更大风险的机会。损失规避不仅仅使人趋于保守，有时反而促使人非理性冒险以避免损失。（决策被表述为损失还是收益也非常重要。）
- 心理会计（Mental accounting）：人们会把资金分门别类对待，好像不同账户中资金彼此不可流通。例如，使用信用卡时花得更多，现金花得更谨慎。投资者可能对继承所得与博彩收益投入方式不同。如果涉及到承认损失或失误，他们可能不愿“关闭”心理账户。损失规避和其他行为偏差（如“沉没成本”处理）常使心理会计更加扭曲。
- 鸵鸟效应（Ostrich effect）：指非理性地回避坏消息，从而避免做出不舒服的决策或采取行动。例如，投资者可能更关注牛市而较少关注平淡或下跌的市场。（相反，过度关注每一次损失也会导致非理性厌恶。）

[1] 2020 FRM 第一部分：《风险管理基础》第十版，Pearson Learning Solutions, 2019 年 10 月

22. The probability graph below illustrates event A (the yellow rectangle) and event B (the blue rectangle). The unconditional probability of event A is 50.0% and the unconditional probability of event B is 44.0%; i.e., $\Pr(A) = 50.0\%$ and $\Pr(B) = 44.0\%$. Their overlap is graphed by the green rectangle such that $\Pr(A \cap B) = 27.0\%$. The orange rectangle conditions on the event C. For example, conditional on event C, there is a 50.0% probability that event A occurs, $\Pr(A|C) = 50.0\%$.



Which of the following is TRUE about, respectively, the unconditional and conditional relationship between events A and B?
事件 A（黄色矩形）和事件 B（蓝色矩形）的概率图说明了事件 A 的无条件概率为 50.0%，事件 B 的无条件概率为 44.0%；即 $\Pr(A) = 50.0\%$ 和 $\Pr(B) = 44.0\%$。它们的重叠用绿色矩形表示，使得 $\Pr(A \cap B) = 27.0\%$。橙色矩形条件于事件 C。例如，条件于事件 C，事件 A 发生的概率为 50.0%，$\Pr(A|C) = 50.0\%$。



哪一项关于事件 A 和 B 的分别在无条件和条件下的关系是正确的?

A. A and B are unconditionally dependent but conditionally (on event C) independent
A 和 B 是无条件依赖的，但在事件 C 的条件下是条件独立的

B. A and B are unconditionally dependent and also conditionally (on event C) dependent
A 和 B 是无条件依赖的，并且在事件 C 的条件下也是条件依赖的

C. A and B are unconditionally independent but conditionally (on event C) dependent
A 和 B 是无条件独立的，但在事件 C 的条件下是条件依赖的

D. A and B are unconditionally independent and also conditionally (on event C) independent
A 和 B 是无条件独立的，并且在事件 C 的条件下也是条件独立的

答案:A
解析:
A 选项正确。正确的说法是：A 和 B 是无条件依赖的，但在事件 C 的条件下是条件独立的。
如果 A 和 B 是无条件独立的，那么必须有 $\Pr(A \cap B) = P(A) \times P(B)$；然而，在这个例子中，$P(A) \times P(B) = 50.0\% \times 44.0\% = 22.0\%$，但 $\Pr(A \cap B) = 27.0\%$。因此，A 和 B 是无条件依赖的。
另一方面，A 和 B 是条件独立的，因为确实有 $\Pr(A \cap B|C) = \Pr(A|C) \times \Pr(B|C)$。$\Pr(A \cap B|C) = 5.0\%/20.0\% = 25.0\%$，而 $\Pr(A|C) \times \Pr(B|C) = (10.0\%/20.0\%) \times (10.0\%/20.0\%) = 0.50 \times 0.50 = 25.0\%$。

23. We are told to assume that an interest rate is 9.00% per annum with quarterly compounding. Each of the following statements is true EXCEPT which is false?
我们被告知假设一个年化利率为 9.00% 的利率，半年付息一次。以下哪一项陈述是正确的，除了哪一项是错误的?

A. The effective annual rate is higher than 9.00%
有效年利率高于 9.00%

B. The equivalent rate with continuous compounding is higher than 9.00%
等价连续利率高于 9.00%

C. The equivalent rate with monthly compounding (i.e., 12 periods per year) is lower than 9.00%
等价月利率低于 9.00%

D. The stated (aka, quoted, nominal) rate is given as 9.00% such that the periodic rate is 2.25%
名义利率为 9.00%，使得周期利率为 2.25%

答案:B
解析:
B 选项错误。实际上，等价的连续复利利率比 9.00% 更低。
关于 (A)、(C) 和 (D)，它们都是正确的。
具体来说，如果年化利率为 9.00%，按季度复利，则:

- 等价连续复利利率为 $4 \times \ln(1 + 9\%/4) = 8.90\%$
- 等价的按月复利年化利率为 $[(1 + 9\%/4)^{4/12} - 1] \times 12 = 8.933\%$
- 有效年利率（即等价的年复利利率）为 $(1 + 9\%/4)^4 - 1 = 9.3083\%$

如果名义利率（又称报价利率）为 9.00%，且每年有 4 个计息期（即季度复利），则每期利率为 $9.00\%/4 = 2.25\%$。

24. Quotecan Corporation has issued a bond. Quotecan's interest coverage ratio is within reasonable limits, providing adequate protection for investors in the near term. The bond is "better than junk" but carries more credit risk than bonds issued by the highest-rated corporations (e.g., Microsoft, Apple, Exxon Mobil, Johnson & Johnson, and Walmart). Quotecan's capacity to repay may be impaired if adverse economic conditions materialize, or if circumstances suddenly change. While the obligations are not speculative in their entirety, they do contain just a few speculative elements which may render the protective elements

unreliable over a longer time horizon. Based on these facts, which credit rating is most likely assigned to Quotecan's bond offering?
Quotecan Corporation 发行了一只债券。Quotecan 的利息覆盖率在合理范围内，为投资者提供了足够的保护。这只债券比垃圾债券好，但比最高评级的公司（例如，微软、苹果、埃克森美孚、强生和沃尔玛）发行的债券风险更大。如果出现不利经济条件或情况突然变化，Quotecan 的偿债能力可能会受到影响。虽然这些义务并非完全投机，但它们确实包含一些投机元素，这些元素可能在更长的时间范围内使保护性元素变得不可靠。基于这些事实，哪一项信用评级最有可能分配给 Quotecan 的债券发行?

A. A-1 or P-1
   A-1 或 P-1

B. AA or Aa
   AA 或 Aa

C. B or B
   B 或 B

D. BBB or Baa
   BBB 或 Baa

答案:D
解析:
D 选项正确。 BBB（标准普尔）或 Baa（穆迪）评级最有可能被分配给该债券；也就是说，这是投资级别中最低的评级。参考 `https://en.wikipedia.org/wiki/Credit_rating`。
根据标准普尔（S&P),“BBB”级别的债务表明具有充分的保护性参数。然而，如果经济条件不利或环境变化，债务人的偿付能力更有可能削弱。
根据穆迪（Moody's),“Baa”评级对应的是具有中等信用风险的债务，这些债务可能带有投机性特征。

25. GARP explains that "perhaps the biggest argument for ERM is that an enterprise-level perspective is the best way to prioritize risks and optimize risk management." Each of the following is one of the four key reasons that enterprise risk might demand the practice of (or, the art and science of) enterprise risk management, ERM, EXCEPT which is NOT one of the four key reasons?
GARP 解释说,“也许对企业风险管理（ERM）的最大论点是，企业层面的视角是优先考虑风险和优化风险管理的最佳方式。”以下哪一项是企业风险可能要求实践（或，企业风险管理（ERM）的艺术和科学）企业风险管理的四个关键原因之一，除了哪一项不是四个关键原因之一?

A. Identify hidden and/or dangerous concentrations including (for example) geographical, industry, product, or among suppliers
   识别隐藏和/或危险集中包括（例如）地理、行业、产品或供应商

B. Vertical vision recognizes potential threats to the whole enterprise ideally in their early stages to reduce the leverage effect of time
   垂直视角认识到整个企业的潜在威胁，理想情况下在早期阶段识别，以减少时间杠杆效应

C. Portfolio thinking permits the business units to specify a more accurate distribution that incorporates the allocation of corporate overhead
   投资组合思维允许业务单位指定更准确的分布，该分布考虑了企业 overhead 的分配

D. Thinking beyond silos acknowledges diversification among risk types and better understanding of how interactions might worsen enterprise threats
   超越 silos 的思考承认风险类型之间的多元化，并更好地理解如何交互可能恶化企业威胁

答案:C
解析:
C 选项错误。正确的说法是：将企业风险的更好理解（特别是可保风险）转化为成本节约或更好的资源分配。
企业风险可能要求实践（或，企业风险管理（ERM）的艺术和科学）企业风险管理的四个关键原因如下:

- 识别隐藏和/或危险集中包括（例如）地理、行业、产品或供应商

- 垂直视角认识到整个企业的潜在威胁，理想情况下在早期阶段识别，以减少时间杠杆效应
- 将企业风险的更好理解（特别是可保风险）转化为成本节约或更好的资源分配
- 超越 silos 的思考承认风险类型之间的多元化，并更好地理解如何交互可能恶化企业威胁

26. Yesterday a web page hosted by Acme received tens of thousands of page views but some were views by malicious bots. Acme utilizes two software applications to detect these malicious "bot-views." It uploads the same data file from yesterday to both applications. The first application detects 200 bot-views and the second application detects 300 bot-views. Among these, only 40 bot-views were detected by both applications. All bot-views are equally likely to be located, but clearly both applications only identify a minority of the bot-views (otherwise there would be a much higher number of identified bot-views common to both applications). Further, the identification of a bot-view by one application is independent of its identification by the other application. How many malicious bot-views did the web page experience on this day?
昨天 Acme 托管的网页收到了数万次页面浏览，但其中一些是由恶意机器人浏览的。Acme 使用两个软件应用程序来检测这些恶意"机器人浏览"。它将昨天的相同数据文件上传到两个应用程序。第一个应用程序检测到 200 个机器人浏览，第二个应用程序检测到 300 个机器人浏览。在这些中，只有 40 个机器人浏览被两个应用程序检测到。所有机器人浏览都同样可能被定位，但显然两个应用程序只识别了机器人浏览的少数（否则会有更多被两个应用程序共同识别的机器人浏览）。此外，一个应用程序识别机器人浏览与另一个应用程序识别机器人浏览是独立的。这个网页在这个日期的恶意机器人浏览数量是多少?

A. 300

B. 460

C. 540

D. 1,500

答案:D
解析:
D 选项正确。正确答案是：1,500
第二个应用程序在第一个应用程序已识别的机器人浏览中发现了 $40/200 = 20.0\%$，因此（考虑到独立性），我们可以推断它自己识别到的 300 个机器人浏览约占总数的 20.0%，据此估算总的机器人浏览数为 $300/20\% = 1,500$。同理，第一个应用程序在第二个应用程序已识别的机器人浏览中发现了 $40/300 = 13.33\%$，所以它自己检测到的 200 个也约占总数的 13.33%，因此 $200/13.33\% = 1,500$。

27. Consider a binary cash-or-nothing call option that pays \$100.00 if the stock is above \$100.00 at maturity in three months. The option's underlying non-dividend-paying stock is currently trading at \$85.00. The riskfree rate is 4.0% per annum with continuous compounding. We are also told that while $S = \$85.00$ and $K = \$100.00$, $N(d_1) = 0.12350$ and $N(d_2) = 0.100$, and the price of a vanilla three-month European call option on this stock is \$0.62. Which is nearest to the price of the cash-or-nothing call?
考虑一个二元现金或无价值看涨期权，如果在三个月后股票价格高于 \$100.00，则支付 \$100.00。该期权的标的非分红股票目前交易价格为 \$85.00。无风险利率为 4.0%，连续复利。我们还被告知，当 $S = \$85.00$ 和 $K = \$100.00$ 时，$N(d_1) = 0.12350$ 和 $N(d_2) = 0.100$，以及该股票的欧式三个月看涨期权的价格为 \$0.62。哪个最接近现金或无价值看涨期权的价值?

A. \$0.62

B. \$9.88

C. \$10.50

D. \$20.38

答案:B
解析:
B 选项正确。正确答案为：\$9.88

$\$100.00 \times 0.10 \times \exp(-4.0\% \times 0.25) = \$9.88$。

或者，我们可以通过资产或无价值期权的价格计算：$\$85.00 \times 0.12350 \times \exp(0\% \times 0.25) = \$10.50$，因此 $\$10.50 - \$0.62 = \$9.88$。

28. Rebecca is evaluating four different countries in an attempt to determine their respective default risk. She evaluates the countries in four categories: degree of indebtedness as measured by debt as a percentage of gross domestic product; social service/pension commitments as estimated by the average age of the population; nature of the economy (e.g., diverse versus concentrated in oil as a natural resource), and monetary policy. The four countries are summarized in the exhibit below:

Country	**Debt as % of GDP**	**Age of Population**	**Economy**	**Monetary Policy**
Country A	$> 100\%$	Old	Diverse	Independent CB
Country B	$\sim 100\%$	Young	Tourism	EU member
Country C	50%	Young	Oil	Autocracy
Country D	30%	Old	Diverse	Semi-independent CB

Which of the four countries is most likely to default?

Rebecca 正在评估四个不同的国家，以确定它们各自的违约风险。她从四个方面进行评估：以政府债务占国内生产总值（GDP）的百分比衡量的债务水平；以人口平均年龄估算的社会服务/养老金承诺；经济性质（如经济多元化还是依赖某一自然资源比如石油）；以及货币政策。下表总结了这四个国家的情况：

国家	**债务占 GDP 比例**	**人口年龄结构**	**经济类型**	**货币政策**
国家 A	$> 100\%$	老龄化	多元化	独立央行
国家 B	$\sim 100\%$	年轻	旅游业	欧盟成员
国家 C	50%	年轻	石油依赖	专制体制
国家 D	30%	老龄化	多元化	半独立央行

在这四个国家中，哪个国家最有可能发生违约?

A. Country A is the most likely to default because debt above 100% is highly predictive of default and overwhelms the other indicators
国家 A 最有可能违约，因为债务超过 100% 是违约的强烈预测指标，其他指标被压倒

B. Country B is the most likely to default because it is the only country with three (out of four) negative indicators
国家 B 最有可能违约，因为它是有四个指标中三个为负的唯一国家

C. Country C is the most likely to default because it has an autocratic government
国家 C 最有可能违约，因为它有一个专制政府

D. It is unclear: each has two negative (and two positive) indicators; further, quantification of this scorecard is an insufficient predictor
不清楚：每个国家都有两个负面（和两个正面）指标；此外，这个评分卡的量化是一个不充分的预测指标

答案:D

解析:

目前尚不清楚：每个国家都有两个负面（和两个正面）指标；此外，对于这个评分卡的量化无法充分预测违约。

Damodaran 指出："总之，要全面评估一个主权国家的违约风险，评估者需要超越表面数据，深入了解该国的经济运作方式、税收体系的强度以及其政府机构的可信度。"[1]

针对（A）、（B）和（C），每个都不是最佳答案：

关于（A），债务程度确实是"最合理的违约风险评估起点"，但"[债务最多的 20 个国家的名单] 表明，这一统计指标（政府债务占 GDP 的百分比）并不能完整衡量违约风险。这个名单上既有高违约风险的国家（希腊、刚果、牙买加、埃及），也有被评级机构和市场认为信用极高的国家（美国、日本、法国和加拿大）。"[1]

关于（B），国家 B 只有两个负面指标：高债务/GDP 和经济结构高度依赖旅游业。但根据 Damodaran 的观点，欧盟成员国身份实际上是正面因素："6. 来自其他实体的隐性支持：当希腊、葡萄牙和西班牙加入欧盟时，投资者、分析师和评级机构都下调了这些国家的违约风险评估。事实上，他们假定更强大的欧盟国家——德国、法国和斯堪的纳维亚国家——会介入保护较弱国家免于违约。当然，风险在于这种支持是隐性的而非明确的，债权人有可能因期望落空而蒙受损失，并且没有法律保障。"[1]

关于（C），Damodaran 指出："5. 政治风险：最终，是否违约既是经济决策也是政治决策。考虑到主权违约通常会给政治领导层带来压力，专制国家（因不太担心政治反弹）实际上更可能违约于民主国家。鉴于违约的另一种选择是印钞，中央银行的独立性和权力也会影响违约风险的评估。"[1]

29. GARP explains that the Basel Committee on Banking Supervision's standard number 239 (aka, BCBS 239) "was a major driver in the rise of the chief data officer (CDO) function. The CDO is typically responsible for standardizing a firm's approach to data management."[1] In addition, which of the following statements is also TRUE about the Basel Committee on Banking Supervision's standard number 239 (aka, BCBS 239)
[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019
GARP 解释说，巴塞尔银行监管委员会第 239 号标准（即 BCBS 239）"是推动首席数据官（CDO）职能崛起的主要驱动力。CDO 通常负责标准化公司对数据管理的处理方式。"[1] 此外，关于巴塞尔银行监管委员会第 239 号标准（即 BCBS 239），下列哪项说法同样是正确的?
[1] 2020 FRM 一级：风险管理基础，第 10 版。培生学习解决方案，2019 年 10 月

A. BCBS 239 offers the advantage of a uniform blueprint for a compliant IT infrastructure
BCBS 239 提供了符合 IT 基础设施的统一蓝图

B. Every systematically important financial institution (SIFI) complied with the original timeline
每个系统性重要金融机构（SIFI）都遵守了最初的时限

C. Compliance with BCBS 239 is a subjective exercise and the standards for each bank are bespoke
遵守 BCBS 239 是一项主观性工作，每家银行的合规标准都是量身定制的

D. While praised for its theory, critics say BCBS 239 forgets to address risk management data and models
虽然赞扬其理论，但批评者认为 BCBS 239 忽略了风险管理数据和模型

答案:C
解析:
C 选项正确。遵守 BCBS 239 是一项主观性工作，每家银行的合规标准都是量身定制的。
关于（A）、（B）和（D），每个都是错误的。相反，以下说法是正确的：

- "没有统一的蓝图适用于符合 BCBS 239 的 IT 基础设施，每家机构的解决方案都是特定的。最优的方法确保银行集团内的所有人员和系统都使用相同的数据、相同的模型和相同的假设。"GARP 如此解释。[1]
- "各家银行在遵守 BCBS 239 方面一直存在困难，没有任何一家银行按最初的时间表实现了完全合规。这主要是因为实现合规所需的 IT 重构极为复杂，原则本身又具有动态性。人工智能技术在大型数据集上的应用呈指数级增长，也使得遵守 BCBS 239 更具挑战性。"GARP 如此解释。[1]
- 尽管执行进度滞后，但 BCBS 239 至少在实践中尝试解决风险管理数据和模型的问题。

[1] 2020 FRM 一级：风险管理基础，第 10 版。培生学习解决方案，2019 年 10 月

30. For her investment firm, Audrey has identified six high-priority machine learning use cases. They are given below in three pairs:

- I. Segment customers into different groups; and perform sentiment analysis of 10K SEC filings
- II. Predict if borrower(s) will default; and forecast future stock price(s) based on fundamental data
- III. Decide how to split large-volume trades to sell quickly while minimizing the adverse price effect, and decide how much of a position to hedge using derivatives.

The three top-level machine learning approaches are supervised learning (S), unsupervised learning (U), and reinforcement learning (R). Which of the following is the best mapping of the advisable machine learning approach to its corresponding pair of use cases above?
Audrey 为其投资公司确定了六个高优先级的机器学习应用案例。它们如下分为三对：

- I. 将客户分为不同的群体；并对 10K SEC 文件进行情感分析
- II. 预测借款人是否会违约；并根据基本面数据预测未来股票价格
- III. 决定如何分割大量交易以快速出售，同时最小化不利价格影响，并决定使用衍生品对多少头寸进行对冲

三个顶级的机器学习方法分别是监督学习（S）、无监督学习（U）和强化学习（R）。哪一项是上述应用案例的最佳机器学习方法映射？

A. SRU: I=Supervised, II=Reinforcement, III=Unsupervised
SRU: I= 监督学习, II= 强化学习, III= 无监督学习

B. RUS: I=Reinforcement, II=Unsupervised, III=Supervised
RUS: I= 强化学习, II= 无监督学习, III= 监督学习

C. RUU: I=Reinforcement, II=Unsupervised, III=Unsupervised
RUU: I= 强化学习, II= 无监督学习, III= 无监督学习

D. USR: I=Unsupervised, II=Supervised, III=Reinforcement
USR: I= 无监督学习, II= 监督学习, III= 强化学习

答案:D
解析:
D 选项正确。正确的映射是：USR: I= 无监督学习, II= 监督学习, III= 强化学习
这三对映射如下：

- I. 无监督学习：客户细分意味着聚类，情感分析意味着自然语言处理（NLP）。
- II. 监督学习：违约预测和资产价格预测是监督学习的经典应用。
- III. 强化学习：正如 GARP 解释的那样，"强化学习在风险管理中有许多应用。例如，它用于确定购买或出售大量股票的最佳方式，确定如何管理投资组合，以及对冲衍生品组合。[第 14.1 节]……正如前面提到的，在金融领域也有很多潜在的应用，包括技术交易规则、确定如何分割大量交易以快速出售，同时最小化不利价格影响，以及确定使用衍生品对多少头寸进行对冲。"[第 14.7 节]

31. A non-dividend-paying stock has a current price of \$10.00 and a volatility of 25.0% per annum. The risk-free rate is 4.0% per annum with continuous compounding. Consider the following four spread trades:

- Trade #1 is a bull spread with calls. The strike prices are: $K_1 = \$9.00$, $K_2 = \$11.00$. The up-front cost is \$1.00.
- Trade #2 is a bull spread with puts. The strike prices are: $K_1 = \$9.00$, $K_2 = \$11.00$. The up-front cash flow is \$0.93.
- Trade #3 is a bear spread with puts. The strike prices are: $K_1 = \$8.00$, $K_2 = \$12.00$. The up-front cost is \$1.84.
- Trade #4 is a bear spread with calls. The strike prices are: $K_1 = \$8.00$, $K_2 = \$12.00$. The up-front cash flow is \$2.00.

In regard to each trade's maximum profit, each of the following statements is true EXCEPT which is false?
考虑以下四个价差交易：

- 交易 #1 是牛市看涨期权价差。行权价格为：$K_1 = \$9.00$，$K_2 = \$11.00$。初始成本为 \$1.00。
- 交易 #2 是牛市看跌期权价差。行权价格为：$K_1 = \$9.00$，$K_2 = \$11.00$。初始现金流量为 \$0.93。
- 交易 #3 是熊市看跌期权价差。行权价格为：$K_1 = \$8.00$，$K_2 = \$12.00$。初始成本为 \$1.84。
- 交易 #4 是熊市看涨期权价差。行权价格为：$K_1 = \$8.00$，$K_2 = \$12.00$。初始现金流量为 \$2.00。

关于每个交易的最大利润，下列陈述中哪一项是正确的，除了哪一项是错误的？

A. Trade #1 offers a maximum profit of \$1.00
交易 #1 提供最大利润为 \$1.00

B. Trade #2 offers a maximum profit of $0.93
交易 #2 提供最大利润为 $0.93

C. Trade #3 offers a maximum profit of $0.84
交易 #3 提供最大利润为 $0.84

D. Trade #4 offers a maximum profit of $2.00
交易 #4 提供最大利润为 $2.00

答案:C
解析:
C 选项错误。正确的说法是：交易 #3 提供最大利润为 $2.16。关于（A）、（B）和（D），每个都是正确的。

32. Political Risk Services (The PRS Group) provides numerical measures of country risk for more than a hundred countries. Its International Country Risk Guide (ICRG) rating "comprises 22 variables in three subcategories of risk: political, financial, and economic. A separate index is created for each of the subcategories. The Political Risk index is based on 100 points, Financial Risk on 50 points, and Economic Risk on 50 points. The total points from the three indices are divided by two to produce the weights for inclusion in the composite country risk score."[1]
As of July 2018, according to Damodaran, Taiwan's composite (ICRG) score was 85. Which of the following is TRUE?
政治风险服务（The PRS Group）为一百多个国家提供了国家风险的数值衡量。其国际国家风险指南（ICRG）评级"包括 22 个变量，分为三个子类别：政治、金融和经济。每个子类别都有一个单独的指数。政治风险指数基于 100 分，金融风险基于 50 分，经济风险基于 50 分。三个指数的总分除以 2，得到包含在综合国家风险得分中的权重。"[1]
截至 2018 年 7 月，根据 Damodaran，台湾的综合（ICRG）得分为 85。下列哪一项是正确的?
[1] 2020 FRM 一级：风险管理基础，第 10 版。培生学习解决方案，2019 年 10 月

A. Taiwan is very low risk
台湾风险非常低

B. This score is irrelevant for investment purposes
这个得分对投资目的无关紧要

C. This score can be neither normalized nor standardized
这个得分既不能标准化也不能标准化

D. This score cannot be used to rank Taiwan against other countries
这个得分不能用于将台湾与其他国家进行排名

答案:A
解析:
A 选项正确。正确的说法是：台湾风险非常低。综合得分，从零到 100，然后分为非常低风险（80 到 100 分）到非常高风险（零到 49.9 分）。
关于（B）、（C）和（D），每个都是错误的。
关于错误的（B），"irrelevant（无关紧要）" 用得太过头了，因为这个指数对金融风险和经济风险各自赋予了 25% 权重（即在 200 分满分中各占 50 分）。Damodaran 的本意并没有这么绝对，他认为："测量模型/方法：许多开发该方法并将其转换为分数的实体并非商业实体，并且会考虑一些与企业关联性较小的风险。实际上，这些服务中的部分分数更多地面向政策制定者和宏观经济学家，而不主要针对企业。"[1]
关于错误的（C），标准化（normalize）是将数据转换到 $(0, 1)$ 区间，标准差标准化（standardize）则是转换为标准正态分布 $N(0, 1)$。我们完全可以对 PRS 分数进行这两种处理。
关于错误的（D），这些分数是有序的（可用于排名），但不是区间尺度或比例尺度。Damodaran 认为："更像排名而不是分值：即使你只使用同一家服务机构的数据，国家风险分数更适合用来对各国进行排名，而不是用来衡量相对风险大小。因此，在 PRS 评分机制下，分数为 80 的国家确实比分数为 40 的国家更安全，但如果认为 80 分就意味着安全性是 40 分国家的两倍，那就是危险的解读。"[1]
[1] Aswath Damodaran, Country Risk: Determinants, Measures and Implications –The 2018 Edition (July 23, 2018). 另见 `https://www.prsgroup.com/explore-our-products/international-country-risk-guide/`

33. Consider the following three portfolios where we know the expected excess returns of Portfolios A and B; i.e., $E[ER(A)] = 6.0\%$ and $E[ER(B)] = 8.4\%$ and these in excess of the riskfree rate. For Portfolios A and B, we also know the factor betas: $\beta(A,1) = 0.40$, $\beta(A,2) = 1.20$, $\beta(B,1) = 0.80$, $\beta(B,2) = 1.50$. The unknowns are the two risk premiums on the two factors, $F(1)$ and $F(2)$:

- Portfolio A: $E[ER(A)] = \beta(A,1) \times F(1) + \beta(A,2) \times F(2) = 0.40 \times F(1) + 1.20 \times F(2) = 6.0\%$
- Portfolio B: $E[ER(B)] = \beta(B,1) \times F(1) + \beta(B,2) \times F(2) = 0.80 \times F(1) + 1.50 \times F(2) = 8.4\%$
- Portfolio C has the following betas: $\beta(C,1) = 1.30$ and $\beta(C,2) = 0.90$.

If Portfolio C has betas as given by $\beta(C,1) = 1.30$ and $\beta(C,2) = 0.90$, what is the expected excess return of Portfolio C?
考虑以下三个投资组合，我们知道投资组合 A 和 B 的预期超额回报，即 $E[ER(A)] = 6.0\%$ 和 $E[ER(B)] = 8.4\%$，这些超额回报高于无风险利率。对于投资组合 A 和 B，我们还知道因子贝塔：$\beta(A,1) = 0.40$，$\beta(A,2) = 1.20$，$\beta(B,1) = 0.80$，$\beta(B,2) = 1.50$。未知的是两个因子风险溢价，$F(1)$ 和 $F(2)$：

- 投资组合 A：$E[ER(A)] = \beta(A,1) \times F(1) + \beta(A,2) \times F(2) = 0.40 \times F(1) + 1.20 \times F(2) = 6.0\%$
- 投资组合 B：$E[ER(B)] = \beta(B,1) \times F(1) + \beta(B,2) \times F(2) = 0.80 \times F(1) + 1.50 \times F(2) = 8.4\%$
- 投资组合 C 的贝塔为：$\beta(C,1) = 1.30$ 和 $\beta(C,2) = 0.90$。

如果投资组合 C 的贝塔为 $\beta(C,1) = 1.30$ 和 $\beta(C,2) = 0.90$，那么投资组合 C 的预期超额回报是多少?

A. 5.00%
   5.00%

B. 7.50%
   7.50%

C. 10.00%
   10.00%

D. Not enough information
   没有足够的信息

答案:B
解析:
B 选项正确。正确的答案是：7.50%。我们有两个方程和两个未知数，因此必须有 $F(1) = 3.0\%$ 和 $F(2) = 4.0\%$。如果投资组合 C 的贝塔为 $\beta(C,1) = 1.30$ 和 $\beta(C,2) = 0.90$，那么投资组合 C 的预期超额回报为 $E[ER(C)] = 1.30 \times 3.0\% + 0.90 \times 4.0\% = 7.50\%$。

34. Robert is trying to price an exotic path-dependent option (Price(E)) for which no analytical solution exists. He uses a Black-Scholes option pricing model (BSM OPM) for vanilla options, referring to its analytical price as BSM(V). He also employs a Monte Carlo simulation (MCS) to estimate the exotic option's price as MCS(E). He re-uses the same simulated paths to get a simulated value for the vanilla option, MCS(V). He then applies an error correction formula:
$Price(E) = MSC(E) + \beta \times [BSM(V) - MCS(V)]$.
The goal is to improve accuracy by reducing variance. The text states that a good control variate should be inexpensive to construct, and asks for the other property that Bob seeks.
Robert 正在尝试为一种不存在解析解的奇异路径依赖期权（Price(E)）定价。他使用 Black-Scholes 期权定价模型（BSM OPM）为普通期权定价，将其解析价格称为 BSM(V)。他还采用蒙特卡洛模拟（MCS）估计奇异期权的价值为 MCS(E)。他重新使用相同的模拟路径得到普通期权的模拟价值 MCS(V)。然后他应用误差修正公式：
$Price(E) = MSC(E) + \beta \times [BSM(V) - MCS(V)]$.
目标是通过减少方差来提高准确性。文本指出一个好的控制变量应该易于构建，并询问 Bob 寻求的另一个属性。

A. High correlation between Price(E) and BSM(V)
   Price(E) 和 BSM(V) 之间的高相关性

B. An analytical function for the price of the path-dependent option, Price(X)
   一个解析函数，用于路径依赖期权的定价，Price(X)

C. A random number generator (RNG) that is neither pseudo-random nor deterministic but genuinely random
一个既不是伪随机也不是确定性的随机数生成器（RNG），而是真正的随机数

D. A second set of random variables that by construction have a negative correlation with the iid variables used in the simulation
一个第二组随机变量，通过构造与模拟中使用的独立同分布（iid）变量具有负相关性

答案:A
解析:
A 选项正确。正确：Price(E) 和 BSM(V) 之间的高相关性。
按照教材中的记号，未知分布为 $g(X)$，蒙特卡洛模拟用于逼近其期望值 $E[g(X)]$，但这个近似实际上可以表示为 $E[g(X(i))]+\eta(i)$，其中 $\eta(i)$ 是均值为零的误差项。控制变量（control variate）是另一个随机变量 $h(X(i))$，它与误差项相关。在本题中，Bob 估算 $E[Price(E)] = g(X)$，而 BSM(V) 是其控制变量。
根据教材，"一个好的控制变量应具备两个特性：

1. 首先，它应该便于从 x_i 构造。如果控制变量计算缓慢，相较于构建控制变量，更大的方差降低可以通过在相同计算成本下增加模拟次数 b 获得。

2. 其次，控制变量应与 $g(X)$ 高度相关。最优的组合系数 β，即能最小化近似误差的 β，可通过回归模型估算：$g(x(i)) = \alpha + \beta h(X(i)) + v(i)$。

35. Every year, Acme Corporation grants stock options to its executives and employees. Like most of its peers, the employee stock options (ESOs) vest over four years and always have a 10-year term. In the event of termination, depending on the particulars, options sometimes accelerate and sometimes are forfeited. Newly recruited executives often receive abnormally large, front-loaded option grants. All option grants always have a fixed strike price, and to date, Acme has never re-priced grants. In the majority of cases, Acme grants at-the-money (ATM) options; in rare cases, Acme grants out-of-the-money (OTM) options. However, Acme never grants in-the-money (ITM) ESOs. To value ESOs, the company has developed basic and advanced versions of both the Black-Scholes-Merton (BSM) and the binomial (aka, lattice) models.
Which of the following statements is TRUE?
每年，Acme Corporation 都会授予其高管和员工股票期权。与大多数同行一样，员工股票期权（ESOs）在四年内行权，并且总是有 10 年的期限。在终止的情况下，根据具体情况，期权有时会加速行权，有时会被放弃。新招聘的高管通常会收到异常大额的前置授予。所有期权授予都始终有一个固定行权价格，到目前为止，Acme 从未重新定价授予。在大多数情况下，Acme 授予平价（ATM）期权；在极少数情况下，Acme 授予价外（OTM）期权。然而，Acme 从未授予价内（ITM）ESOs。为了对 ESOs 进行估值，公司开发了 Black-Scholes-Merton（BSM）模型和二项式（aka，格子）模型的基本和高级版本。下列哪一项陈述是正确的？

A. In theory, the binomial model is better than the BSM model to price the company's ESOs
理论上，二项式模型比 BSM 模型更适合为公司定价 ESOs

B. The ESO grants do not dilute the company's shareholders because the company never grants in-the-money options
ESO 授予不会稀释公司的股东，因为公司从未授予价内（ITM）期权

C. The ESO grants do not reduce the company's reported profit because the ESOs have zero intrinsic value at grant
ESO 授予不会减少公司的报告利润，因为 ESOs 在授予时没有内在价值

D. The fair market value (FMV) of the ESOs is equal to the standard BSM value assuming a 10-year term; i.e., ESO FMV = $c = c(S, K, \sigma, 10, r, q)$
ESO 的公平市场价值（FMV）等于假设 10 年期限的标准 BSM 价值；即，ESO FMV = $c = c(S, K, \sigma, 10, r, q)$

答案:A
解析:
A 选项正确。理论上，二项式模型（Binomial model）比 BSM 模型更适合为公司定价 ESOs，因为 ESOs 为美式期权（带有归属/失效等美式期权特性，二项式模型可以自然处理），而不是欧式期权。需要注意的是，这里说的只是理论上二项式模型"更好"；实际上，许多公司也会采用 BSM 模型给 ESOs 定价，并且（例如）用更短的期望剩余期限代替 10 年期作为期权期限参数。
针对选项 B、C 和 D，均为错误表述。相应正确的说法如下：

- ESO 的授予会稀释公司的股东权益；
- ESO 的授予会减少公司的报告利润，因为按照会计准则需要计入费用；
- ESO 的价值低于欧式期权 BSM 公式计算的价值 $c = c(S, K, \sigma, 10, r, q)$，主要因为（一）归属限制，和（二）其相对缺乏流动性（与普通可交易看涨期权相比）。

36. A bond portfolio with a face value of \$7.0 million is exposed to the following two key rates:

- The 2-year key rate, KR01(2-year) is equal to \$0.030 per 100 face amount and has a daily volatility, σ(bps), of 12.0 basis points
- The 5-year key rate, KR01(5-year) is equal to \$0.090 per 100 face amount and has a daily volatility, σ(bps), of 25.0 basis points

As the curve experiences significant non-parallel shifts, the correlation between the key rates is only 0.440. Which of the following is nearest to the daily volatility of the portfolio (in dollars)?
一个面值为 \$7.0 百万的债券组合暴露于以下两个关键利率：

- 2 年期关键利率，KR01(2-year) 等于 \$0.030 每 100 面值，每日波动率，σ(bps)，为 12.0 基点
- 5 年期关键利率，KR01(5-year) 等于 \$0.090 每 100 面值，每日波动率，σ(bps)，为 25.0 基点

当曲线经历显著的非平行移动时，两个关键利率之间的相关性仅为 0.440。下列哪一项最接近于该组合的每日波动率（以美元为单位）？

A. \$159,500

B. \$170,100

C. \$182,700

D. \$219,300

答案:B
解析:
B 选项正确。正确的答案是：\$170,100。
以美元计算，投资组合波动率 $= \sqrt{(\$2,100.0)^2 \times (12.0)^2 + (\$6,300.0)^2 \times (25.0)^2 + 2 \times \$2,100.0 \times \$6,300.0 \times 12.0 \times 25.0 \times 0.440} =$ $\$170,000.00$，其中 $\$2,100 = \$0.030 \times \$7,000,000/100$ 和 $\$6,300 = \$0.090 \times \$7,000,000/100$。
以百分比计算，投资组合波动率 $= \sqrt{(0.00030)^2 \times (12.0)^2 + (0.00090)^2 \times (25.0)^2 + 2 \times 0.00030 \times 0.00090 \times 12.0 \times 25.0 \times 0.440} =$ 2.430%，因此 $2.430\% \times \$7,000,000 = \$170,100.00$。

37. During a 36-month period during which the risk-free rate was 2.0%, consider a comparison between the market portfolio and two fund managers, Betty and Peter:

- The market portfolio's excess return was +6.0% (its gross return was +8.0%) with a volatility of 24.0% per annum
- Peter's Portfolio: Excess return = +7.0% (gross return = +9.0%), Volatility = 36.0% per annum, Beta $\beta(P, M) = 0.750$
- Betty's Portfolio: Excess return = +11.0% (gross return = +13.0%), Volatility = 44.0% per annum, Beta $\beta(B, M) = 1.650$

If the capital asset pricing model (CAPM) is valid, then each of the following statements is true EXCEPT which is false?
在风险自由利率为 2.0% 的 36 个月期间，考虑市场投资组合和两个基金经理，Betty 和 Peter：

- 市场投资组合的超额回报为 +6.0%（总回报为 +8.0%），波动率为 24.0% 每年
- Peter 的投资组合：超额回报 = +7.0%（总回报 = +9.0%），波动率为 36.0% 每年，贝塔 $\beta(P, M) = 0.750$
- Betty 的投资组合：超额回报 = +11.0%（总回报 = +13.0%），波动率为 44.0% 每年，贝塔 $\beta(B, M) = 1.650$

如果资本资产定价模型（CAPM）有效，那么下列陈述中哪一项是正确的，除了哪一项是错误的?

A. If the investor is diversified (or the portfolio represents only a fraction of the investor's entire wealth), the Treynor (TPI) is a good measure and Peter's TPI is better than Betty's
如果投资者是分散的（或投资组合仅代表投资者总财富的一部分），Treynor（TPI）是一个好的衡量标准，Peter 的 TPI 比 Betty 的更好

B. If the investor is not diversified (or the portfolio represents the investor's entire wealth), the Sharpe (SPI) is a good measure and Betty's SPI is better than Peter's
如果投资者不是分散的（或投资组合代表投资者的总财富），Sharpe（SPI）是一个好的衡量标准，Betty 的 SPI 比 Peter 的更好

C. Betty's portfolio already lies on the efficient frontier such that we do not expect her to sustain further improvement in the reward-to-variability ratio
Betty 的投资组合已经位于有效前沿，因此我们不期望她在奖励-波动比率上进一步改善

D. If Betty's and Peter's portfolio belong to different peer groups, Jensen's alpha is a good measure for ranking them and Betty's (Jensen's) alpha is better than Peter's
如果 Betty 和 Peter 的投资组合属于不同的同行组，Jensen's alpha 是一个好的衡量标准来排名他们，Betty 的 (Jensen's) alpha 比 Peter 的更好

答案:D
解析:
D 选项错误。 False, doubly: if the portfolios belong in SIMILAR peer groups (i.e., similar risk levels), Jensen's alpha is good for ranking and Peter's is better than Betty's.
In regard to (A), (B), and (C), each is TRUE.
For (A), If the investor is diversified (or the portfolio represents only a fraction of the investor's entire wealth), the Treynor (TPI) is a good measure. Peter's TPI is 9.33% and Betty's TPI is 6.67%.
For (B), If the investor is not diversified (or the portfolio represents the investor's entire wealth), the Sharpe (SPI) is a good measure. Peter's SPI is 0.194 and Betty's is 0.250.
For (C), Betty's portfolio already lies on the efficient frontier because her Sharpe ratio equals the Market's, which is also 6.0%/24% = 0.250; we do not expect portfolios to be located beyond the efficient portfolio.

38. Emily listed the daily returns last week for two cryptocurrencies, bitcoin (BTC) and Ethereum (ETH). These are very small samples of only five returns for each currency. The returns are displayed below along with their ranks; e.g., Wednesday saw the worst performance for both cryptocurrencies, while Tuesday saw the best performance. Albeit a very small sample, the ranks do match on three of the days (Monday, Tuesday, and Wednesday) and differ on only two days (Thursday and Friday).

Day (sort by BTC)	Return (as %) of BTC	Return (as %) of ETH	Rank of BTC	Rank of ETH
Wed	-3.50	-4.70	1	1
Thur	-2.90	0.75	2	3
Fri	1.40	0.30	3	2
Mon	4.70	2.80	4	4
Tue	5.80	9.90	5	5

What are, respectively, the Spearman's rank and Kendall's tau?
Emily 列出了上周两个加密货币，比特币（BTC）和以太坊（ETH）的每日回报。这些是每个加密货币只有五个回报的非常小的样本。回报显示如下，包括它们的排名；例如，周三对于两种加密货币都是最差的表现，而周二则是最佳表现。尽管这是一个非常小的样本，但三天（周一、周二和周三）的排名匹配，只有两天（周四和周五）不同。

Day (sort by BTC)	Return (as %) of BTC	Return (as %) of ETH	Rank of BTC	Rank of ETH
Wed	-3.50	-4.70	1	1
Thur	-2.90	0.75	2	3
Fri	1.40	0.30	3	2
Mon	4.70	2.80	4	4
Tue	5.80	9.90	5	5

分别是什么是斯皮尔曼排名和肯德尔 tau?

A. -0.70 (rank) and -0.55 (Kendall's)
   -0.70 (排名) and -0.55 (肯德尔's)

B. 0.33 (rank) and 0.67 (Kendall's)
   0.33 (排名) and 0.67 (肯德尔's)

C. 0.90 (rank) and 0.80 (Kendall's)
   0.90 (排名) and 0.80 (肯德尔's)

D. 1.40 (rank) and 1.60 (Kendall's)
   1.40 (排名) and 1.60 (肯德尔's)

答案:C
解析:
Option C is correct. True: 0.90 (rank) and 0.80 (Kendall's)
The Spearman's rank correlation is given by $1 - \frac{6\times 2}{((5^2-1)\times 5)} = 0.90$. Given there are nine (9) concordant pairs and only one (1) discordant pair, the Kendall's tau is given by $\frac{9-1}{(5\times 4/2)} = 0.80$.
考察: Quantitative Methods: Correlation　**重要程度:** [■■□]
科目: Quantitative Analysis

39. The spot price of oil is \$100.000 per barrel while the riskfree rate is 3.0% and its storage cost is 2.0% per annum. Oil has positive systematic risk. Specifically, its beta with respect to the market, $\beta(\text{Oil, M}) = 1.10$. The market's expected return is 14.0%; i.e., the market's expected excess return is +11.0%. Finally, the expected growth rate of the oil price (aka, price appreciation) is +8.0%. All rates are per annum with continuous compounding. We are interested in the traded price of the fifteen-month (i.e., 1.25 year) forward contract, and we'll assume the traded price equals the theoretical price. Which of the following statements is true about (i) the slope of the forward curve and (ii) whether we expect normal contango or normal backwardation?
石油的即期价格为每桶 \$100.000，无风险利率为 3.0%，存储成本为每年 2.0%。石油具有正的系统性风险。具体来说，其相对于市场的贝塔系数，$\beta(\text{Oil, M}) = 1.10$。市场预期回报率为 14.0%；即，市场的预期超额回报为 +11.0%。最后，石油价格的预期增长率（aka，价格增值）为 +8.0%。所有利率均为年化，并采用连续复利。我们感兴趣的是十五个月（即 1.25 年）远期合约的交易价格，我们假设交易价格等于理论价格。下列陈述中哪一项是关于（i）远期曲线的斜率和（ii）我们预期正常正向市场还是正常反向市场?

A. The forward curve is inverted, $F(1.25) = \$97.50$, and we perceive normal contango, $F(1.5) > E[S(1.25)]$
   远期曲线是反向的，$F(1.25) = \$97.50$，我们预期正常正向市场，$F(1.5) > E[S(1.25)]$

B. The forward curve is inverted, $F(1.25) = \$95.00$, and we perceive normal backwardation, $F(1.5) < E[S(1.25)]$
   远期曲线是反向的，$F(1.25) = \$95.00$，我们预期正常反向市场，$F(1.5) < E[S(1.25)]$

C. The forward curve is upward-sloping, $F(1.25) = \$105.00$, and we perceive normal contango, $F(1.5) > E[S(1.25)]$
   远期曲线是向上的，$F(1.25) = \$105.00$，我们预期正常正向市场，$F(1.5) > E[S(1.25)]$

D. The forward curve is upward-sloping, $F(1.25) = \$102.50$, and we perceive normal backwardation, $F(1.5) < E[S(1.25)]$
远期曲线是向上的，$F(1.25) = \$102.50$，我们预期正常反向市场，$F(1.5) < E[S(1.25)]$

答案:B
解析:
B. True: The forward curve is inverted, $F(1.25) = \$95.00$, and we perceive normal backwardation, $F(1.5) < E[S(1.25)]$
The expected future spot price is given by, $E[S(1.25)] = S(0) * \exp(g * T) = \$100.0 * \exp(8.0\% * 1.25) = \$110.517$; i.e., the current spot projected at its growth rate.
The discount rate is given by $k = 3.00\% + 11.00\% * 1.10 = 15.10\%$; i.e., per the CAMP where $\beta(\text{Oil, M}) = 1.10$. The theoretical futures price is given by $F(1.25) = E[S(1.25)] * \exp[(RF - k) * T] = \$100.517 * \exp[(3.00\% - 15.10\%) * 1.25] = \$95.004$.
考察: Finance: Derivatives **重要程度:** [■■□]
科目: Quantitative Analysis

40. Peter is a Financial Analyst whose boss asked him to develop a model, or if necessary a set of models, that are multi-factor risk models for the firm's fixed income and derivative investment portfolios. Specifically, there are four objectives as follows:
I. To measure and hedge the risk of bond portfolios in terms of the relatively small number of liquid bonds available, in particular, on-the-run government bonds
II. To measure and hedge the risk of a portfolio of swaps in terms of the highly liquid money market and swap instruments, which are more numerous
III. To measure the risk of mixed portfolios not terms of other securities but instead to model direct changes to the shape of the term structure
IV. To measure portfolio volatility by assuming a term structure of volatility but doing so without an excessive "curse of dimensionality;"i.e., avoiding too many correlation pairs
For these four objectives, Peter has three basic multi-factor risk metrics: key-rate exposures, partial '01s, and forward-bucket '01s. For which of the four objectives is key-rate exposures best suited?
彼得是一名金融分析师，他的老板要求他开发多因子风险模型，用于公司固定收益和衍生品投资组合的风险衡量。具体来说，有四个目标：

- I. 衡量和对冲债券组合风险，特别是可用流动性债券数量相对较少的情况，特别是近期发行的政府债券
- II. 衡量和对冲互换组合风险，特别是高流动性货币市场和互换工具数量较多的情况
- III. 衡量混合投资组合的风险，而不是其他证券，而是直接改变期限结构
- IV. 假设期限结构波动，但避免过多的"维度诅咒"；即，避免过多的相关性对

彼得有三个基本多因子风险指标：关键利率暴露，部分'01s，和前向桶'01s。对于哪一项目标，关键利率暴露最适合?

A. I. only
只有 I.

B. III. only
只有 III.

C. I. and IV. only
只有 I. 和 IV.

D. I., II., III. and IV.; i.e., all four
I., II., III. 和 IV.; 即，所有四个

答案:C
解析:
C 选项正确。

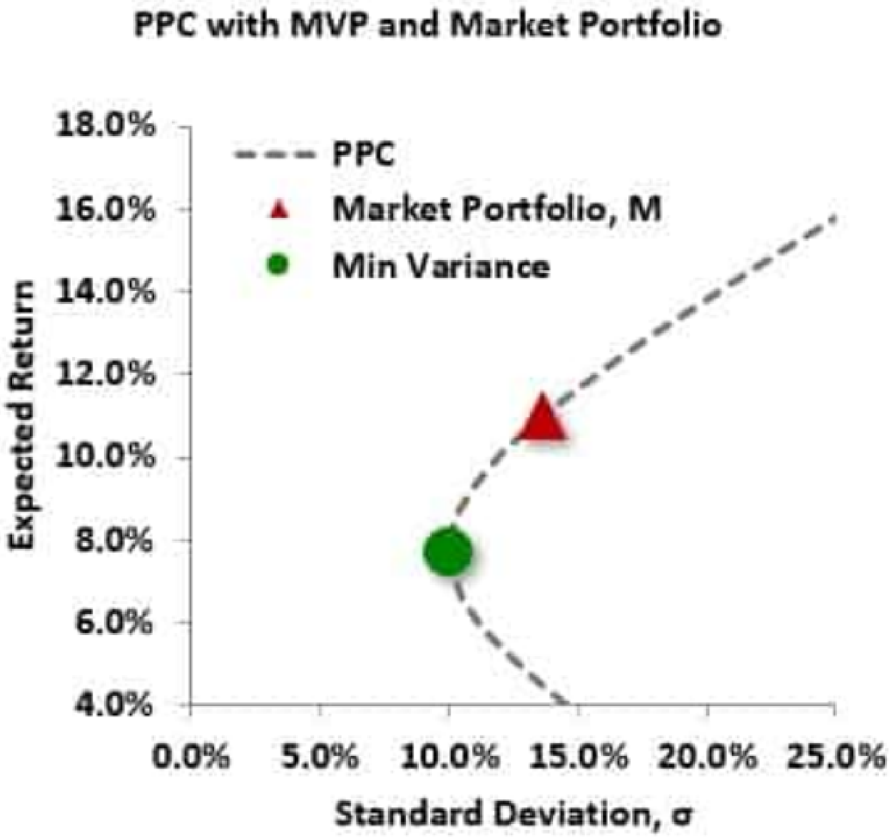
- *Key-rate exposures* are used for measuring and hedging the risk of bond portfolios in terms of a relatively small number of the most liquid bonds available, usually the most recently issued, near-par, government bonds.

- *Partial '01s* are used for measuring and hedging the risk of portfolios of swaps or portfolios that contain both bonds and swaps in terms of the most liquid money market and swap instruments. As these instruments are almost always those whose prices are used to build a swap curve, the number of securities used in this methodology is usually greater than the number used in a key-rate framework.

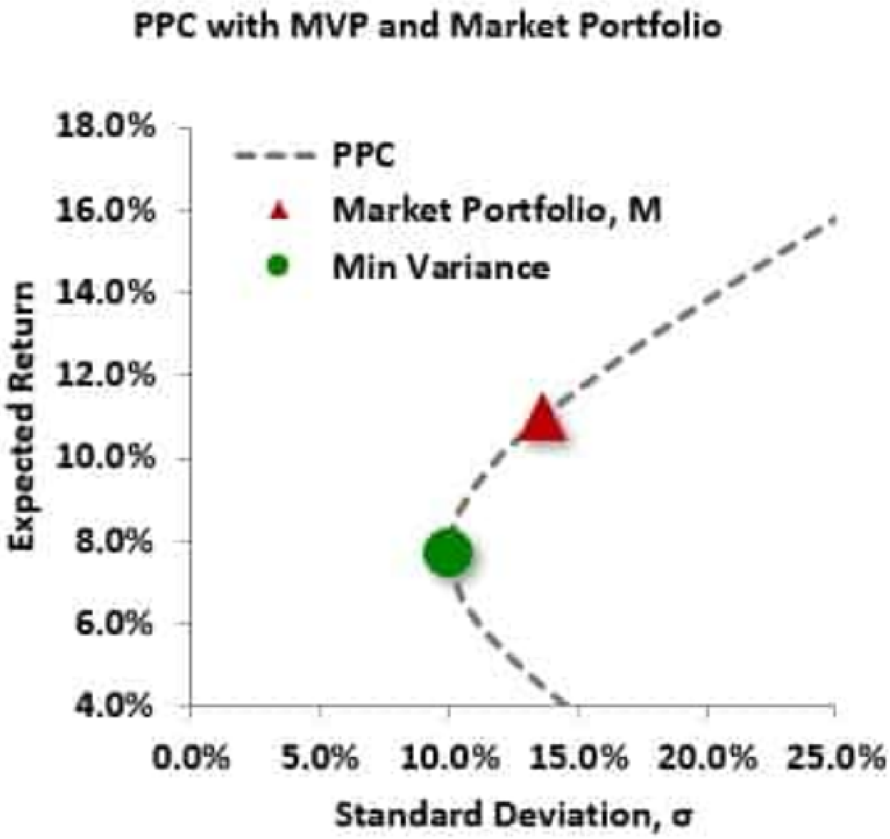
- Finally, *forward-bucket '01s*, mostly used in the swap or combined bond and swap contexts as well, measure the risk of a portfolio not in terms of other securities but in terms of direct changes in the shape of the term structure. As a result, forward-bucket '01s are often the most intuitive way to understand the curve risks of a portfolio, but not the quickest way to see which hedges are required to neutralize such risks. This chapter concludes with a comment on the use of these methods to measure the volatility of a portfolio.

考察: Finance: Fixed Income Analysis　　**重要程度:** [■■□]
科目: Quantitative Analysis

41. The plot below illustrates an actual portfolio possibilities curve (PPC, dashed line). Two prominent portfolios (minimum variance and market portfolio) are also plotted.



The green point is the minimum variance portfolio, MVP, which has an expected return of 7.70% and standard deviation, σ, of 10.0%. The red triangle is the Market Portfolio that has an expected return of 11.0% and standard deviation, σ, of 13.7%. If the riskfree rate is 4.0%, what is the formula for the capital market line (CML)?
下图显示了一个实际的投资组合可能性曲线（PPC，虚线）。图中还绘制了两个重要的投资组合（最小方差组合和市场投资组合）。



绿色点表示最小方差投资组合（MVP），其期望收益为 7.70%，标准差 σ 为 10.0%。红色三角形表示市场投资组合，其期望收益为 11.0%，标准差 σ 为 13.7%。如果无风险利率为 4.0%，那么资本市场线（CML）的公式是什么?

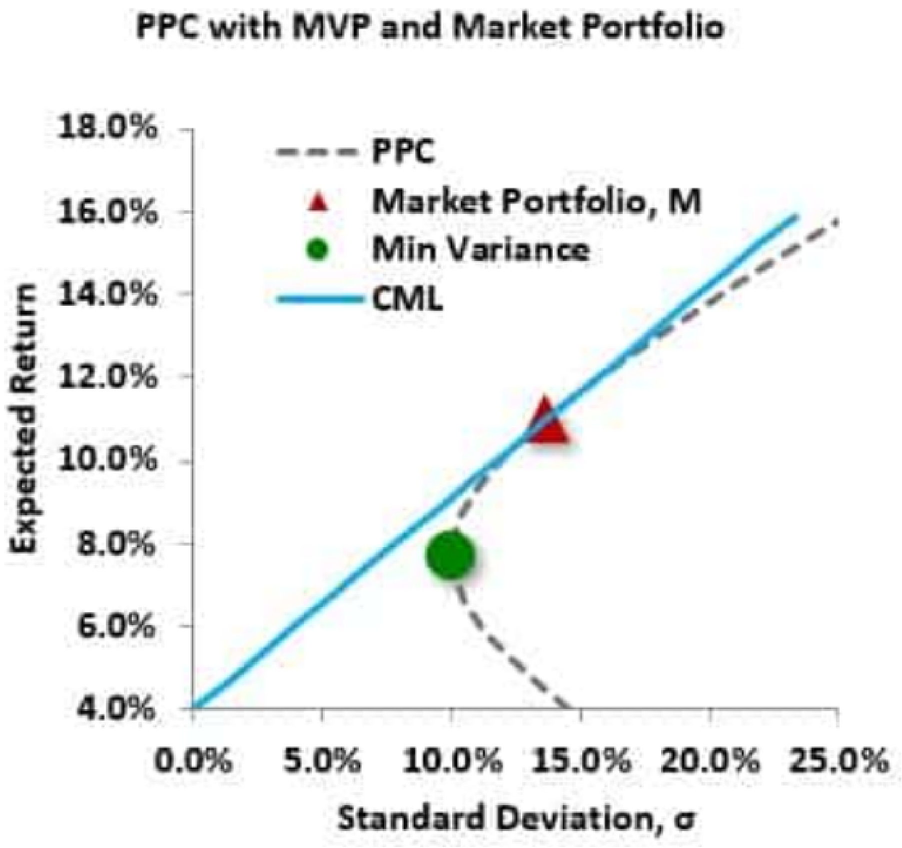
A. $E(R) = 0.040 + 0.511 \times \sigma(P)$
   资本市场线公式：$E(R) = 0.040 + 0.511 \times \sigma(P)$

B. $E(R) = 0.137 + 0.511 \times \sigma(P)$
   资本市场线公式：$E(R) = 0.137 + 0.511 \times \sigma(P)$

C. $E(R) = 0.040 + 0.375 \times \beta(P, M)$
   资本市场线公式：$E(R) = 0.040 + 0.375 \times \beta(P, M)$

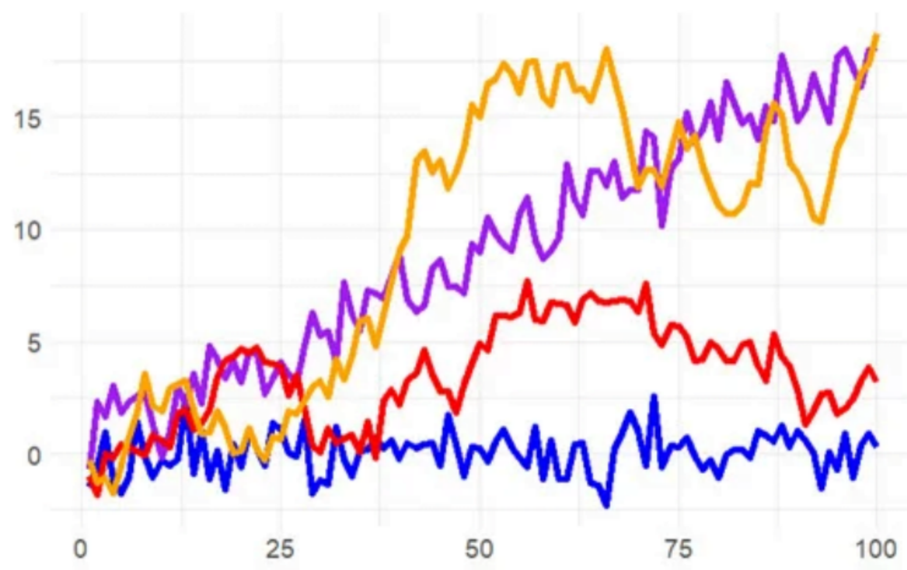
D. $E(R) = 0.077 + 0.511 \times \beta(P, M)$
   资本市场线公式：$E(R) = 0.077 + 0.511 \times \beta(P, M)$

答案:A
解析:
Option A is correct. A. True: the CML is here given by $E(R) = 0.040 + 0.511 \times \sigma(P)$
The capital market line (CML) must have the y-intercept intercept at the riskfree rate, which was given as 4.0%. Its tangent is the market portfolio. The slope of the CML is the market portfolio's Sharpe ratio, which is $(11.0\% - 4.0\%)/13.7\%$. Therefore, the CML is given by $E(R) = 0.040 + (0.110 - 0.040)/0.1370 \times \sigma(P) = 0.040 + 0.511 \times \sigma(P)$.



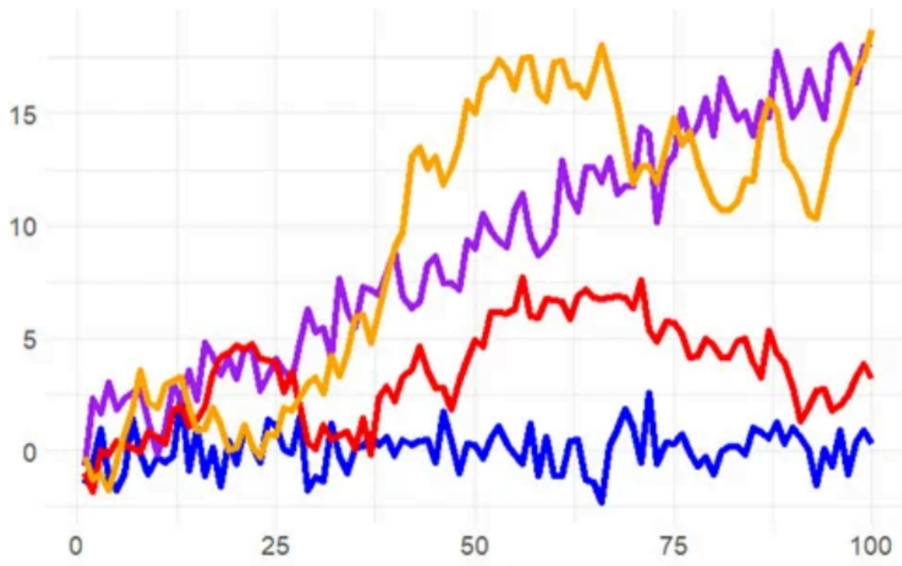
考察: Finance: Portfolio Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

42. Jeremy simulates four time series: white noise, white noise with trend, a random walk, and a random walk with drift. They are plotted below.



Jeremy conducts an Augmented Dickey-Fuller (ADF) test with the R function `adf.test()`. According to GARP, the ADF is the most widely used unit root test. Its null hypothesis is that the series is a unit root process and its corresponding alternative hypothesis is that the series is stationary. Importantly, this `adf.test()` does allow an intercept and trend: it evaluates a trend-stationary process as stationary but evaluates a difference-stationary process (prior to transformation) as non-stationary. If Jeremy performs an ADF test on each of these four series, what is the most likely outcome?

Jeremy 模拟了四个时间序列：白噪声，白噪声加趋势，随机游走，随机游走加漂移。它们绘制如下。



Jeremy 进行了一个增广迪基-富勒（ADF）测试，使用 R 函数 `adf.test()`。根据 GARP，ADF 是最广泛使用的单位根检验。其零假设是序列是一个单位根过程，其对应的备择假设是序列是平稳的。重要的是，这个 `adf.test()` 允许截距和趋势：它将趋势平稳过程评估为平稳，将差分平稳过程（在变换之前）评估为非平稳。如果 Jeremy 对这四个序列进行 ADF 测试，最可能的结果是什么?

A. All four have low p-values; e.g., p-values below 0.050
   所有四个都有低 p 值；即，p 值低于 0.050

B. All four have high p-values; e.g., p-values above 0.100
   所有四个都有高 p 值；即，p 值高于 0.100

C. The random walks have low p-values but the white noise processes have high p-values
   随机游走有低 p 值，但白噪声过程有高 p 值

D. The white noise processes have low p-values but the random walks have high p-values
   白噪声过程有低 p 值，但随机游走有高 p 值

答案:D
解析:
D 选项正确。 The white noise processes have low p-values but the random walks have high p-values

43. The current level of a stock index is 7,790 while the risk-free rate is 4.0% per annum with annual compounding. The dividend yield assumption is 1.50% with continuous compounding. We can observe that the nine-month futures contract is currently trading at a price of 8,000. If an index arbitrage is possible, what is the index arbitrage trade?
当前股票指数的水平为 7,790，无风险利率为 4.0% 每年，采用年复利。股息收益率假设为 1.50%，采用连续复利。我们可以观察到九个月远期合约目前交易价格为 8,000。如果指数套利是可能的，那么指数套利交易是什么?

A. Buy the underlying stocks and short the futures contract
   买入底层股票并卖出远期合约

B. Sell the underlying stocks and buy the futures contract
   卖出底层股票并买入远期合约

C. Cannot figure because compound frequencies differ
   无法计算，因为复利频率不同

D. The implied reverse cash-and-carry may not be possible if the ease lease rate on the borrowed stocks exceeds the risk-free rate
   如果借入股票的易租利率超过无风险利率，则隐含的反向现金和 carry 可能不可行

答案:A

解析:

A 选项正确。 Buy the underlying stocks and short the futures contract. The theoretical price of the futures contract is given by $7,790 \times (1 + 4.0\%)^{0.75} \times \exp(-1.50\% \times 0.75) = 7,932.80$. The observed futures price of 8,000 is trading rich such that the arbitrage is a cash and carry variation: buy the stocks and short the futures contract on the index.

考察: Derivatives: Forwards and Futures　　**重要程度:** [■■□]

科目: Quantitative Analysis

44. On May 28, 2019, Robin the Portfolio Manager considers the purchase of \$100.0 million face amount of the U.S. Treasury 2 3/8%s (2.375%) due November 15, 2028, at a cost of \$90,250,000. She refers to this as her "bullet" investment. After an analysis of the interest rate environment, she is comfortable with the pricing of the bond at a yield of 3.60% and with its duration of 8.32 years.

Data on Three U.S. Treasury Bonds as of May 28, 2019

Coupon	Maturity	Years	Price	Yield	Duration, D	Convexity
2.500%	3/31/24	4.8	\$102.300	2.00%	4.55	23.6
2.375%	11/15/28	9.5	\$90.250	3.60%	8.32	78.4
4.625%	11/15/48	29.5	\$98.810	4.70%	15.96	365.9

However, upon further inspection, Robin considers an alternative to the above-mentioned bullet investment in the 10-year 2 3/8%s. Specifically, the alternative is a "barbell" investment in the shorter maturity 5-year 2 1/2%s and the longer maturity 30-year 4 5/8%s. The barbell would be constructed to have the same COST and DURATION as the bullet investment. Which of the following is nearest to the convexity of the alternative barbell portfolio?

在 2019 年 5 月 28 日，Robin the Portfolio Manager 考虑购买面值为 \$100.0 百万的美国财政部 2 3/8%s（2.375%）债券，到期日为 2028 年 11 月 15 日，成本为 \$90,250,000。她将其称为她的"子弹"投资。在利率环境分析后，她对债券的定价为 3.60%，期限为 8.32 年。

Data on Three U.S. Treasury Bonds as of May 28, 2019

Coupon	Maturity	Years	Price	Yield	Duration, D	Convexity
2.500%	3/31/24	4.8	\$102.300	2.00%	4.55	23.6
2.375%	11/15/28	9.5	\$90.250	3.60%	8.32	78.4
4.625%	11/15/48	29.5	\$98.810	4.70%	15.96	365.9

然而，进一步检查后，Robin 考虑了一种替代上述子弹投资的 10 年期 2 3/8%s。具体来说，替代是一种"哑铃"投资，在较短到期 5 年期 2 1/2%s 和较长到期 30 年期 4 5/8%s。哑铃将构造为具有与子弹投资相同的成本和期限。下列哪一项最接近替代哑铃投资组合的凸性?

A. 50.4 years^2
   哑铃投资组合的凸性最接近 50.4 年 ^{2}

B. 78.4 years^2
   哑铃投资组合的凸性最接近 78.4 年 ^{2}

C. 136.7 years^2
   哑铃投资组合的凸性最接近 136.7 年 ^{2}

D. 305.9 years^2
   哑铃投资组合的凸性最接近 305.9 年 ^{2}

答案:C

解析:

Option C is correct. True: 136.7 years^2

We need to solve for two unknowns simultaneously:

$$V(5) + V(30) = 90.250, \text{ and}$$

$$\frac{V(5)}{90.250} \times 4.55 + \frac{V(30)}{90.250} \times 15.96 = 8.32.$$

Solving, $V(5) = \$60.43$ and $V(30) = \$29.82$, such that the barbell's duration is given by

$$D(\text{barbell}) = \frac{\$60.43}{90.250} \times 4.55 + \frac{\$29.82}{90.250} \times 15.96 = 8.32 \text{ years}$$

(by definition of the solution!); and the barbell's convexity is given by:

$$C(\text{barbell}) = \frac{\$60.43}{90.250} \times 23.6 + \frac{\$29.82}{90.250} \times 365.9 = 136.70 \text{ years}^2$$

考察: Fixed Income　　**重要程度:** [■■□]
科目: Quantitative Analysis

45. Securitization is a trend that is over fifty years old: The Housing and Urban Development Act of 1968 gave birth to Ginnie Mae (https://www.ginniemae.gov/) in an effort to promote homeownership by way of guaranteeing residential mortgage-backed securities (MBS). As GARP explains (see Figure 4.2 of Chapter 4) there have been several milestones along the way. Each of the following statements about securitization is true EXCEPT which is false?
Securitization 是一个已经存在了五十年的趋势:1968 年的住房和城市发展法案催生了 Ginnie Mae(https://www.ginniemae.gov/),旨在通过保证住宅抵押贷款支持证券（MBS）来促进住房所有权。根据 GARP 的解释（见第 4 章的图 4.2），一路上有几个里程碑。下列关于证券化的陈述中哪一项是正确的，除了哪一项是错误的?

A. Securitization offers at least some diversification of credit risk
证券化提供了至少一些信用风险的分散

B. Securitization transfers a substantial amount of risk, including but not limited to credit risk, from the originator's balance sheet to investors
证券化将大量的风险，包括但不限于信用风险，从发起人的资产负债表转移到投资者

C. The 2007-2009 global financial crisis (GFC) discredited the principles of securitization as evidenced by the subsequent evaporation of asset-backed securities (ABS)
2007-2009 全球金融危机（GFC）动摇了证券化的原则，正如随后资产支持证券（ABS）的蒸发所表明的那样

D. Securitization enables originate-to-distribute (OTD) but the 2007-2009 crisis witnessed poor securitization risk management that allowed too much systemic concentration of risk
证券化允许发起-分销（OTD），但 2007-2009 危机见证了糟糕的证券化风险管理，允许过多的系统性风险集中

答案:C
解析:
C 选项正确。 The crisis neither discredited securitization nor saw the evaporation of asset-backed securities.
In regard to (A), (B) and (D), each is TRUE.
The first two true statements (i.e., A and B) capture the two essential features of classical securitization:

- by pooling credit-sensitive assets, it provides diversification; and
- By issuing liabilities (notes) to investors, it transfers credit risk.

考察: Credit Risk: Securitization　　**重要程度:** [■■□]
科目: Quantitative Analysis

46. Consider the following seasonal time series model that is represented with lag notation:

$$(1-\phi_1 L)(1-\phi_S L^{12})Y_t = \delta + \varepsilon_t$$

$$\downarrow \qquad \downarrow$$

$$(1-0.53L)(1-0.26L^{12})Y_t = 0.17 + \varepsilon_t$$

Note that $\phi(1) = 0.530$ and $\phi(s) = 0.260$. In regard to this seasonal model, each of the following statements is true EXCEPT which is false?
考虑以下季节性时间序列模型，用滞后表示法表示：

$$(1-\phi_1 L)(1-\phi_S L^{12})Y_t = \delta + \varepsilon_t$$

$$\downarrow \qquad \downarrow$$

$$(1-0.53L)(1-0.26L^{12})Y_t = 0.17 + \varepsilon_t$$

注意 $\phi(1) = 0.530$ 和 $\phi(s) = 0.260$。关于这个季节性模型，下列陈述中哪一项是正确的，除了哪一项是错误的?

A. This seasonal model is a restricted AR(13) model
这个季节性模型是一个受限的 AR(13) 模型

B. The coefficient on lag 13 is -0.13780
滞后 13 的系数为-0.13780

C. The coefficients on lags 2 through 11 are zero
滞后 2 到 11 的系数为零

D. This seasonal model is deterministic because lag 12 is a dummy variable
这个季节性模型是确定的，因为滞后 12 是一个虚拟变量

答案:D
解析:
D 选项错误。虚拟变量是分类的 0 或 1，但这是一个随机（且平稳）的 ARMA 模型，其系数 0.26 对滞后 12 项加权；因此，这是 ARMA 模型的一个特例，可以用符号表示为 $\text{ARMA}(1,0)\times(1,0)_{12}$。
关于 (A)、(B) 和 (C)，每个都是正确的。具体而言：这个季节性模型是一个受限的 AR(13) 模型。滞后 13 的系数为 -0.13780，因为 $[1-\phi(1)L]\times[1-\phi(s)L^{12}]\times Y(t) = \delta+\varepsilon(t) \rightarrow [1-\phi(1)L-\phi(s)L^{12}+\phi(1)\times\phi(s)L^{13}]\times Y(t) = \delta+\varepsilon(t) \rightarrow Y(t) = \delta+\phi(1)\times Y(t-1)+\phi(s)\times Y(t-12)-\phi(1)\times\phi(s)\times Y(t-13)+\varepsilon(t)$。滞后 2 到 11 的系数为零。
考察: Quantitative Methods: Time Series **重要程度:** [■■□]
科目: Quantitative Analysis

47. The current spot exchange rate between the euro (EUR) and quote currency YYY is 1.3000 EURYYY. The EUR risk-free rate is 4.0% and the YYY risk-free rate is 2.0%; each interest rate is per annum with annual compounding. According to covered interest rate parity (CIRP), which is nearest to the forward rate in nine months ($T = 0.75$ years) quoted in points?
当前欧元（EUR）和报价货币 YYY 的即期汇率是 1.3000 EURYYY。欧元无风险利率为 4.0%，YYY 无风险利率为 2.0%；每个利率均为年化，采用年复利。根据覆盖利率平价（CIRP），九个月（$T = 0.75$ 年）的远期汇率最接近多少点?

A. -12,576

B. -188.0

C. +33.0

D. +291.0

答案:B
解析:
B 选项正确。 -188.0 forward rate points
According to CIRP, 1.30 EURYYY $\times$ $[(1.02/1.04)]^{(9/12)} = 1.28120$, and $(1.28120 - 1.3000) \times 10,000 = -188.0$.
考察: Finance: Foreign Exchange **重要程度:** [■■□]
科目: Quantitative Analysis

48. A speculative (aka, junk) bond has 15.0 years to maturity and pays a semi-annual 6 1/8 coupon; i.e., its coupon rate is 6.125% payable semi-annually. Its yield is 11.00%. If we assume a reasonable shock value (i.e., less than 100 basis points), which of the following is **nearest** to the bond's effective convexity?
一个投机性（即，垃圾）债券有 15 年到期，每年支付两次 6 1/8 的利息；即，其票面利率为 6.125%，每年支付两次。其收益率为 11.00%。如果我们假设一个合理的冲击值（即，小于 100 个基点），下列哪一项最接近债券的有效凸性?

A. 12.4 years^2
债券的有效凸性最接近 12.4 年 ^{2}

B. 98.0 years^2
债券的有效凸性最接近 98.0 年 ^{2}

C. 124.3 years^2
债券的有效凸性最接近 124.3 年 ^{2}

D. Cannot answer because face value is not given
无法回答，因为面值未给出

答案:B
解析:
B 选项正确。 98.0 years^2 $= (1/\$64.574) * [(\$61.996 + \$67.310 - 2 * \$64.574)/0.0050^2]$ where we've selected 50 basis points as the shock. In this case, the initial price $= -\text{PV}(0.1100/2, 15 * 2, 100 * 0.06125/2, 100) = \$64.574$, and if the yield shock is $+/-50$ bps, then:

- $\text{Price}(\text{yield} + 0.0050) = -\text{PV}(0.1150/2, 15 * 2, 100 * 0.06125/2, 100) = \$61.996$, and
- $\text{Price}(\text{yield} - 0.0050) = -\text{PV}(0.1050/2, 15 * 2, 100 * 0.06125/2, 100) = \$67.310$.

In regard to false (D), please note that the convexity will be unchanged if we (consistently) change the face value to any amount. Further, while the convexity will vary very slightly as we alter the shock value, it is approximately 98.0 years for shock values below 1.0%.
考察: Fixed Income: Bond Convexity **重要程度:** [■■□]
科目: Quantitative Analysis

49. In regard to limits policies, GARP explains that "optimal risk governance requires the ability to link risk appetite and limits to specific business practices. Accordingly, appropriate limits need to be developed for each business as well as for the specific risks associated with the business (as well as for the entire portfolio of the enterprise)." Most institutions set two types of limits, tier 1 (one) and tier 2 (two) limits. About these limits, which of the following is TRUE?
关于限制政策，GARP 解释说"最佳风险治理需要将风险偏好与特定业务实践联系起来。因此，需要为每个业务以及与业务相关的特定风险（还包括整个企业的投资组合）制定适当的限制。"大多数机构设置两种类型的限制，一级（一级）和二级（二级）限制。关于这些限制，哪一项是正确的?

A. Firms should choose and either adopt tier 1 or tier 2 limits but not both simultaneously
公司应该选择并采用一级或二级限制，但不能同时采用

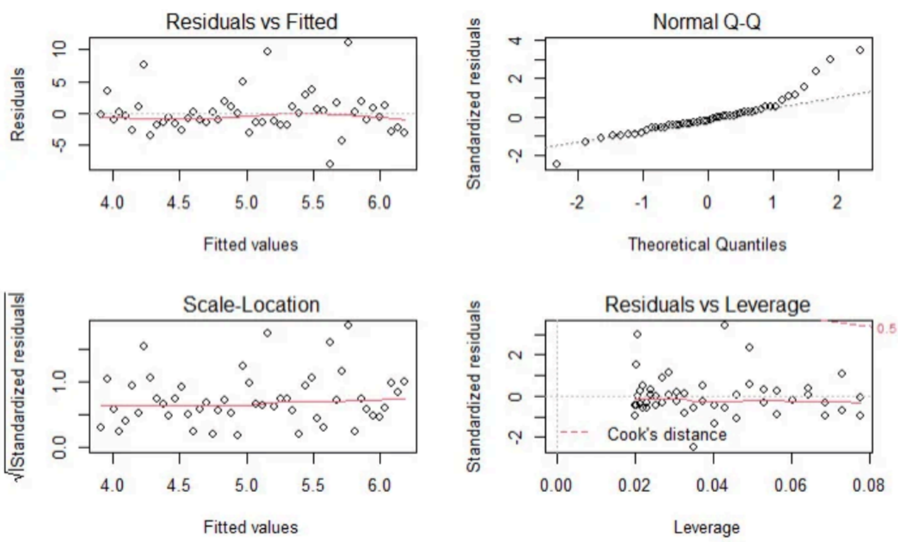
B. Tier 1 (tier one) limit exceedances must be cleared or corrected immediately, while tier 2 (tier two) exceedances are less urgent and can be cleared within a few days or a week.
一级（一级）限制超过必须立即清除或纠正，而二级（二级）限制超过不那么紧急，可以在几天或一周内清除。

C. Tier 2 (tier two) limits are specific and often include an overall limit by asset class, an overall stress-test limit, and a maximum drawdown limit
二级（二级）限制是特定的，通常包括资产类别的总体限制、总体压力测试限制和最大回撤限制

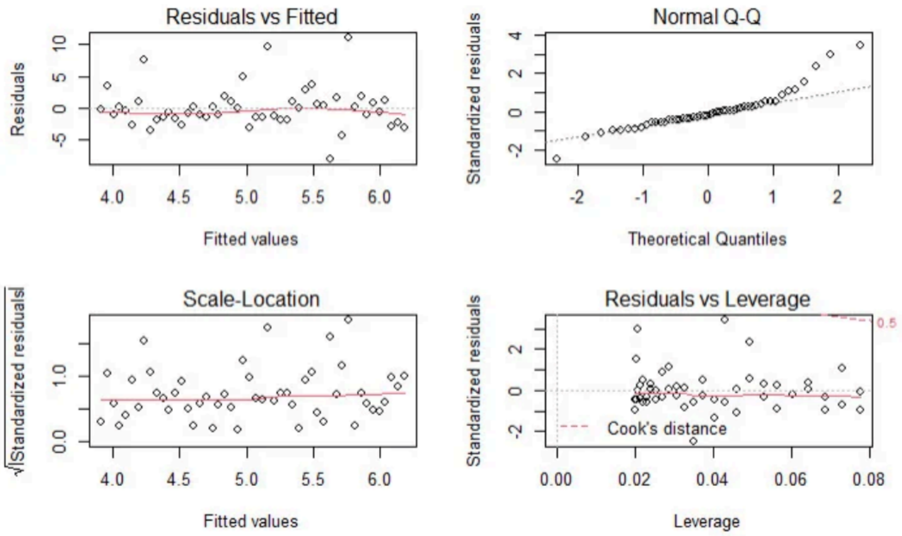
D. Tier 1 (tier one) limits are more generalized and relate to areas of business activity as well as aggregated exposures categorized by credit rating, industry, maturity, and region
一级（一级）限制更通用，与业务活动区域以及按信用评级、行业、期限和地区分类的聚合暴露有关

答案:B
解析:
B 选项正确。 Tier 1 (tier one) limit exceedances must be cleared or corrected immediately, while tier 2 (tier two) exceedances are less urgent and can be cleared within a few days or a week.
Explains GARP, "Most institutions set two types of limits. 1. Tier 1 limits are specific and often include an overall limit by asset class, an overall stress-test limit, and a maximum drawdown limit. 2. Tier 2 limits are more generalized and relate to areas of business activity as well as aggregated exposures categorized by credit rating, industry, maturity, and region, and so on."[1]
In regard to (A), (C) and (D), each is FALSE.
[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions
考察: Foundations of Risk Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

50. Patrick generated a simple regression line for a sample of 50 pairwise observations. After generating the regression model, he ran R's built-in plot(model) function which produces a standard set of regression diagnostics. These four plots are displayed below.



About these diagnostic plots, which of the following statements is TRUE?
Patrick 生成了一组 50 个成对观测值的简单回归线。在生成回归模型后，他运行 R 的内置 plot(model) 函数，生成一组标准的回归诊断图。这些四个图显示如下。关于这些诊断图，哪一项是正确的?



关于这些诊断图，哪一项是正确的?

A. There are many outliers
数据中有很多异常值

B. The data is significantly heteroskedastic
数据具有显著的异方差性

C. The residuals are a bit heavy-tailed (non-normal) on the right side
残差在右侧略微重尾（非正态）

D. The residuals reveal that the relationship between the explanatory and response variable is non-linear
残差揭示解释变量和响应变量之间的关系是非线性的

答案:C
解析:
Option C is correct. The residuals are a bit heavy-tailed (non-normal) on the right side.
In regard to (A), (B), and (D) each is false. We do see potential outliers in the Residual vs Fitted plot, however the Residuals vs. Leverage plot shows there are no observations with a Cook's distance greater than 0.5 (i.e., no observations above the dotted red line). Heteroskedasticity is not demonstrated, and the plots do support an approximately linear relations. In regard to the specific plots:

- *Residuals vs Fitted*: This plots residuals against the fitted values. We would like to see the residuals randomly scattered across the zero (which these are). The scatter pattern is relatively even suggesting homoskedasticity; i.e., we do not see a pattern that suggests heteroskedasticity. There are not many outliers. This is pretty good-looking residuals vs fitted plot suggestive of a decent linear regression.
- *Normal Q-Q*: If the distribution is normal, the plot will approximate along the straight line. But notice how this plot contains an obvious heavy-tail on the right side.
- *Scale-location*: This plot is similar to the Residuals vs Fitted plot, but the residuals are standardized. It is also used to evaluate heteroskedasticity. But, again, we do not perceive strong evidence of a non-constant variance.
- *Residuals vs Leverage*: The red dashed line represents a Cook's distance of 0.5, but there are not observations outside of this line (i.e., in the upper-left) such that we do not have a case for outlier(s).

考察: Machine Learning: Linear Regression　　**重要程度:** [■■□]
科目: Quantitative Analysis

51. On January 15th of Year 1, a company decides to hedge the planned purchase of 100,000 bushels of corn thirteen months later (on February 15 of Year 2). Below are displayed the per-bushel futures prices of three selected contracts on four different dates.

Bushels	**100,000**
Per Contract (Size)	**5,000**

	Year 1			Year 2
	Jan-15	Apr-15	Aug-15	Feb-15
May, Year 1 Futures Price	\$5.85	\$6.00		
Sep, Year 1 Futures Price		\$6.10	\$6.30	
March, Year 2 Futures Price			\$6.40	\$6.50

Assume the following: at the time of ultimate purchase (February 15 of Year 2), the spot price is \$6.47 per bushel when the final contract's basis is $\$6.47 - \$6.50 = -\$0.03$. If the company employs a stack-and-roll strategy, what is the company's net (i.e., after hedging) cost to acquire the corn? [inspired by GARP's EOC Question 8.20]

在 1 月 15 日，一家公司决定对 13 个月后（即 2 月 15 日）计划购买的 100,000 蒲式耳玉米进行对冲。下面显示了三个选定合约在四个不同日期的每蒲式耳期货价格。

Bushels	**100,000**
Per Contract (Size)	**5,000**

	Year 1			Year 2
	Jan-15	Apr-15	Aug-15	Feb-15
May, Year 1 Futures Price	\$5.85	\$6.00		
Sep, Year 1 Futures Price		\$6.10	\$6.30	
March, Year 2 Futures Price			\$6.40	\$6.50

假设：在最终购买时（2 月 15 日），现货价格为每蒲式耳 \$6.47，最终合约的基差为 $\$6.47 - \$6.50 = -\$0.03$。如果公司采用堆叠和滚动策略，公司购买玉米的净成本（即对冲后）是多少？[inspired by GARP's EOC Question 8.20]

A. $\$45,000$

B. $\$602,000$

C. $\$647,000$

D. $\$650,000$

答案:B

解析:

B 选项正确。 In this scenario, because the company plans to purchase corn in the future, the hedge is a stack-and-roll of long futures contracts:

- The May of Year 1 contracts are bought at \$5.85 and sold at \$6.00 in April of Year 1.
- The September of Year 1 contracts are bought at \$6.10 sold at \$6.30 in August of Year 1.
- The March of Year 2 contracts are bought at \$6.40 and sold at \$6.50 in February of Year 2.

The gain per contract is $5,000 * [(\$6.00 - \$5.85) + (\$6.30 - \$6.10) + (\$6.50 - \$6.40)] = 5,000 * \$0.45 = \$2,250.00$; and given 20 contracts are used (i.e., 100,000 bushels divided by 5,000 per contract), the total gain on the stack-and-roll hedge is $20 * \$2,250.00 = \$45,000$; put another way, \$0.45 per bushel * 100,000 bushels = $\$45,000.00$. The net cost is therefore $(\$6.47 * 100,000) - \$45,000.00 = \$602,000.00$

考察: Financial Markets and Products　**重要程度:** [■■□]

科目: Financial Markets and Products

52. Consider a 9-year bond with a semi-annual 10.0% coupon that has a current price of \$119.780 and a yield of 7.000%. If the yield drops to 6.850%, the bond's price increases to \$120.900. If this is the case, then which of the following is **nearest** to

the bond's dollar value of '01 (DV01)?
考虑一个 9 年期的债券，每年支付两次 10.0% 的利息，当前价格为 \$119.780，收益率为 7.000%。如果收益率下降到 6.850%，债券的价格增加到 \$120.900。如果是这种情况，那么哪一项最接近债券的'01 美元价值（DV01）?

A. -0.0121

B. 0.0121

C. 0.0747

D. 1.1200

答案:C
解析:
C 选项正确。 True: DV01 = 0.0747. The price change is \$120.900 − \$119.780 = \$1.120 given a 15-basis point drop in the yield, which is \$1.120/15 = 0.0746667 per basis point. Note that strictly (technically) the DV01 is the price change associated with a decrease of one basis point. This will be an almost identical price change associated with an increase of one basis point, but it won't be exactly the same.
考察: Fixed Income: Risk Measures **重要程度:** [■■□]
科目: Quantitative Analysis

53. GARP explains that "risk management must be implemented across the entire enterprise under a set of unified policies and methodologies. (This is called enterprise risk management ...). The infrastructure of risk management, which includes both physical resources and clearly defined operational processes, must be up to the task of an enterprise-wide scope."[1] However, it is a difficult and daunting task to evaluate the fitness of a firm or bank's risk management system. Among the following features, indicators, or characteristics, which is MOST LIKELY to signify that the firm is serious about its risk management process?
[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019.
GARP 解释道:"风险管理必须在整个企业范围内，根据一套统一的政策和方法来实施（这被称为企业风险管理……）。风险管理的基础设施，包括物理资源和明确定义的操作流程，必须能够满足企业范围管理的需求。"[1] 然而，评估一家公司或银行风险管理体系的适用性是一项困难且艰巨的任务。在下列特征、指标或特性中，哪一项最有可能表明该公司对其风险管理流程非常重视?
[1] 2020 年 FRM 一级考试《风险管理基础》第 10 版，培生学习解决方案，2019 年 10 月。

A. The firm's risk manager(s) is a well-compensated member of the executive staff with compelling career opportunities
公司风险经理（们）是享有丰厚薪酬的执行人员，拥有诱人的职业发展机会

B. At least one-third of executive compensation plans consist of stock options, or in the case of private firms, phantom stock options
至少三分之一的执行薪酬计划包括股票期权，或者在私人公司的情况下，包括虚幻的股票期权

C. The board has quantified the firm's agency costs and can show that agency costs are within a quantifiable confidence interval, or at least below an upper numeric bound
董事会已经量化了公司的代理成本，并可以证明代理成本在可量化的置信区间内，或者至少低于一个上界数字

D. The board's audit committee reports to the board's risk management committee and every member of the board is also a member of the board's risk management committee
董事会审计委员会向董事会风险管理委员会报告，并且董事会每个成员也是董事会风险管理委员会的成员

答案:A
解析:
A 选项正确。 The firm's risk manager(s) is a well-compensated member of the executive staff with compelling career opportunities.
We think GARP makes an excellent point that a true test of the firm's view of its risk managers is revealed by how it treats the human capital. Explains GARP, "One way to measure the seriousness of a risk management process is to examine the

human capital employed and the risk managers'standing within the corporate hierarchy: Is the risk manager considered to be a member of the executive staff and can this position lead to other career opportunities? How independent is the risk manager? What authority does he or she hold? To whom does he or she report? Are risk managers paid well relative to other employees who are rewarded for performance (e.g., traders)? To what extent can one characterize the enterprise's ethical culture as being strong and resilient against the actions of bad actors? Has the firm set clear-cut ethical standards and are these standards actively enforced?"[1]

In regard to (B), (C) and (D), these are false!

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019.

考察: Foundations of Risk Management **重要程度:** [■■□]

科目: Quantitative Analysis

54. Mary works for an insurance company and she has regressed medical costs (aka, the response or dependent variable) for a sample of patients against four independent variables: AGE, BMI, SMOKER, and CHARITY. The sample's average age is 38.51 years. Body mass index (BMI) is mass divided by height squared and the sample's average BMI is 22.16. SMOKER is a dummy variable where zero indicates a non-smoker and 1 indicates a smoker; the sample's average SMOKER value is 0.163 which indicates that 16.3% of the sample are smokers. CHARITY is the dollar amount of charitable spending in the last year; the sample average is $490.70 donated to charity in the last year. Mary's regression results are displayed below.

Medical COST regressed against AGE + BMI + SMOKER(1/0) + CHARITY($)
Simulated dataset

Coefficient	Estimate	Std Error	t-stat	p value
(Intercept)	**−324.21**	**415.77**	**−0.78**	**4.40×10^{-1}**
AGE	**56.95**	**7.55**	**7.54**	**4.60×10^{-9}**
BMI	**111.76**	**11.94**	**9.36**	**2.06×10^{-11}**
SMOKER	**454.08**	**125.83**	**3.61**	**8.84×10^{-4}**
CHARITY	**−0.55**	**0.18**	**−3.06**	**4.06×10^{-3}**

Residual standard error: 295.2 on 38 degrees of freedom

Multiple R-squared: 0.8343, Adjusted R-squared: 0.8168

F-statistic: 47.82 on 4 and 38 DF, p-value: 2.486e-14

Each of the following statements is true about these regression results EXCEPT which is false?

Mary 对一个样本的病人进行了医疗成本(aka, 响应或因变量)的回归, 使用四个自变量: AGE, BMI, SMOKER, 和 CHARITY。样本的平均年龄是 38.51 岁。身体质量指数(BMI)是体重除以身高平方, 样本的平均 BMI 是 22.16。SMOKER 是一个虚拟变量, 其中零表示非吸烟者, 1 表示吸烟者; 样本的平均 SMOKER 值是 0.163, 这意味着样本中有 16.3% 的人是吸烟者。CHARITY 是去年慈善支出的金额; 样本的平均值是 $490.70 捐赠给慈善机构。Mary 的回归结果如下。

Medical COST regressed against AGE + BMI + SMOKER(1/0) + CHARITY($) Simulated dataset				
Coefficient	Estimate	Std Error	t-stat	p value
(Intercept)	**−324.21**	**415.77**	**−0.78**	**4.40×10^{-1}**
AGE	**56.95**	**7.55**	**7.54**	**4.60×10^{-9}**
BMI	**111.76**	**11.94**	**9.36**	**2.06×10^{-11}**
SMOKER	**454.08**	**125.83**	**3.61**	**8.84×10^{-4}**
CHARITY	**−0.55**	**0.18**	**−3.06**	**4.06×10^{-3}**

Residual standard error: 295.2 on 38 degrees of freedom

Multiple R-squared: 0.8343, Adjusted R-squared: 0.8168

F-statistic: 47.82 on 4 and 38 DF, p-value: 2.486e-14

关于这些回归结果，哪一项是正确的，除了哪一项是错误的?

A. The sample size is 43 patients
样本大小是 43 个病人

B. Mary can reject a null hypothesis that all explanatory variables (jointly) have zero coefficients
Mary 可以拒绝所有解释变量（jointly）具有零系数的零假设

C. Mary can infer that patient medical cost is positively associated with each of AGE, BMI, and, on average, is greater for a smoker
Mary 可以推断出病人医疗成本与 AGE，BMI 和平均来说，吸烟者的医疗成本更高

D. Mary should suspect problematic multicollinearity because the intercept is suspiciously negative and the adjusted R-squared is too near to the unadjusted R-squared
Mary 应该怀疑多重共线性，因为截距是可疑的负值，并且调整后的 R 平方与未调整的 R 平方太接近

答案:D
解析:
D 选项正确。False (triple false!). The telltale symptom of problematic multicollinearity is a high R^2 with few significant t-stats (but all t-stats are significant). The negative intercept is not itself suspicious: it is simply out-of-sample. Finally, given $n = 43$ and $k = 4$, per adjusted $R^2 = 1 - [(1 - R^2) * (n - 1)/(n - k - 1)]$, the adjusted R^2 is by definition near to the unadjusted R^2. In regard to (A), (B) and (C), each is TRUE

- The sample size is 43 patients
- Mary can reject a null hypothesis that all explanatory variables (jointly) have zero coefficients
- Mary can infer that patient medical cost is positively associated with each of AGE, BMI and, on average, is greater for a smoker

考察: Quantitative Methods **重要程度:** [■□□□□]
科目: Quantitative Analysis

55. Today an exchange introduced a new futures contract. The open interest started at zero. Because it is a new contract, there were only the following four trades:

- Trader Adam (TA) buys (aka, takes a long position in) 1,000 contracts from Trader Betty (TB) who is the seller (aka, takes a short position) and both are new positions

- Trader Charles (TC) buys 300 contracts, in a new position, from Trader Adam (TA) who closes out 300 of his contracts
- Trader Denise (TD) sells 225 contracts, in a new position, to Trader Betty (TB), the buyer who closes out 225 of her contracts
- Trader Betty (TB) delivers on 175 of her short contracts to Trader Adam (TA) who takes delivery on 175 of his long contracts

What is the open interest, after the above four trades, at the end of the day?
今天一个交易所推出了一种新的期货合约。初始的未平仓合约数量为零。因为这是一个新的合约，只有以下四笔交易：

- Trader Adam (TA) 从 Trader Betty (TB) 手中买入（aka，建立多头头寸）1,000 份合约，Trader Betty (TB) 是卖家（aka，建立空头头寸），双方都是新的头寸
- Trader Charles (TC) 买入 300 份合约，建立新的多头头寸，从 Trader Adam (TA) 手中买入，Trader Adam (TA) 平仓 300 份合约
- Trader Denise (TD) 卖出 225 份合约，建立新的空头头寸，卖给 Trader Betty (TB)，Trader Betty (TB) 平仓 225 份合约
- Trader Betty (TB) 向 Trader Adam (TA) 交付 175 份空头合约，Trader Adam (TA) 接收 175 份多头合约

在上述四笔交易后，当天的未平仓合约数量是多少?

A. Zero
零

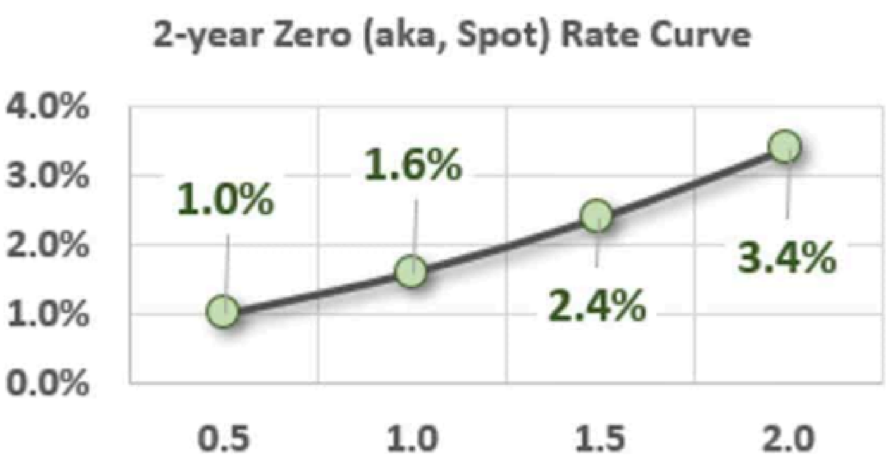
B. 300 contracts
300 份合约

C. 825 contracts
825 份合约

D. 1,000 contracts
1,000 份合约

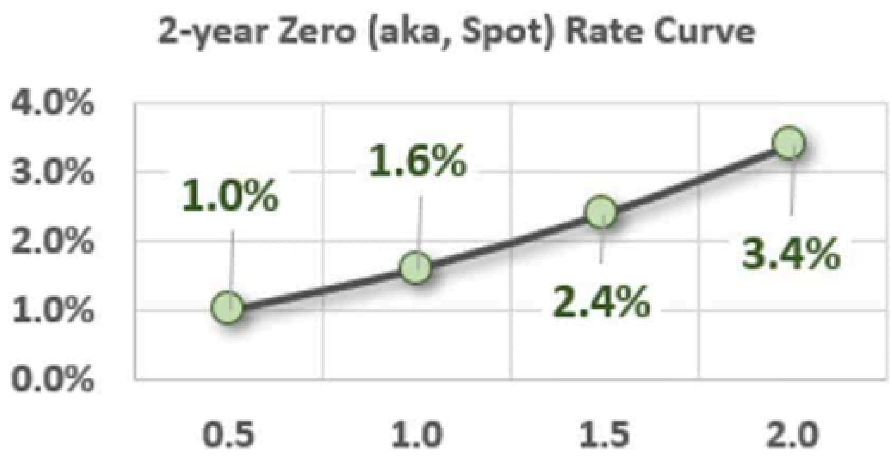
答案:C
解析:
C 选项正确。 TRUE: 825 contracts
考察: Futures Markets　**重要程度:** [■■□]
科目: Quantitative Analysis

56. Assume that the following upward-sloping zero rate (aka, spot rate) curve prevails:



Tuckman introduces the concept of a coupon effect in Chapter 3. Each of the following statements relates to this coupon effect and is necessarily true EXCEPT which statement is false?

假设目前市场上存在如下上升趋势的零利率（即即期利率）曲线：



Tuckman 在第三章介绍了债券的息票效应（coupon effect）概念。下列每个陈述都与这种息票效应相关，除了哪一项是错误的以外，其余均正确？请选出唯一不正确的说法。

A. Given this spot rate curve, the two-year par rate (aka, par yield) must be less than 3.4%
给定这个即期利率曲线，两年期票面利率（aka，票面收益率）必须小于 3.4%

B. Given this spot rate curve, a zero-coupon bond with a two-year maturity must have a yield (yield to maturity) of 3.4%
给定这个即期利率曲线，两年期零息债券必须有一个收益率（到期收益率）为 3.4%

C. Given this spot rate curve, a coupon-bearing bond with a two-year maturity must have a yield (yield to maturity) that is less than 3.4%
给定这个即期利率曲线，两年期附息债券必须有一个收益率（到期收益率）小于 3.4%

D. Given this spot rate curve, the yields (yields to maturity) of all bonds with two-year maturities must be identical due to the Law of One Price
给定这个即期利率曲线，所有两年期债券的收益率（到期收益率）必须相同，因为根据一价定律

答案:D
解析:
Option D is correct. The Law of One Price effectively requires that there is only one discount factor at each maturity, which is equivalent to saying that each maturity can have only one zero rate.
This statement (D) is false precisely because of the coupon effect: the yields of two-year bonds will vary as their coupons vary. Yield is a complex average of all the relevant spot rates and, importantly, is a function of the bond's cash flows.
In regard to (A), (B), and (C), each is TRUE. In regard to true (C), the 2-year par yield implied by this zero rate curve is 3.3621% because a 2-year bond with a coupon rate of 3.3621% has a theoretical price of exactly $100.00 under this curve, and therefore it's yield is also 3.3621%.
考察: Finance: Fixed Income Analysis　**重要程度:** [■■□]
科目: Quantitative Analysis

57. To hedge risk, the firm's toolbox includes derivatives such as swaps, futures, forwards, and options. About the use of derivatives to hedge, each of the following is true EXCEPT which is inaccurate?
为了对冲风险，公司的工具箱包括衍生品，如互换、期货、远期和期权。关于使用衍生品对冲，下列每个陈述都是正确的，除了哪一项是不准确的？

A. Derivatives represent a decision to transfer (or mitigate) the risk(s) when the firm neither wants to retain nor avoid the risk
衍生品代表了一种决策，当公司既不想保留风险，也不想避免风险时，转移（或减轻）风险

B. Because hedging can only mute the volatility of accrual-based earnings, hedging cannot increase (or even alter) the firm's cash flows and therefore cannot influence agency risks
因为对冲只能减弱基于应计的收入波动，对冲不能增加（甚至改变）公司的现金流，因此不能影响代理风险

C. A large conglomerate (i.e., a firm with multiple business units operating in different industries/sectors) is more likely to create natural hedges than a small, focused firm
一个大型多元化企业（即，一个在不同行业/部门运营多个业务单位的企业）更有可能创造自然对冲，而不是一个小型、专注的企业

D. Airlines can cross-hedge the cost of jet fuel with futures contracts on (crude or heating) oil, but if the commodity's price drops rapidly, unhedged airlines are likely to outperform their hedged competitors
航空公司可以使用期货合约对冲航空燃油成本，但如果商品价格迅速下跌，未对冲的航空公司可能比对冲的竞争对手表现更好

答案:B
解析:
B 选项正确。 Inaccurate. Hedging can alter cash flows and can leverage agency risks
Explains GARP, "[H]edging can make sense for investors if it is used as a tool to increase the firm's cash flows (rather than to reduce equity investor risk). For example, firms may need to offer their customers a stable price over the next three years, which may be impossible without hedging a key cost input. If hedging like this increases customer demand, then equity investors are happy ... Equity investors are also happy if the firm uses hedging to reduce its tax bill (e.g., by stabilizing revenues from one year to the next). Again, hedging has the effect of increasing after-tax revenues. Finally, equity investors are not the only stakeholders, and certainly not the only decision-makers. Managers, regulators, and general staff expect the firm to be financially sound and protected from sudden mishaps. Less legitimately, managers may use hedging to ensure their firm meets key short-term targets (e.g., stock analyst expectations) that affect their prestige and compensation. Risk managers need to pay close attention to how derivatives can leverage agency risks."[1]
In regard to (A), (C), and (D), each is TRUE
[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019
考察: Foundations of Risk Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

58. Debra is an economist who is interested in the relationship between consumer spending and the gross domestic product (GDP). From the FRED database at the Fed's Bank of St. Louis (https://fred.stlouisfed.org/) she collects quarterly data from 1980 through the first quarter of 2020; her series includes $n = 161$ quarters of data. She regresses consumer spending (C_SPEND), as the response (aka, dependent) variable against GDP as the explanatory (aka, independent) variable. Each series is not a level, but rather a seasonally adjusted percent change. The regression results are displayed below.

Consumer Spending (C_SPEND) regressed against Gross Domestic Product (GDP)
Quarterly Growth (Seasonally adjusted), 1980 to 2020:Q1, n = 161

Coefficient	Estimate	Std Error	t-stat	p value
(Intercept)	**0.9000**	**0.3156**	**2.8520**	**4.92×10^{-3}**
GDP	**0.3659**	**0.0499**	**7.3285**	**1.11×10^{-11}**

Source: FRED at https://fred.stlouisfed.org/

Additionally, Debra calculates the (univariate) standard deviation of each variable over the period: $\sigma(\text{C\_SPEND}) = 2.484$ and $\sigma(\text{GDP}) = 3.412$. In regard to these regression results, each of the following statements is true EXCEPT which is false?
Debra 是一位经济学家，她对消费者支出和国内生产总值(GDP)之间的关系感兴趣。她从联邦储备银行圣路易斯(https://fred.stlou
的 FRED 数据库收集了 1980 年至 2020 年第一季度的季度数据；她的系列包括 161 个季度的数据。她将消费者支出（C_SPEND）作为响应变量（aka，因变量），将 GDP 作为解释变量（aka，自变量）进行回归。每个系列都不是水平，而是季节调整后的百分比变化。回归结果如下。

Consumer Spending (C_SPEND) regressed against Gross Domestic Product (GDP) Quarterly Growth (Seasonally adjusted), 1980 to 2020:Q1, n = 161				
Coefficient	Estimate	Std Error	t-stat	p value
(Intercept)	**0.9000**	**0.3156**	**2.8520**	**4.92×10^{-3}**
GDP	**0.3659**	**0.0499**	**7.3285**	**1.11×10^{-11}**
Source: FRED at https://fred.stlouisfed.org/				

此外，Debra 计算了每个变量在时期内的（单变量）标准差：$\sigma(\text{C\_SPEND}) = 2.484$ 和 $\sigma(\text{GDP}) = 3.412$。关于这些回归结果，下列每个陈述都是正确的，除了哪一项是错误的?

A. The coefficients are jointly insignificant because each two-sided 95.0% confidence interval contain zero
系数联合不显著，因为每个双侧 95.0% 的置信区间都包含零

B. The GDP's t-statistic ("t-stat") of 7.3285 is equal to the coefficient estimate (0.3659) divided by its standard error (0.0499)
GDP 的 t 统计量（"t-stat"）为 7.3285，等于系数估计值（0.3659）除以其标准误差（0.0499）

C. Given the variables' cross-volatility, $\sigma(\text{C\_SPEND})/\sigma(\text{GDP}) = 2.484 \div 3.412 = 0.7280$, the correlation between the variables is approximately 0.50
给定变量之间的交叉波动性，$\sigma(\text{C\_SPEND})/\sigma(\text{GDP}) = 2.484 \div 3.412 = 0.7280$，变量之间的相关性约为 0.50

D. The coefficient of determination (R-squared; aka, R^2) is about 0.25 such that we can say about 25% in the variation in consumer spending (C_SPEND) is explained by gross domestic product (GPD)
判定系数（R-squared；aka，R^2）约为 0.25，因此我们可以认为消费者支出（C_SPEND）变异的 25% 可以由国内生产总值（GPD）解释

答案:A
解析:
A 选项正确。 False: Each (aka, both) coefficients are SIGNIFICANT (as we can immediately observe by their small p-values) such that neither two-sided 95.0% confidence interval contains zero.
At 95.0% confidence, the CI for the intercept coefficient is given by $0.9000 \pm 1.96 \times 0.3156 = (0.281, 1.518)$. At 95.0% confidence, the CI for the beta (aka, slope coefficient) is given by $0.3659 \pm 1.96 \times 0.0499 = (0.268, 0.464)$.
In regard to (B), (C) and (D), each is TRUE. Specifically:

- True: The GDP's t-statistic ("t-stat") of 7.3285 is equal to the coefficient estimate (0.3659) divided by its standard error (0.0499)
- True: Given the variables' cross-volatility, $\sigma(\text{C\_SPEND})/\sigma(\text{GDP}) = 2.484 \div 3.412 = 0.7280$, the correlation between the variables is approximately 0.50. Because $\beta(\text{C\_SPEND, GDP}) = \rho(\text{C\_SPEND, GPD}) \times \sigma(\text{C\_SPEND})/\sigma(\text{GDP})$, it follows that $\rho(\text{C\_SPEND, GPD}) = \beta(\text{C\_SPEND, GDP}) \times \sigma(\text{GDP})/\sigma(\text{C\_SPEND})$; in this case, $\rho(\text{C\_SPEND, GPD}) = 0.3659 \times 3.412/2.484 = 0.5026$ or about 0.50.
- True: The coefficient of determination (R^2) is about 0.25 such that we can say about 25% in the variation in consumer spending (C_SPEND) is explained by gross domestic product (GPD). In a univariate regression, the R^2 is the square of the correlation coefficient; in this case, the $R^2 = 0.50^2 = 0.25$ (indeed the dataset's regression produces an exact R-squared of 0.2525, and an adjusted R-squared of 0.2478, both of which round to 0.25).

考察: Quantitative Methods: Linear Regression　　**重要程度:** [■■□]
科目: Quantitative Analysis

59. Derivatives can be cleared through a central counterparty (CCP) or they can be cleared bilaterally; i.e., uncleared transactions are cleared bilaterally. If the derivatives are cleared through a CCP, they can be either exchange-traded or over-the-market (OTC). In this way, we distinguish between the exchange and the clearinghouse (which may be owned by an exchange). While exchanges operate CCPs, we can also refer to the emergent OTC CCPs. In regard to central counterparties (CCPs), which of

the following statements is TRUE?
衍生品可以通过中央对手方（CCP）清算，也可以通过双边清算；即，未清算的交易通过双边清算。如果衍生品通过 CCP 清算，它们可以是交易所交易（exchange-traded）或场外交易（OTC）。通过这种方式，我们可以区分交易所和清算所（可能由交易所拥有）。虽然交易所运营 CCP，我们也可以指出现场场外 CCP。关于中央对手方（CCPs），下列哪个陈述是正确的?

A. Banks are easier to regulate than both OTC CCP and exchange CCP
银行比场外 CCP 和交易所 CCP 更容易监管

B. An exchange CCP is more dependent on valuation models than an OTC CCP
交易所 CCP 比场外 CCP 更依赖估值模型

C. CCP member defaults are negatively correlated such that CCPs are counter-cyclical
CCP 成员违约是负相关的，因此 CCP 是反周期的

D. In the cleared market, CCPs want initial margin that covers a 5-day 99.0% price value at risk (VaR)
在清算市场中，CCP 希望初始保证金覆盖 5 天的 99.0% 价格价值风险（VaR）

答案:D
解析:
D 选项正确。 In the cleared market, CCPs want initial margin that covers a 5-day 99.0% price value at risk (VaR)
In regard to (A), (B), and (C), each is FALSE. Instead, the following are true statements:

- CCPs are easier to regulate than banks
- An OTC CCP is more dependent than an exchange CCP on valuation models
- CCP member defaults are positively correlated and CCPs are pro-cyclical

考察: Clearing and Settlement　　**重要程度:** [■■□]
科目: Quantitative Analysis

60. Exactly one year ago, Sally purchased a $100.00 face value bond that pays a semi-annual coupon with a coupon rate of 9.0% per annum. When she purchased the bond, it had a maturity of 10.0 years and its yield to maturity (YTM; aka, yield) was 6.00%. If the bond's price today happens to be unchanged from one year ago (when she purchased the bond), which of the following is nearest to the bond's yield (yield to maturity) today?
一年前，Sally 购买了一张面值为 $100.00 的债券，每年支付两次 9.0% 的利息。当她购买债券时，它的期限为 10 年，收益率（YTM；aka，收益率）为 6.00%。如果今天债券的价格与一年前相同（当她购买债券时），那么下列哪个选项最接近今天的债券收益率（yield to maturity）?

A. 5.57%

B. 5.78%

C. 6.00%

D. 6.22%

答案:B
解析:
B 选项正确。5.78% is the yield to maturity if the bond's price remains unchanged, but its maturity shortens to 9.0 years. When Sally initially purchased the bond its price was equal to $= -PV(0.060/2, 10*2, \$100*0.090/2, 100) = \$122.3162$. If the price is unchanged, then one year later (i.e., $N-2$ periods), the yield is given by $RATE(9*2, \$100*0.090/2, -122.3162, 100)*2 = 5.7843\%$. Using the TI BA II+ calculator: 20 $N, 3\ I/Y, 4.5\ PMT, 100\ FV$ and $CPT\ PV = -122.3162$, then we only need to change the periods with 18 $N, CPT\ I/Y$ [display: $I/Y = 2.8922$] $\times 2 = 5.7843\%$.
考察: Fixed Income　　**重要程度:** [■■□]
科目: Quantitative Analysis

61. One of the risk management building blocks is the need to balance risk and reward. Specifically, GARP says, "Economic capital provides the firm with a conceptually satisfying way to balance risk and reward. For each activity, firms can compare the revenue and profit they are making from an activity to the amount of economic capital required to support that activity." Each of the following statements is true about RAROC EXCEPT which is inaccurate?
风险管理的一个关键组成部分是平衡风险和回报。具体来说，GARP 说："经济资本提供了一种概念上令人满意的方式来平衡风险和回报。对于每个活动，公司可以比较从活动中获得的收入和利润与支持该活动的经济资本量。"下列每个陈述都是关于 RAROC 的，除了哪一项是不准确的?

A. For an activity to increase shareholder value, its RAROC should be higher than the cost of equity capital
为了增加股东价值，活动的 RAROC 应该高于股权资本的成本

B. Four applications of RAROC include business comparison, investment analysis, pricing strategies, and risk management cost/benefit analysis
RAROC 的四个应用包括业务比较、投资分析、定价策略和风险管理成本/效益分析

C. Advantages of RAROC include (i) it has one universal regulatory definition (without credible variants), such that benchmarking against peers is easy; and (ii) it is easy to implement in practice
RAROC 的优势包括（i）它有一个通用的监管定义（没有可信的变体），因此与同行比较很容易；和（ii）它在实践中很容易实施

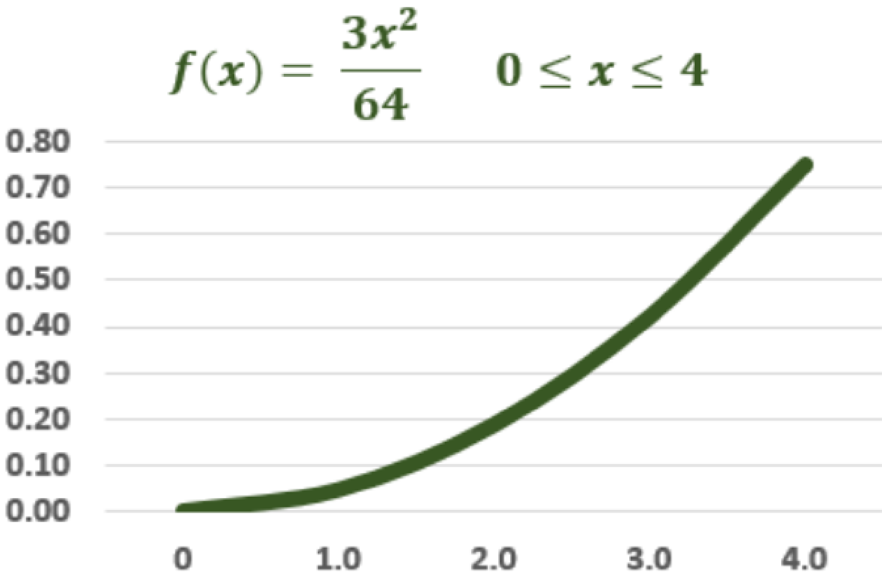
D. If RAROC's denominator is economic capital, which is typical, then its numerator should be an after-tax risk-adjusted expected return where the risk-adjusted refers to an adjustment for expected losses
如果 RAROC 的分母是经济资本，这是典型的，那么它的分子应该是一个税后风险调整后的预期回报，其中风险调整指的是对预期损失的调整

答案:C
解析:
C 选项正确。 C. FALSE, doubly. Instead, RAROC is notorious for the lack of a universal definition and several variations (says GARP: "There are many variants on the RAROC formula, applied across many different industries and institutions."). Also, RAROC is not easy to apply: "There are many practical difficulties in applying RAROC, including its dependence on the underlying risk calculations. Business lines often dispute the validity of RAROC numbers, sometimes for self-interested reasons,"explains GARP.
In regard to (A), (B) and (D), each is TRUE. Specifically, each of the following is TRUE:

- For an activity to increase shareholder value, its RAROC should be higher than the cost of equity capital
- Four applications of RAROC include business comparison, investment analysis, pricing strategies, and risk management cost/benefit analysis
- If RAROC's denominator is economic capital, which is typical, then its numerator should be an after-tax risk-adjusted expected return where the risk-adjusted refers to an adjustment for expected losses

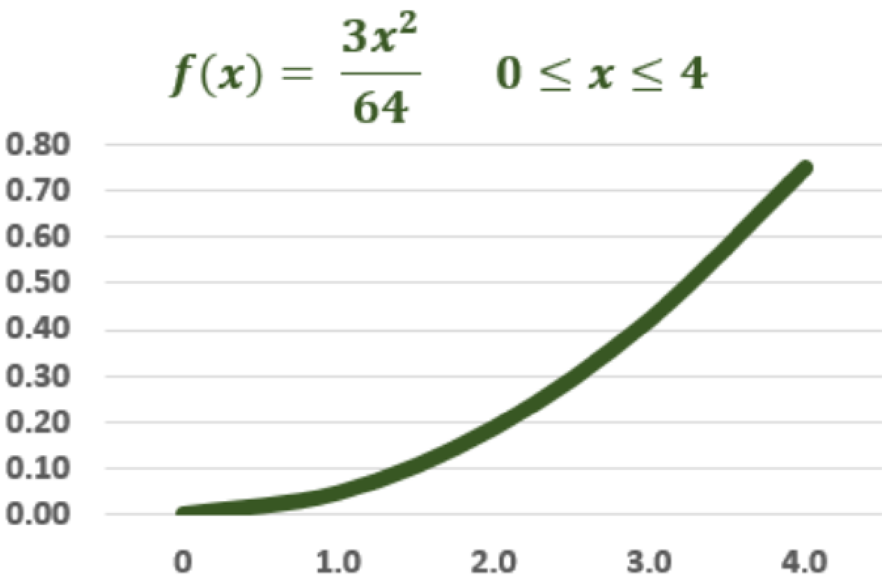
考察: Foundations of Risk Management: RAROC　　**重要程度:** [■■□]
科目: Quantitative Analysis

62. Consider a random variable that represents the loss severity of a risky asset and is given by the continuous probability distribution $f(x) = 3x^2/64$ on the domain from zero to four:



What are this variable's mean and variance?

考虑一个代表风险资产损失严重性的随机变量，由连续概率分布 $f(x) = 3x^2/64$ 在零到四的域上给出：



这个变量的均值和方差是多少?

A. $\mu = 1.0$ and $\sigma^2 = 3.50$

B. $\mu = 2.0$ and $\sigma^2 = 2.75$

C. $\mu = 3.0$ and $\sigma^2 = 0.60$

D. $\mu = 3.6$ and $\sigma^2 = 0.25$

答案:C

解析:

C 选项正确。 Correct: $\mu = 3.0$ and $\sigma^2 = 0.60$

- To retrieve the mean μ, we integrate $x \cdot f(x)$
- To retrieve the variance, we integrate $(x - \mu)^2 \cdot f(x)$

考察: Quantitative Methods: Probability Distributions **重要程度:** [■■□]

科目: Quantitative Analysis

63. A trader shorts 500 shares when the stock price is \$70.00. The initial and maintenance margin, respectively, are 150.0% and 125.0%. At what share price will further margin be required?
一个交易者以 \$70.00 的价格卖出 500 股股票。初始和维持保证金分别是 150.0% 和 125.0%。在什么股票价格时需要进一步保证金?

A. \$65.00

B. \$72.00

C. \$79.00

D. \$84.00

答案:D
解析:
Option D is correct. Because the initial margin is 150.0%, the trader is required to contribute $150.0\% \times 500 \times \$70.00 = \$52,500.00$ which is comprised of \$35,000.00 from selling the shares plus \$17,500.00. If the share price increases to X, the maintenance margin becomes $500 \times 1.25 \times X = 625 \times X$. This is greater than the \$52,500.00 when $625 \times X > \$52,500.00$, or when $X > \$52,500.00/625 = \$84.00$.
考察: Finance: Margin Trading **重要程度:** [■■□]
科目: Quantitative Analysis

64. As a market maker, Toughgreen Financial has written (aka, taken a short position in) a single contract for 100 call options on a non-dividend-paying stock whose volatility is 36.0% per annum when the riskless rate is 3.0%. Their strike price is \$100.00 but the options are underwater because the current stock price is \$90.00. The option term is six months. At the current stock price, each option has a value of \$5.88 and each option's percentage delta, $\Delta = +0.410$. From Toughgreen's perspective, if the stock price increases by +\$3.00 to \$93.00, which is nearest to the impact on the position's value?
作为一个做市商,Toughgreen Financial 以 \$100.00 的执行价格写(aka,做空)了一份合约,合约是针对一个没有分红支付的股票的 100 份看涨期权。该股票的波动率为 36.0%,无风险利率为 3.0%。他们的执行价格是 \$100.00,但期权是亏本的,因为当前股票价格是 \$90.00。期权期限为六个月。在当前股票价格下,每个期权价值为 \$5.88,每个期权的百分比 delta,$\Delta = +0.410$。从 Toughgreen 的角度来看,如果股票价格增加 \$3.00 到 \$93.00,哪个最接近对头寸价值的影响?

A. a) A loss of few dollars less than \$8.00; i.e., $100 * [\$5.88 * (\$3.00/\$90.00) * 0.410]$ minus (-) gamma adjustment
a) 几美元的损失,小于 \$8.00;即,$100 * [\$5.88 * (\$3.00/\$90.00) * 0.410]$ 减去 (-) gamma 调整

B. b) A loss of few dollars more than \$8.00; i.e., $100 * [\$5.88 * (\$3.00/\$90.00) * 0.410]$ plus (+) gamma adjustment
b) 几美元的损失,大于 \$8.00;即,$100 * [\$5.88 * (\$3.00/\$90.00) * 0.410]$ 加上 (+) gamma 调整

C. c) A loss of few dollars less than \$123.00; i.e., $100 * (\$3.00 * 0.410)$ minus (-) gamma adjustment
c) 几美元的损失,小于 \$123.00;即,$100 * (\$3.00 * 0.410)$ 减去 (-) gamma 调整

D. d) A loss of few dollars more than \$123.00; i.e., $100 * (\$3.00 * 0.410)$ plus (+) gamma adjustment
d) 几美元的损失,大于 \$123.00;即,$100 * (\$3.00 * 0.410)$ 加上 (+) gamma 调整

答案:D
解析:
D. True: A loss of few dollars more than \$123.00; i.e., $100 * (\$3.00 * 0.410)$ plus (+) gamma adjustment.
As a partial first derivative, delta is the rate of change of the option price with respect to the price of the underlying asset (stock, in this case): $\Delta = \partial c/\partial S$. A percentage delta of 0.410 signifies the expectation of a linear change of \$0.410 in the call option value for each \$1.00 change in the stock price. However, this excludes the gamma adjustment: the actual price increase in the call option is GREATER than the change estimated by the linear approximation. Consequently, the short position (who writes the call) is exposed to a greater price drop.
Specifically, in this example (although the problem does NOT require calculating gamma!), the percentage gamma is about 0.0170 such that the loss due to a +\$3.00 stock price change is $\$3.00 * 0.410 + 0.5 * 0.0170 * \$3.00^2 = \$1.3065$. Indeed, the actual option price increases from \$5.882 to \$7.188, for an exact difference of \$1.3060. In this way, the actual loss on the position (of 100 options) is \$130.60 while the delta-gamma estimate is \$130.65

考察: Finance: Option Greeks　　**重要程度:** [■■□]
科目: Quantitative Analysis

65. According to GARP, one of the building blocks in risk management is a proper understanding of the difference between expected loss, unexpected loss, and extreme risk; aka, tail risk. In regard to this building block, which of the following statements is TRUE?
根据 GARP，风险管理的一个关键组成部分是正确理解预期损失、意外损失和极端风险（aka，尾部风险）之间的区别。关于这个关键组成部分，下列哪个陈述是正确的?

A. Effective risk management should reduce a credit portfolio's expected loss (EL) to approximately zero
有效的风险管理应该将信用组合的预期损失（EL）减少到大约零

B. Expected loss is a product of (i) the probability of the risk event occurring; (ii) the severity of the loss if the risk event occurs, and (iii) the expected recovery rate
预期损失是一个产品，包括（i）风险事件发生的概率；（ii）风险事件发生时的损失严重程度；和（iii）预期回收率

C. While expected loss (EL) is a function of default correlation, unexpected loss (UL) is NOT influenced by portfolio granularity
虽然预期损失（EL）是违约相关性的函数，但意外损失（UL）不受投资组合粒度影响

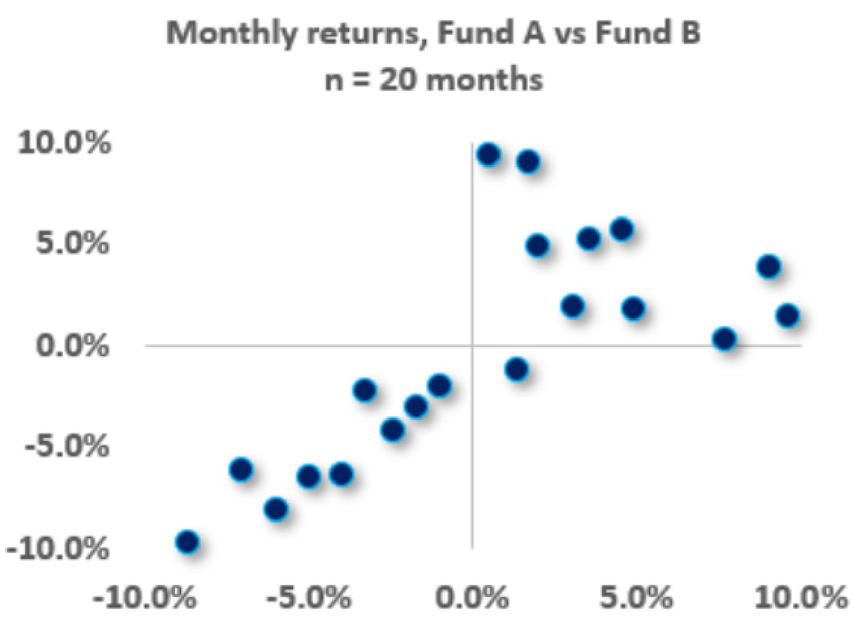
D. Although banks theoretically do not need to set aside provisions when loan products are accurately priced, in realistic practice, banks should provision for expected losses
尽管理论上银行在贷款产品准确定价时不需要计提准备金，但在实际实践中，银行应该计提预期损失

答案:D
解析:
D 选项正确。 TRUE: Although banks theoretically do not need to set aside provisions when loan products are accurately priced, in realistic practice, banks should provision for expected losses.
Each of (A), (B), and (C) is FALSE. Instead, the following are true:

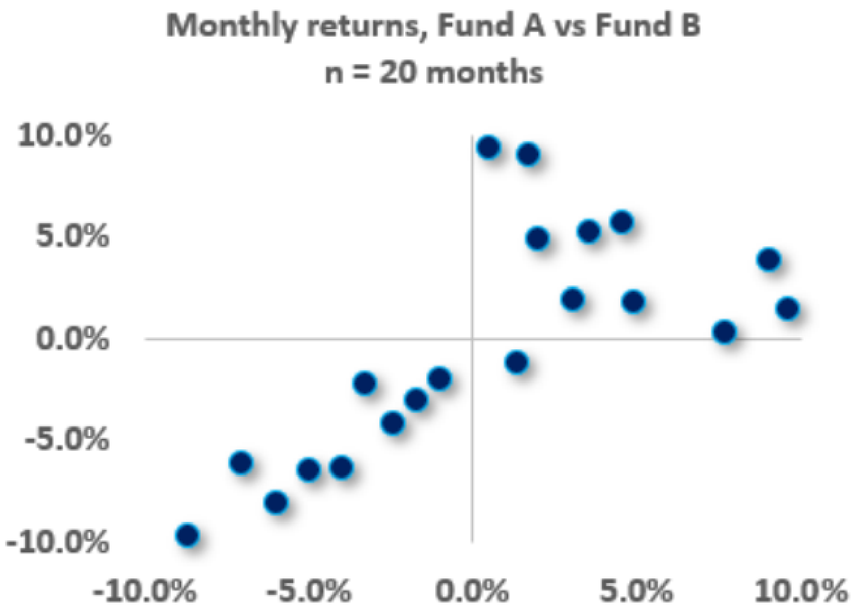
- All credit portfolios have non-negative expected loss (EL)
- Expected loss is a product of (i) the probability of the risk event occurring; (ii) the severity of the loss if the risk event occurs, and (iii) exposure at default.
- While unexpected loss (UL) is a function of default correlation, expected loss (EL) is not influenced by portfolio granularity

考察: Foundations of Risk Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

66. Below is a scatterplot of monthly returns, Fund A versus Fund B, over twenty months ($n = 20$).



Skewness is a trivial (special) case of coskewness; e.g., $S(AAA)$ and $S(BBB)$ are the trivial instances of coskewness. Similarly, Kurtosis is the trivial case of coskewness; e.g., $K(AAAA)$ and $K(BBBB)$ are the trivial instances of cokurtosis. In regard to this scatterplot (this two-variable application), how many nontrivial coskewness measures apply; are they positive or negative; and how many nontrivial cokurtosis measures exist?
下图展示了 A 基金与 B 基金在二十个月（月度回报，$n = 20$）中的散点图。



偏度（Skewness）是共偏度（coskewness）的一个特殊（平凡）情况，例如 $S(AAA)$ 和 $S(BBB)$ 就是共偏度的平凡实例。同样地，峰度（Kurtosis）是共峰度（cokurtosis）的平凡情况，例如 $K(AAAA)$ 和 $K(BBBB)$ 是共峰度的平凡实例。针对本题中的散点图（两变量的应用），满足多少个非平凡共偏度？它们是正的还是负的？非平凡共峰度又有多少个？

A. There are two nontrivial coskewness measures and they are negative (aka, negative coskewness); there are three nontrivial cokurtosis measures
有两项非平凡共偏度，它们是负的（aka，负共偏度）；有三项非平凡共峰度

B. There are three nontrivial coskewness measures and they are negative (aka, negative coskewness); there are four nontrivial cokurtosis measures
有三项非平凡共偏度，它们是负的（aka，负共偏度）；有四项非平凡共峰度

C. There are two nontrivial coskewness measures and they are positive (aka, positive coskewness); there are five nontrivial cokurtosis measures
有两项非平凡共偏度，它们是正的（aka，正共偏度）；有五项非平凡共峰度

D. There are three nontrivial coskewness measures and they are positive (aka, positive coskewness); there are six nontrivial cokurtosis measures
有三项非平凡共偏度，它们是正的（aka，正共偏度）；有六项非平凡共峰度

答案:A
解析:
A. True: There are two nontrivial coskewness measures and they are negative (aka, negative coskewness); there are three nontrivial cokurtosis measures
When there are two variables, there are two non-trivial coskewness moments and three non-trivial cokurtosis moments. In regard to skew, the cross-moments are: $S(AAA)$, $S(AAB)$, $S(ABB)$, and $S(BBB)$ but $S(AAA)$ and $S(BBB)$ are the (trivial) univariate skewness moments. In regard to cokurtosis, the nontrivial cross-moments are $K(AAAB)$, $K(AABB)$, and $K(ABBB)$. Visually, the pair has negative coskewness because their negative returns tend to occur in the same period (see lower-left quadrant of the scatterplot where the relationship is fairly linear), but their positive returns tend to occur in different periods (see upper-right quadrant where the relationship is more scattered). If the funds were combined into one portfolio, their negative coskewness reflects their higher risk when returns are negative (as opposed to their somewhat diversified dynamic when returns are positive).

The spreadsheet below is located if you want to take a deeper dive (but further understanding is NOT necessary for the exam!): https://www.dropbox.com/s/i7bx2iewfrezgph/t2-20-13-3-sample-moments-3rd-coskewness.xlsx

n =	20						
			Coskew		CoKurt		
	Fund A	Fund B	S(AAB)	S(ABB)	K(AAAB)	K(AABB)	K(ABBB)
1	-6.0%	-8.1%	-0.0003	-0.0004	0.0000	0.0000	0.0000
2	-8.7%	-9.6%	-0.0008	-0.0008	0.0001	0.0001	0.0001
3	-7.1%	-6.1%	-0.0003	-0.0003	0.0000	0.0000	0.0000
4	-4.0%	-6.3%	-0.0001	-0.0002	0.0000	0.0000	0.0000
5	-5.0%	-6.4%	-0.0002	-0.0002	0.0000	0.0000	0.0000
6	-2.4%	-4.2%	0.0000	0.0000	0.0000	0.0000	0.0000
7	-3.3%	-2.2%	0.0000	0.0000	0.0000	0.0000	0.0000
8	-1.7%	-3.0%	0				00
9	-1.0%	-2.0%	0				00
10	1.3%	-1.1%	0				00
11	7.7%	0.4%	0				00
12	9.6%	1.5%	0				00
13	4.9%	1.8%	0				00
14	3.1%	2.0%	0				00
15	9.0%	3.9%	0				00
16	2.0%	5.0%	0				00
17	3.5%	5.2%	0				00
18	4.6%	5.8%	0.0001	0.0002	0.0000	0.0000	0.0000
19	0.5%	9.4%	0.0000	0.0000	0.0000	0.0000	0.0000
20	1.7%	9.1%	0.0000	0.0001	0.0000	0.0000	0.0000
Average	0.4%	-0.2%					
Sample Variance	0.3%	0.3%					
Sample StdDev	5.3%	5.6%					
			S(AAB)	S(ABB)	K(AAAB)	K(AABB)	K(ABBB)
		cross-central moments	-0.0001	-0.0001	0.0000	0.0000	0.0000
		standardized	-0.411	-0.457	1.634	1.387	1.423

考察: Quantitative Analysis　　**重要程度:** [■■□]
科目: Quantitative Analysis

67. Ralph creates a portfolio with two positions: a short call and a short put with identical maturities on the same stock that has a current price, $S(0) = \$20.00$. His short call with a strike price, $K_1 = \$22.00$ has a premium, $c_1 = \$1.75$. His short put with a strike price, $K_2 = \$18.00$ has a premium, $p_2 = \$1.25$. Both options are out-of-the-money by the same amount of \$2.00. Recall that option profit is equal to the payoff net of the premium and we (typically) ignore the impact of discounting. On the future date with both options mature, under which future stock price scenario is Ralph's trade profitable?
Ralph 创建了一个投资组合，包括两个头寸：一个短卖看涨期权和一个短卖看跌期权，这两个期权具有相同的到期日，并且针对同一只股票，当前价格为 $S(0) = \$20.00$。他的短卖看涨期权，执行价格为 $K_1 = \$22.00$，期权费为 $c_1 = \$1.75$。他的短卖看跌期权，执行价格为 $K_2 = \$18.00$，期权费为 $p_2 = \$1.25$。这两个期权都处于相同的金额，即 \$2.00 的虚值。记住，期权利润等于净收益减去期权费，我们（通常）忽略折现的影响。在未来日期，当两个期权都到期时，哪个未来股票价格场景使得 Ralph 的交易盈利?

A. If the stock price falls within \$15.00 and \$25.00
如果股票价格在 \$15.00 和 \$25.00 之间

B. His portfolio will always generate a small profit
他的投资组合总是会产生小额利润

C. If the stock price is less than \$18.00 or greater than \$22.00
如果股票价格小于 \$18.00 或大于 \$22.00

D. If realized volatility increases; i.e., is higher on the future date
如果实际波动率增加；即，在未来日期更高

答案:A

解析:
Option A is correct. If the stock price falls within \$15.00 and \$25.00.
This portfolio is a trading strategy called a short strangle or top vertical combination which is a variation on the more common straddle. In a straddle, the investor buys a (European) call and put with identical strike prices and maturities. This is not a directional trade but rather a sort of volatility trade because it gains if the stock prices increase or decrease dramatically. The strangle is a variation because in the strangle the call's strike price is higher than the put's strike price. Writing the strangle, as illustrated in this question, is a short volatility trade because it is exposed to losses if the stock price increases or decreases greatly. Specifically, the initial cash inflow (the two premiums) is $\$1.75 + \$1.25 = \$3.00$. Therefore breakeven occurs when the payoff is \$3.00 which occurs when the stock price drops below \$15.00 (because the exercised put will cost \$3.00 at this price) or when the stock price increases above \$25.00 (because the exercised call will cost \$3.00 at this price). This short strangle generates a net profit if the stock price falls within $\{\$15.00, \$25.00\}$.
In regard to false (D), a straddle purchase (aka, bottom straddle) is a long volatility strategy, and a straddle write (aka, top straddle) is a short volatility strategy. Because the strike prices are different, the combination in this question is a short strangle and is similar to the straddle write because is a short volatility trade: it profits if the realized volatility is less than the initial implied volatility, but experiences a loss if the realized volatility is greater than the initial implied volatility.
考察: Finance: Options Strategies　**重要程度:** [■■□]
科目: Quantitative Analysis

68. Sally owns a non-dividend-paying stock that is currently trading at \$60.00. As a hedge, she buys an out-of-the-money three-month European put option on the stock with a fixed strike price of \$57.00, which is a 5.0% discount to the current stock price. The stock's volatility is 36.0% per annum. The riskfree rate is 4.0%. According to the Black-Scholes-Merton model, what is the value of the put?

A. \$1.55

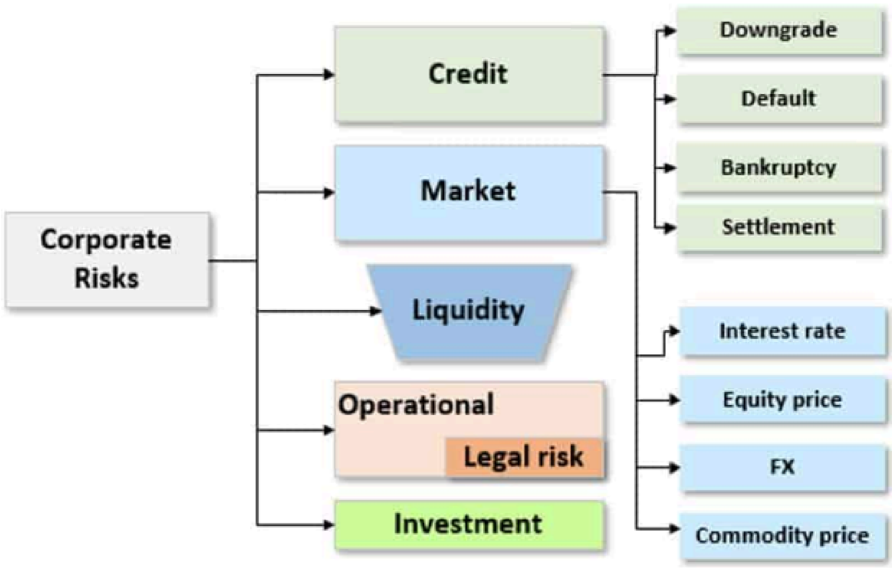
B. \$2.63

C. \$3.89

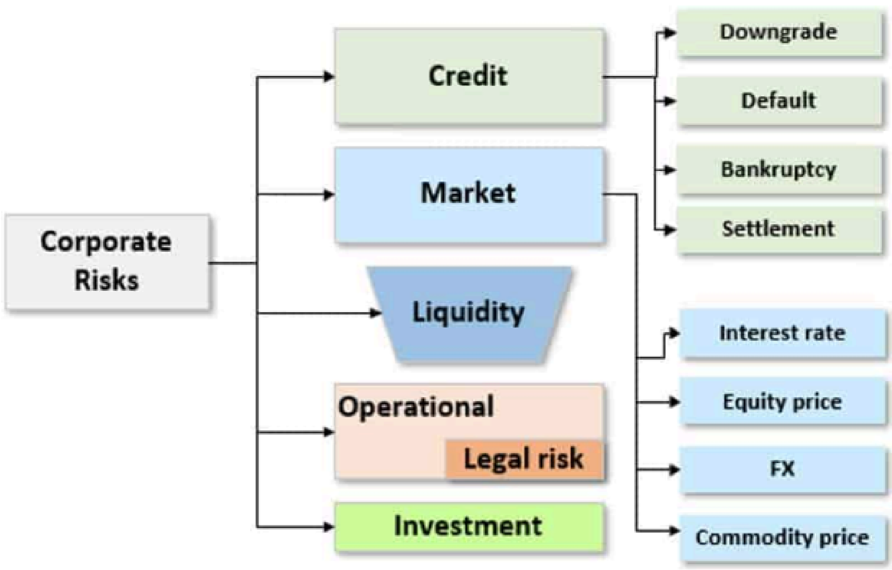
D. \$4.40

答案:B
解析:
B 选项正确。Per the BSM option pricing model, the value of a put is given by: $p = K \times \exp(-rT) \times N(-d_2) - S(0) \times N(-d_1)$. If we use the rounded lookup values, then $p = \$57.00 \times \exp(-0.040 \times 3/12) \times 0.4013 - \$60.00 \times 0.336 = \$2.630$. By design, this is also the exact value of the put (i.e., with exact normal Z value), per $c = \$57.00 \times \exp(-0.040 \times 3/12) \times 0.4011 - \$60.00 \times 0.334 = \$2.630$.
考察: Finance: Option Pricing　**重要程度:** [■■□]
科目: Quantitative Analysis

69. Galaxy Financial is a large bank whose risk department developed the following risk typology after surveying key stakeholders within the bank:



Which of the following is most likely TRUE about their typology?
Galaxy Financial 是一个大型银行，其风险部门在调查银行内部的关键利益相关者后开发了以下风险分类:



下列哪个陈述最可能是关于他们的分类的?

A. This typology is incorrect because it does not comport with any of the typologies approved by regulators, including Basel
这个分类是错误的，因为它不符合监管机构批准的任何分类，包括巴塞尔协议

B. If the typology is good then any and all newly identified risks should be slotted into an existing risk type within the typology
如果分类是好的，那么所有新识别的风险都应该被放入分类中的现有风险类型

C. Although not illustrated directly here, many of the risk types interact with one another so that risk might flow or cascade from one type to another
尽管这里没有直接说明，但许多风险类型相互作用，因此风险可能从一个类型流向另一个类型

D. It is incorrect to locate Liquidity risk as a top-level risk (i.e., alongside Credit and Market risk) because it should be a sub-risk to operational risk
将流动性风险定位为顶级风险（即，与信用风险和市场风险并列）是错误的，因为它应该是操作风险的一个子风险

答案:C
解析:
Option C is correct. Although not illustrated in the typology, many of the risk types interact with one another so that risk might flow from one type to another
As GARP explains, "The risk types interact with one another so that risk flows. During a severe crisis, for example, risk can flow from credit risk to liquidity risk to market risk, (which was the case during the global financial crisis of 2007–2009). The same

can occur within an individual firm: the fat finger of an unlucky trader (operational risk) creates a dangerous market position (market risk) and potentially ruins the standing of the firm (reputational risk). That is why a sophisticated understanding of risk types and their interactions is an essential building block of risk management."[1]
In regard to (A), (B), and (D), each is FALSE

- In regard to false (A), we cannot deem the typology incorrect, especially without further information, as there is no single set of so-called correct typologies; nor do regulators qualify banks'typologies.
- In regard to false (B), the risk typology is dynamic, not static. New risks (e.g., cyber risk) create new types, or subdivide current types. Explains GARP, "Risk typologies must be flexible because new risks are always emerging."[1]
- In regard to false (D), consistent with flexibility, liquidity risk might be top-level, or often, it as sub-risk to Market Risk. Interestingly, in 2020 GARP introduced "Liquidity and Treasury Risk Measurement and Management"as a new top-level topic in Part 2 of the FRM. Previously Liquidity Risk was somewhat embedded into the Operational Risk Topic.

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019
考察: Foundations of Risk Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

70. An asset has a one-month volatility of 6.0%. If the asset's returns are i.i.d., the asset has a three-month volatility of $6.0\% \times \sqrt{3/1} = 10.4\%$ and a one-year volatility of $6.0\% \times \sqrt{12/1} = 20.8\%$. In general, by convention (and to avoid confusion), we should prefer the annualized measure: interest rate and volatility inputs and outputs are generally expressed in per annum terms. However, let's assume we are interested in the three-month value-at-risk (VaR) of this asset. If we assume that the asset's arithmetic returns, $[S(t)-S(t-1)]/S(t-1)$, have a normal distribution and i.i.d., then per the square root rule (SRR) the 95.0% confident three-month relative VaR of this asset is $6.0\% \times \sqrt{3/1} \times 1.645 = 17.10\%$; the "relative" adjective signifies that we are excluding the positive expected return which would mitigate this worst expected loss.
Now let's violate the i.i.d. assumption. Specifically, let's retain the "identical" but edit the "independence" to make an assumption of +0.50 auto- or serial-correlation between the returns; i.e., if the returns are correlated, they cannot be independent and i.i.d. is not satisfied such that we cannot rely on the square root rule (SRR). Now we are interested in the three-period (i.e., three-month) VaR based on a three-month volatility but where the variable (log return) has a positive correlation (with itself) of 0.50. Hint: per variance properties $\sigma^2(X+Y+Z) = \sigma^2(X+Y) + \sigma^2(Z) + 2 \times \mathrm{COV}(X+Y, Z)$ or equivalently $\sigma^2(R1+R2+R3) = \sigma^2(R1+R2) + \sigma^2(R3) + 2 \times \mathrm{COV}(R1+R2, R3)$.
If each one-month return has a standard deviation of 6.0% and the serial correlation between returns is +0.50, which is **nearest** to the three-month 95.0% normal relative VaR?
一项资产的单月波动率为 6.0%。如果该资产的收益是 i.i.d. 的（一致且独立同分布），则该资产的三个月波动率为 $6.0\% \times \sqrt{3/1} = 10.4\%$，一年期波动率为 $6.0\% \times \sqrt{12/1} = 20.8\%$。一般来说，按照惯例（并且为了避免混淆），我们应优先采用年化指标：利率和波动率的输入与输出通常以每年为单位表示。然而，假设我们关心该资产的三个月价值-at-风险（VaR）。如果我们假定资产的算术收益 $[S(t)-S(t-1)]/S(t-1)$ 服从正态分布且 i.i.d.，则根据平方根法则（SRR），该资产在 95.0% 置信三个月的相对 VaR 为 $6.0\% \times \sqrt{3/1} \times 1.645 = 17.10\%$；"相对"一词意味着我们未计入正的预期收益，这些预期收益会缓和这种最坏预期损失。
现在我们违反 i.i.d. 假设。具体来说，我们保留"一致性"，但调整"独立性"，假设收益之间的自相关（序列相关）为 +0.50；即如果收益相关，它们就不是独立的，此时不满足 i.i.d. 假设，因此不能依赖平方根法则（SRR）。现在我们关心基于三个月波动率且变量（对数收益）自相关为 0.50 情况下的三期（即三个月）VaR。提示：根据方差性质 $\sigma^2(X+Y+Z) = \sigma^2(X+Y) + \sigma^2(Z) + 2 \times \mathrm{COV}(X+Y, Z)$，或等价地 $\sigma^2(R1+R2+R3) = \sigma^2(R1+R2) + \sigma^2(R3) + 2 \times \mathrm{COV}(R1+R2, R3)$。
如果每个月的标准差为 6.0%，且月收益之间的序列相关系数为 +0.50，哪一个最接近于三个月期 95.0% 正态分布下的相对 VaR?

A. 17.1%

B. 24.2%

C. 29.6%

D. 51.5%

答案:B
解析:

B 选项正确。 Taking the $\sigma^2(R1+R2+R3) = \sigma^2(R1+R2) + \sigma^2(R3) + 2 \times \mathrm{COV}(R1+R2, R3)$, and extending:
Because $2 \times \mathrm{COV}(R1+R2, R3) = 2 \times [\mathrm{COV}(R1, R3) + \mathrm{COV}(R2, R3)] = 2 \times \mathrm{COV}(R1, R3) + 2 \times \mathrm{COV}(R2, R3)$, and plugging back in:
$\sigma^2(R1+R2+R3) = \sigma^2(R1) + \sigma^2(R2) + \sigma^2(R3) + 2 \times \mathrm{COV}(R1, R2) + 2 \times \mathrm{COV}(R1, R3) + 2 \times \mathrm{COV}(R2, R3)$; this is the three-variable variance formula
Per the "identical" assumption, $\sigma^2(R1) = \sigma^2(R2) = \sigma^2(R3) = 6.0\%^2$; and further $\rho(R1, R2) = \rho(R1, R3) = \rho(R2, R3) = 0.50$ such that:
$\sigma^2(R1+R2+R3) = 3 \times 6\%^2 + 3 \times 2 \times 6\% \times 6\% \times 0.5 = 0.02160$ and $\sigma(R1+R2+R3) = \sqrt{0.02160} = 14.70\%$
The three-month 95.0% normal relative VaR is given by $14.70\% \times 1.645 = 24.18\%$.
考察: Quantitative Analysis **重要程度:** [■■□]
科目: Quantitative Analysis

71. Joe is a 29-year-old single male who prefers to invest in a fund because he has neither the time nor interest to spend researching and selecting individual stocks. His financial advisor Sally is experience in the following fund categories: closed- and open-end mutual funds, hedge funds, and index funds. When Sally asks for Joe's preferences, his only strongly felt opinion is that he prefers to avoid paying high expenses or fees (reading the financial news gives him the impression that he should seek low fees because fees are trending toward zero). Which of the following is she MOST LIKELY to recommend to Joe?
Joe 是一个 29 岁的单身男性，他更喜欢投资基金，因为他没有时间和兴趣去研究并选择单个股票。他的财务顾问 Sally 在以下基金类别方面有经验：封闭式和开放式共同基金、对冲基金和指数基金。当 Sally 询问 Joe 的偏好时，他唯一的强烈感受是，他更喜欢避免支付高额费用或费用（阅读金融新闻使他相信，由于费用趋向于零，他应该寻求低费用）。下列哪一项是她最有可能推荐给 Joe 的?

A. An index fund (or exchange-traded fund, ETF) to achieve diversification with low fees
一个指数基金（或交易所交易基金，ETF），以实现多元化并控制费用

B. An active closed-end mutual fund that trades below its net asset value (NAV) because it will outperform as it pulls to maturity
一个活跃的封闭式共同基金，其交易价格低于净资产价值（NAV），因为它将在到期时表现出色

C. A fund of hedge funds (FOHF) in order to overcome the principal-agent problem and ensure alignment between Joe's interest and the managers
一个对冲基金的基金（FOHF），以克服主要-代理问题并确保 Joe 的利益与经理之间的对齐

D. An open-end mutual fund that has beaten the market in the prior three consecutive years because mutual fund performance tends to be persistent
一个开放式共同基金，在过去三年中击败了市场，因为共同基金的表现倾向于持续

答案:A
解析:
A 选项正确。 An index fund (or exchange-traded fund, ETF) to achieve diversification while controlling fees
I regard to false (B), (C) and (D), please note:

- The closed-end fund will not necessarily "pull to par": closed-end fund tend to trade at a discount
- The fund of hedge funds has two layers of fees (notoriously expensive!) and will not meet Joe's preference to keep fees low
- Past performance is no guarantee of future performance, and unfortunately, mutual fund performance tends not to be persistent. According to GARP (see "Mutual Fund Research"), actively managed funds on average do not outperform the market; and research has found that *persistence* is an elusive, uncommon feature among the vast majority of mutual funds.

考察: Finance: Investment Funds **重要程度:** [■■□]
科目: Quantitative Analysis

72. Given a universe of 500 stocks and an initial assumption for the risk-free rate, an analyst generates the efficient frontier. She assumes the mean-variance framework. The investment committee compliments her analysis and makes a request: they ask her to decrease the risk-free rate (the rate at which an investor can both borrow and lend) but otherwise make no changes (aka, ceteris paribus) to the expected returns or variance matrix for the equities. Which of the following is most likely to be correct?

给定 500 只股票的 universe 和一个初始的无风险利率假设，分析师生成有效前沿。她假设均值-方差框架。投资委员会补充了她的分析，并提出了一个请求：他们要求她降低无风险利率（投资者可以借贷和放贷的利率），但除此之外不做任何改变（aka，ceteris paribus），即对股票的预期回报或方差矩阵不做任何改变。下列哪一项最可能是正确的?

A. The expected return on optimal portfolios with low risk & high risk will both decrease
低风险和高风险的最优投资组合的预期回报都会减少

B. The expected return on optimal portfolios with low risk will decrease, and the expected return on optimal portfolios with high risk will increase
低风险的最优投资组合的预期回报会减少，而高风险的最优投资组合的预期回报会增加

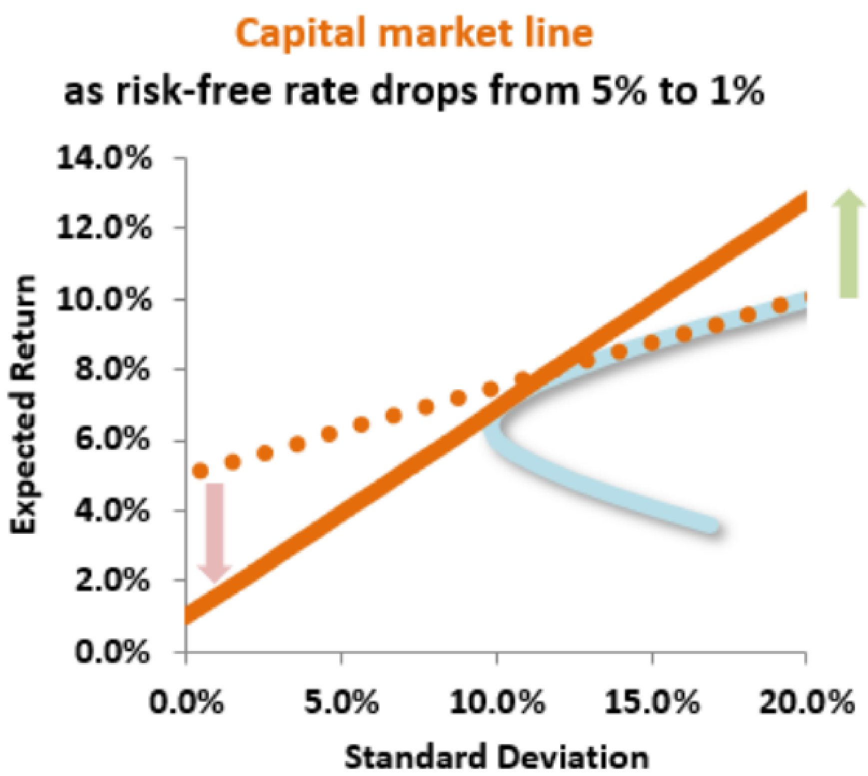
C. The expected return on optimal portfolios with low risk will increase, and the expected return on optimal portfolios with high risk will decrease
低风险的最优投资组合的预期回报会增加，而高风险的最优投资组合的预期回报会减少

D. The expected return on optimal portfolios with low risk & high risk will both increase
低风险和高风险的最优投资组合的预期回报都会增加

答案:B

解析:

B 选项正确。 True: The expected return on optimal portfolios with low risk will decrease, and the expected return on optimal portfolios with high risk will increase Observe the following diagram:



As the risk-free rate moves downward, the new efficient frontier line now has a lower intercept and higher slope −so investors who pursue a rational strategy of selecting the maximum return for a given level of risk will have lower returns if they opt for a very low-risk portfolio but higher returns if they choose to be more aggressive on the risk dimension (standard deviation).

考察: Finance: Portfolio Theory **重要程度:** [■■□]

科目: Quantitative Analysis

73. In a single-factor economy, each of the following portfolios (A, B, and C) is well-diversified:

Riskfree rate	3.0%
Volatility of market index, σ[M]	30.0%

Portfolio	Beta	E[R(i)]	σ[e(i)]
A	0.60	12.0%	10.0%
B	0.80	15.0%	25.0%
C	1.20	???	42.0%

You discover there is NOT an arbitrage strategy among these three portfolios. In this case, what should be the expected return of Portfolio (C)?
在一个单因子经济中，以下每个投资组合（A、B 和 C）都是充分分散的：

Riskfree rate	3.0%
Volatility of market index, σ[M]	30.0%

Portfolio	Beta	E[R(i)]	σ[e(i)]
A	0.60	12.0%	10.0%
B	0.80	15.0%	25.0%
C	1.20	???	42.0%

你发现这三个投资组合之间不存在套利策略。在这种情况下，投资组合 C 的预期收益率应该是多少?

A. 13.3%

B. 16.3%

C. 18.5%

D. 21.0%

答案:D
解析:
D 选项正确。 All three portfolios must lie on the same SML such that Portfolio C's Treynor ratio must be the same as the others. Treynor(Portfolio A) $= (12.0\% - 3.0\%)/0.60 = 0.15$; Treynor(Portfolio B) $= (15.0\% - 3.0\%)/0.80 = 0.15$, also. Therefore, Treynor(Portfolio C) $= (R - 3.0\%)/1.20 = 0.15$ such that its return must be 21.0%.
考察: Finance **重要程度:**
科目: Valuation and Risk Models

74. Let's assume two normal variables: $X \sim N(0, 3.0^2)$ and $Y \sim N(4.0, 9.0^2)$. Let V be a mixture distribution where X has a component weight of 80% and Y has a component weight of 20%. Let $W = 0.80 * X + 0.20 * Y$ with the assumption that X and Y are independent; that is, this W is the sum of independent, random variables. Which is more likely (the mixture or the convolution) to realize a negative outcome; that is, is $\Pr(V < 0) > \Pr(W < 0)$, or on the other hand, is $\Pr(W < 0) > \Pr(V < 0)$?
假设两个正态分布变量：$X \sim N(0, 3.0^2)$ 和 $Y \sim N(4.0, 9.0^2)$。令 V 为一个混合分布，其中 X 的成分权重为 80%，Y 的成分权重为 20%。令 $W = 0.80 * X + 0.20 * Y$，假设 X 和 Y 是独立的；即，W 是独立随机变量的和。哪一种更可能实现负结果；即，$\Pr(V < 0) > \Pr(W < 0)$，还是 $\Pr(W < 0) > \Pr(V < 0)$?

A. The mixture (V) has a greater probability of a negative outcome; i.e., $\Pr(V<0)>\Pr(W<0)$
混合分布（V）有更大的概率实现负结果；即，$\Pr(V<0)>\Pr(W<0)$

B. The convolution (W) has a greater probability of a negative outcome; i.e., $\Pr(W<0)>\Pr(V<0)$
卷积（W）有更大的概率实现负结果；即，$\Pr(W<0)>\Pr(V<0)$

C. They have the same probability of realizing a negative outcome; i.e., $\Pr(W<0)=\Pr(V<0)$
它们有相同的可能性实现负结果；即，$\Pr(W<0)=\Pr(V<0)$

D. It cannot be answered because it depends on correlation in the case of (W)
它无法回答，因为它取决于卷积（W）的情况中的相关性

答案:A
解析:
Option A is correct. The mixture (V) has a greater probability of a negative value; i.e., $\Pr(V<0)>\Pr(W<0)$

- In regard to the mixture distribution, the $\Pr(V<0)=80\%*50.0\%+20\%*32.84\%=40.0\%+6.567\%=46.57\%$.
- In regard to the convolution (i.e., sum of random variables which happen to be independent per given assumption), W has mean of 0.80 and standard deviation of $\sqrt{0.80^2*3.0^2+0.20^2*9.0^2}=3.0$ such that $\Pr(W<0)=39.49\%$.

考察: Quantitative Analysis **重要程度:** [■■□]
科目: Quantitative Analysis

75. Customers of insurance companies do not like to see their premiums increase, obviously. At the same time, most customers care about the claims process and quality: if the insurance doesn't pay out when it is needed, so the cheapest policy is not always the best! In regard to an increase in the premium(s), which of the following is MOST LIKELY to be TRUE?
保险公司的客户不喜欢看到他们的保费增加，显然。同时，大多数客户关心索赔过程和质量：如果保险在需要时没有支付，那么最便宜的策略并不总是最好的！关于保费的增加，哪一项最可能是正确的?

A. A whole life insurance premium increases in the middle of the policy because the insurance company's shareholders have demanded an increase in the dividend
一个终身保险的保费在政策中间增加，因为保险公司的股东要求增加股息

B. A variable life insurance premium increases because new genetic information (i.e., family tree insights) reduces the policy holder's life expectancy
一个变额终身保险的保费增加，因为新的遗传信息（即家族树洞察）减少了政策持有人的预期寿命

C. A health insurance premium increases due to a higher cost for physician services, new technologies and/or hospital upgrades
一个健康保险的保费增加，因为医生服务的成本更高，新的技术和/或医院升级

D. A health insurance premium increases because the policyholder develops unexpected health problems that were unanticipated when the policy was written
一个健康保险的保费增加，因为政策持有人在政策写作时未预料到的发展意外健康问题

答案:C
解析:
Option C is correct. A health insurance premium increases due to a higher cost for physician services, new technologies, and/or hospital upgrades.
In regard to (A), (B) and (D), each is FALSE. Instead,

- The life insurance company cannot increase the premium during the policy; they can increase the premium when the policy renews
- The Affordable Care Act modified the treatment of pre-existing conditions (https://en.wikipedia.org/wiki/Pre-existing_cond
The problem with pre-existing conditions is that the insurance company is exposed to adverse selection which limits their ability to properly pool risk. However, the development of a health problem subsequent to the policy's inception is precisely the situation that insurance is meant for!

考察: Finance: Insurance　　**重要程度:** [■■□]
科目: Quantitative Analysis

76. In the wake of the global financial crisis (GFC), stress testing has received prominent media attention. Siddique and Hasan explain that "stress testing has many approaches. For example, at many institutions, finance and treasury have had ownership of the budgeting process. Hence, centralized stress testing could be housed in such a central function. Other institutions have taken a more decentralized approach, wherein the central function has only acted as an aggregator. Effective stress testing does not need a given organisational form or approach. However, a given organisational form will require certain aspects to ensure effectiveness. For example, in a decentralized approach towards stress testing, consistency across the organisation becomes important, and the institution will need to find ways of ensuring that consistency."[1]
Each of the following statements is true about stress testing EXCEPT which is false?
[1] Siddique and Hasan, Stress Testing: Approaches, Methods, and Applications (London: Risk Books, 2013)
在金融危机（GFC）之后，压力测试受到了媒体的高度关注。Siddique 和 Hasan 解释说，"压力测试有多种方法。例如，在许多机构中，财务和 Treasury 拥有预算过程的所有权。因此，集中化的压力测试可以放在这样一个中心职能中。其他机构采取了更加分散的方法，即中心职能只充当一个聚合器。有效的压力测试不需要特定的组织形式或方法。然而，特定的组织形式需要确保某些方面是有效的。例如，在分散的压力测试方法中，组织的一致性变得重要，机构需要找到方法来确保这种一致性。"[1] 下列每个陈述都是关于压力测试的，除了哪一个是错误的？ [1] Siddique 和 Hasan，压力测试：方法和应用（伦敦：风险图书，2013 年）

A. Stress testing is a non-statistical risk measure that may employ scenario analysis as one approach
压力测试是一种非统计风险度量，可能采用情景分析作为一种方法

B. An acceptable proxy for stress testing is to generate value at risk (VaR) with a series of progressively increasing confidence levels; e.g., 99.0%, 99.50%, 99.75%, 99.90%, and 99.99%
一个可接受的代理压力测试是生成一系列逐渐增加的置信水平的价值风险（VaR）；即，99.0%，99.50%，99.75%，99.90%，和 99.99%

C. While backtesting validates model parameter(s) using a large number of actual, historical observations; stress testing on the other hand uses a small number of extreme movements in market variables
回溯测试使用大量实际的历史观察来验证模型参数；另一方面，压力测试使用市场变量中的少量极端运动

D. Unlike value at risk (VaR) which indicates neither the magnitude nor shape of extreme losses beyond the VaR threshold, stress testing DOES attempt to characterize the magnitude/shape of the extreme loss tail
与价值风险（VaR）不同，它既不指示极端损失的大小，也不指示极端损失的形状超过 VaR 阈值，压力测试确实试图表征极端损失尾部的大小/形状

答案:B
解析:
Option B is correct. False, or at least it is not necessarily true that high levels of VaR confidence are good proxies for stress testing. VaR might be based on historical observations that do not anticipate (or imagine) discontinuous shifts. Related, this is also false for the same reason that choice (D) is true. In regard to (A), (C), and (D) each is TRUE.
考察: Risk Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

77. Suppose that there are two independent economic factors, F1 and F2. The risk-free rate is 1.0%, and all stocks have independent firm-specific components with a standard deviation of 25%. The following are well-diversified portfolios; e.g., Portfolio (A) has a beta sensitivity to factor the first factor, $\beta(F1)$, of 1.20 and an expected return of 13.0%:

Portfolio	β(F1)	β(F2)	E(R)
A	1.20	-0.30	13.00%
B	0.60	-0.40	5.00%

Which is the correct return-beta relationship in this economy? 假设有两个独立的经济因素，F1 和 F2。无风险利率为 1.0%，所有股票都有独立的特定公司成分，标准差为 25%。以下是充分分散的投资组合；例如，投资组合（A）对第一个因素 F1 的敏感度为 1.20，预期回报为 13.0%：

Portfolio	β(F1)	β(F2)	E(R)
A	1.20	-0.30	13.00%
B	0.60	-0.40	5.00%

在这个经济体中，哪个是正确的回报-beta 关系?

A. $E[R(P)] = 1.0\% - \beta(F1) * 6.0\% - \beta(F2) * 4.0\%$

B. $E[R(P)] = 1.0\% - \beta(F1) * 5.0\% + \beta(F2) * 2.0\%$

C. $E[R(P)] = 1.0\% + \beta(F1) * 9.0\% + \beta(F2) * 3.0\%$

D. $E[R(P)] = 1.0\% + \beta(F1) * 12.0\% + \beta(F2) * 8.0\%$

答案:D
解析:
D 选项正确。 $E[R(P)] = 1.0\% + \beta(F1) * 12.0\% + \beta(F2) * 8.0\%$.
We need to solve for two equations with two unknowns:

- $0.13 = 0.01 + 1.20 * RP(1) - 0.30 * RP(2)$
- $0.05 = 0.01 + 0.60 * RP(1) - 0.40 * RP(2)$

One way is to double the second equation and subtract it from the first in order to eliminate RP(1):

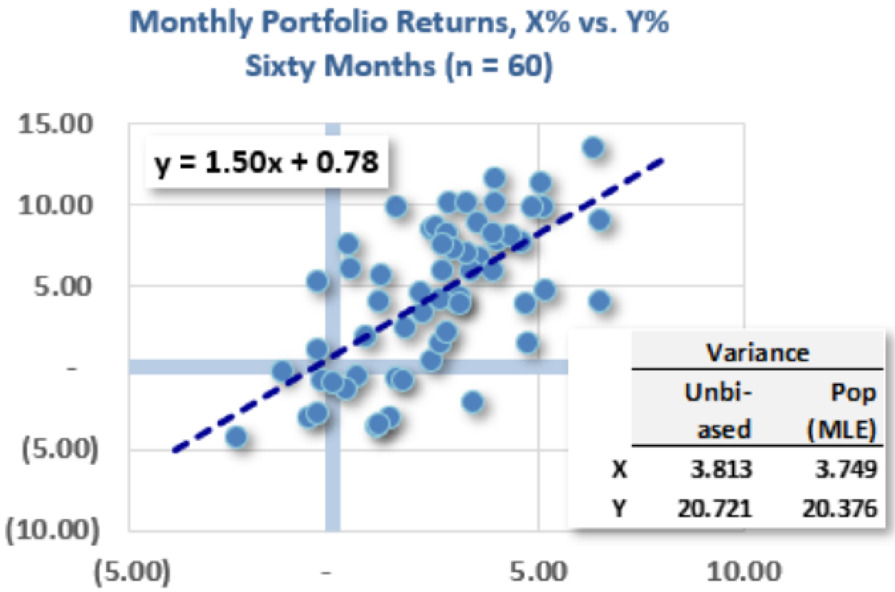
$$0.13 = 0.01 + 1.20 * RP(1) - 0.30 * RP(2)$$

$$-2 * [0.05 = 0.01 + 0.60 * RP(1) - 0.40 * RP(2)]$$

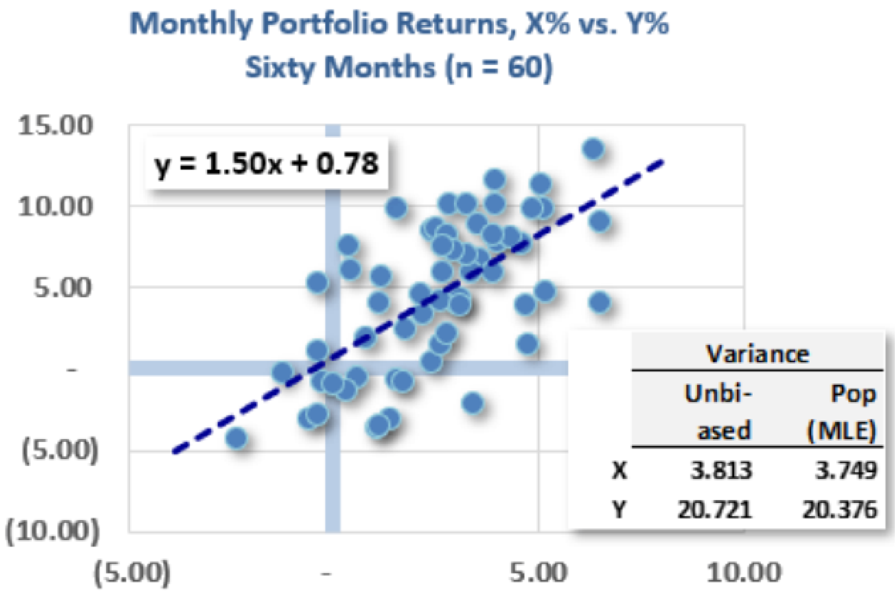
$$= 0.03 = -0.01 + 0.5 * RP(2) \rightarrow RP(2) = 0.04/0.5 = 0.08.$$

考察: Finance: Multifactor Models **重要程度:** [■■□]
科目: Quantitative Analysis

78. The scatterplot below plots sixty pairwise monthly returns over the last five years; each dot represents the return of Y (on the y-axis) and the return of X (on the x-axis) in a given month. Also shown is the ordinary least squares (OLS) regression line generated by Excel: $Y = 1.50 \times X + 0.78$. Finally, the univariate variances for each series are shown.



Which of the following is **nearest** to the correlation between X and Y? (Bonus question: if we flipped the axes, then what is the correlation?) 下图展示了过去五年内六十对月度回报的散点图；每个点代表给定月份中 Y 的回报（在 y 轴上）和 X 的回报（在 x 轴上）。还显示了 Excel 生成的普通最小二乘（OLS）回归线：$Y = 1.50 \times X + 0.78$。最后，每个系列的单变量方差也显示了。



下列哪个最接近 X 和 Y 之间的相关性？（附加问题：如果我们翻转轴，那么相关性是多少？）

A. 0.28

B. 0.41

C. 0.64

D. 1.50

答案:C

解析:

C 选项正确。 True: 0.64 because $\rho(X,Y) = 1.50 \times \sqrt{3.813}/\sqrt{20.721} = 0.6435$

Because $\beta(Y,X) = \sigma(X,Y)/\sigma^2(X) = [\rho(X,Y)\sigma(X)\sigma(Y)]/\sigma^2(X) = \rho(X,Y)\sigma(Y)/\sigma(X)$ such that $\rho(X,Y) = \beta(Y,X)\sigma(X)/\sigma(Y)$; in this case, $\rho(X,Y) = 1.50 \times \sqrt{3.813}/\sqrt{20.721} = 0.643457$. Notice that we get the same result (!) if we use the biased variances, because the $(n-1)$ cancels: $\rho(X,Y) = 1.50 \times \sqrt{3.749}/\sqrt{20.376} = 0.643457$.

In regard to the bonus question, the correlation does not change if we flip the axes, although beta does change. If we flip the axes, the regression line is $y = 1.237 + 0.276x$, in which case $\rho(X,Y) = 0.276 \times \sqrt{20.376}/\sqrt{3.749} = 0.643457$.

考察: Quantitative Methods: Linear Regression   **重要程度:** [■■□]
科目: Quantitative Analysis

79. The originate-to-distribute (OTD) banking model has both advantages and disadvantages. In regard to the OTD banking model, which of the following statements is TRUE?
originate-to-distribute（OTD）银行模式既有优势也有劣势。关于 OTD 银行模式，哪一项陈述是正确的?

A. OTD depresses the originating bank's reported return on equity (ROE)
OTD 降低了发起银行的报告股本回报率（ROE）

B. OTD implies that investors should be aware of the adverse selection problem implied by information asymmetry
OTD 意味着投资者应该意识到由信息不对称引起的逆向选择问题

C. OTD has an inherent, unavoidable design bug and public relations problem: mezzanine and equity tranches absorb losses before senior tranches
OTD 有一个内在的、不可避免的设计缺陷和公关问题：夹层和股权分层在优先分层之前吸收损失

D. OTD was banned after the global financial crisis (GFC 2007-08) such that it can no longer be conducted without incurring grave regulatory risk
OTD 在 2007-08 年全球金融危机（GFC）之后被禁止，因此现在无法在没有严重监管风险的情况下进行

答案:B
解析:
Option B is correct. OTD implies that investors should be aware of the adverse selection problem implied by information asymmetry
In regard to (A), (C), and (D), each is FALSE.

- In regard to (A), OTD shrinks the balance sheet and tends to increase ROE
- In regard to (C), subordination is one of the two key features of securitization (the other is credit risk transfer)
- In regard to (D), this greatly overstates the post-crisis reaction to OTD. As GARP explains, "The originate-to-distribute model played a role in the 2007–2008 crisis. Banks relaxed their mortgage lending standards so that the quality of the mortgages being originated declined. Despite this decline in quality, however, banks still managed to securitize them. In fact, they re-securitized mortgages by creating tranches from tranches. As defaults on the mortgages grew higher, losses were experienced by tranche holders. Unsurprisingly, for a period after the crisis, the originate-to-distribute model could not be used because it was not trusted by investors." — Source: Education, Pearson. Financial Markets and Products. Pearson Learning Solutions, 2020. VitalBook file.

考察: Financial Markets and Products   **重要程度:** [■■□]
科目: Quantitative Analysis

80. Gigh Inc., a publicly traded company, had a wild swing today after OPEC cut production. Gigh ended the day with a -16% loss. Prior to this (most recently), the daily volatility was estimated to be 4.40% according to the exponentially weighted moving average (EWMA) model. The EWMA model had originally assumed a lambda, λ, of 0.70, but David wants to adjust the model's responsiveness to recent market conditions. David notices that his error is too high, so he wants to change the model to respond more quickly to changes in volatility, so he decides to decrease λ by 0.10. After David properly adjusts lambda, what is the new volatility estimate using EWMA?
Gigh Inc.，一家上市公司，在 OPEC 减产后经历了一次大幅波动。Gigh 当天收盘时下跌了-16%。在此之前（最近），根据指数加权移动平均（EWMA）模型，每日波动率估计为 4.40%。EWMA 模型最初假设 lambda，λ，为 0.70，但 David 希望调整模型对近期市场条件的响应。David 注意到他的误差过高，因此他希望改变模型，更快地响应波动率的变化，因此他决定将 λ 减少 0.10。在 David 正确调整 lambda 后，使用 EWMA 计算的新波动率估计是多少?

A. $\sigma = 8.16\%$

B. $\sigma = 0.66\%$

C. $\sigma = 10.67\%$

D. $\sigma = 1.14\%$

答案:C
解析:
C 选项正确。$\sigma = 10.68\%$ A lower λ assigns more weight to the recent observations, making the EWMA respond more quickly to changes in volatility. Therefore, to make the model respond quicker to upticks in volatility David should decrease λ. Therefore,

$$\sigma_n^2 = \lambda * \sigma^2 + (1-\lambda) * (\text{return})^2$$

$$\text{New variance} = 0.6 * 4.40\%^2 + (1 - .6) * -16\%^2 = 0.0114016$$

$$\text{Volatility} = \text{SQRT}(0.0114016) = 10.68\%$$

考察: Quantitative Analysis　　**重要程度:** [■■□]
科目: Quantitative Analysis

81. Assume the riskfree rate is 3.0% while the market portfolio has a return of 13.0% (i.e., excess return is 10.0%) with volatility of 15.0%. Further, the following four portfolios are observed:

- Portfolio (A) has an expected return of 11.0% with volatility of 8.0%
- Portfolio (B) has an expected return of 7.0% with volatility of 20.0% and its correlation to the market is 0.30
- Portfolio (C) has a beta with respect to the market $\beta(C, M) = 1.50$ and its realized return of +20.0% implies an alpha of +2.0%
- Portfolio (D) has a beta with respect to the market $\beta(C, M) = 0.40$ and its realized return of +9.0% implies an alpha of +2.0%

Under the conditions of modern portfolio theory (MPT; mean-variance framework) and the capital asset pricing model (CAPM), each of the above portfolios is valid and plausible EXCEPT which is not possible?
假设无风险利率为 3.0%，市场投资组合的收益率为 13.0%（即超额收益为 10.0%），波动率为 15.0%。此外，观察到以下四个投资组合：

- 投资组合（A）的预期收益率为 11.0%，波动率为 8.0%
- 投资组合（B）的预期收益率为 7.0%，波动率为 20.0%，且与市场的相关系数为 0.30
- 投资组合（C）相对于市场的贝塔系数 $\beta(C, M) = 1.50$，其实现收益为 +20.0%，意味着阿尔法为 +2.0%
- 投资组合（D）相对于市场的贝塔系数 $\beta(C, M) = 0.40$，其实现收益为 +9.0%，意味着阿尔法为 +2.0%

在现代投资组合理论（MPT，均值-方差框架）和资本资产定价模型（CAPM）的条件下，上述每个投资组合都是有效且合理的，除了哪一个是不可能的?

A. Portfolio A
投资组合 A

B. Portfolio B
投资组合 B

C. Portfolio C
投资组合 C

D. Portfolio D
投资组合 D

答案:A

解析:

A 选项正确。 Portfolio A is unrealistic because it is "more efficient" than the Market portfolio: it has a Sharpe ratio of 1.0 versus the market portfolio's Sharpe ratio of $(13\% - 3\%)/15\% = 0.667$.

In regard to (B), (C), and (D), each is plausible

- Portfolio B has $\beta(B, M) = 0.30 * 20.0\%/15.0\% = 0.40$ with a CAPM expected return, $E(B) = 3.0\% + 0.40 * 10.0\% = 7.0\%$
- Portfolio C's Jensen's alpha, $\alpha(J, C) = 20.0\% - 3.0\% - 1.50 * 10.0\% = +2.0\%$
- Portfolio D's Jensen's alpha, $\alpha(J, D) = 9.0\% - 3.0\% - 0.40 * 10.0\% = +2.0\%$

考察: Finance: Portfolio Management　　**重要程度:** [■■□]

科目: Quantitative Analysis

82. In comparing and contrasting the law of large numbers (LLN) and the central limit theorem (CLT), each of the following statements is true EXCEPT which is false?

在比较和对比大数定律（LLN）和中心极限定理（CLT）时，下列每个陈述都是正确的，除了哪一个是错误的?

A. The LLN does not require independence but it does require normally distributed data
大数定律不需要独立性，但需要正态分布的数据

B. The CLT extends the LLN by requiring one additional assumption (the variance is finite)
中心极限定理扩展了大数定律，需要一个额外的假设（方差是有限的）

C. LLN says that as the sample size increases $(n \to \infty)$ the $\Pr(X_{\text{avg}} = \mu)$ approaches 100%
大数定律说，随着样本量增加（$n \to \infty$），$\Pr(X_{\text{avg}} = \mu)$ 接近 100%

D. CLT says that as the sample size increases the distribution of the sample mean converges on the normal distribution $N(\mu, \sigma^2/n)$
中心极限定理说，随着样本量增加，样本均值的分布收敛于正态分布 $N(\mu, \sigma^2/n)$

答案:A

解析:

A 选项正确。 A. False. The inverse is true: the LLN requires independent (i.i.d.) random variables but is useful in large because (like the CLT) it makes no assumption about their distribution; it certainly does not require normally distributed random variables.

In regard to (B), (C) and (D), each is TRUE. Specifically:

- The Law of Large Numbers (LLN) says that as the sample size increases $(n \to \infty)$ the $\Pr(X_{\text{avg}} = \mu)$ approaches 100%. This is the *essence* of the LLN. As GARP writes, "The Law of Large Numbers (LLN) establishes the large sample behavior of the mean estimator and provides conditions where an average converges to its expectation. The simplest LLN for iid random variables is the Kolmogorov Strong Law of Large Numbers." — 2020 FRM Part I: Quantitative Analysis, 10th Edition.
- The central limit theorem (CLT) extends the LLN by requiring one additional assumption (the variance is finite). While the LLN only requires a finite mean, the CLT adds one additional assumption that the variance also be finite.
- CLT says that as the sample size increases the distribution of the sample mean converges on the normal distribution $N(\mu, \sigma^2/n)$. This is the *essence* of the CLT: its magical superpower is that we require no knowledge of the random variable's distribution, yet if the sample is i.i.d. random (with finite mean and variance), then the sample mean is asymptotically normal.

考察: Quantitative Analysis　　**重要程度:** [■■□]

科目: Quantitative Analysis

83. For its most recent fiscal year, Acehouse Property-Casualty (APC) reports revenue of $300.0 million. Losses due to payouts (aka, loss and loss adjustment expense) were $213.0 million. Expenses (aka, underwriting expenses) were $69.0 million. Dividends paid were $6.0 million. Investment income was $9.0 million. What was the Operating ratio; aka, Combined ratio after dividends (CRAD) adjusted for Investment Income?

对于其最近的财政年度，Acehouse Property-Casualty（APC）报告收入为 $300.0 million。由于 payout（aka，loss and loss adjustment expense）导致的损失为 $213.0 million。费用（aka，underwriting expenses）为 $69.0 million。股息支付为 $6.0 million。投资收入为 $9.0 million。运营比率是多少；aka，调整投资收入后的综合比率（CRAD）？

A. 7.0%

B. 23.0%

C. 93.0%

D. 106.0%

答案:C

解析:

C 选项正确。 TRUE: 93.0%. See below.

Revenue	**$300.0**	
Losses due to payouts / Loss ratio	**$213.0**	**71.0%**
Expenses / Expense ratio	**$69.0**	**23.0%**
Combined ratio	**$282.0**	**94.0%**
Dividends	**$6.0**	**2.0%**
Combined ratio after dividends (CRAD)	**$288.0**	**96.0%**
Investment Income	**$9.0**	**3.0%**
	$279.0	**93.0%**

考察: Finance: Insurance Ratios　　**重要程度:** [■■□]

科目: Quantitative Analysis

84. Scott is a risk manager for an investment manager. He is interested in estimating the potential loss the portfolio can expect over a specific time frame at a given confidence level (VaR) and the average of all potential losses beyond that potential loss (aka, expected shortfall, ES). Over the past 120 days, the portfolio's ten worst daily returns were the following:

-6.90%	-7.55%	-7.43%	-4.85%	-15.69%	-14.48%	-0.97%	-18.38%	-16.54%	-24.10%

Find the 1-day 5% VaR and ES for Scott (per Ch 2 Calculating and Applying VaR).

斯科特是一家投资管理公司的风险经理。他希望估算投资组合在特定时间范围和置信水平下可能遭受的最大损失（VaR），以及超过该最大损失的所有潜在损失的平均值（即，预期损失，ES）。在过去的 120 天里，该投资组合最差的 10 天日收益率如下：

-6.90%	-7.55%	-7.43%	-4.85%	-15.69%	-14.48%	-0.97%	-18.38%	-16.54%	-24.10%

请根据第一卷第二章"VaR 的计算与应用"，为斯科特计算单日 5% 的 VaR 和 ES。

A. VaR 17.84%, ES 7.55%

B. VaR 7.43%, ES 16.12%

C. VaR 14.48%, ES 16.12%

D. VaR 14.48% ES 17.84%

答案:B

解析:

B 选项正确。 Step 1. Arrange values from worst to least bad day.

120	119	118	117	116	115	114	113	112	111
-24.10%	-18.38%	-16.54%	-15.69%	-14.48%	-7.55%	-7.43%	-6.90%	-4.85%	-0.97%

Step 2. Calculate the 5th percentage = $(5\% * 120)$ = 6th worst day + 1 = 7th worst day (per P2 Dowd). However, it should be noted that other chapters will show another way of calculating VaR, including using the 6th worst, 7th worst, or multiple interpolations, including using the median between the 6th and 7th worst.
Step 3. Identify the 7th worst day in our example it is -7.43%. This means that there is a 95% chance of having a loss of less than 7.43%.
Calculating ES Conditional is the average of losses that exceed VaR, not ES. However, ES is not ambiguous and is always the average of the 5% tail, which in this case is the 6 worst losses.

$$\text{ES} = \frac{(-24.1\% -18.38\% -16.54\% -15.69\% - 14.48\% -7.55\%)}{6} = 16.12\%$$

考察: Risk Management: VaR and ES　　**重要程度:** [■■□]
科目: Quantitative Analysis

85. A portfolio with a volatility of 30.0% has a Treynor measure of 0.080. The portfolio has a correlation of 0.50 with the market index which itself has a volatility of 20.0%. What is the portfolio's Sharpe measure?
一个波动率为 30.0% 的投资组合的 Treynor 测度为 0.080。该投资组合与市场指数的相关系数为 0.50，市场指数本身的波动率为 20.0%。该投资组合的 Sharpe 测度是多少?

A. 0.095

B. 0.200

C. 0.330

D. 0.475

答案:B
解析:
B 选项正确。 The beta(P, M) = correlation$(P, M) \times$ volatility(P)/volatility$(M) = 0.50 \times 30\%/20\% = 0.750$. Because Treynor = (Portfolio's excess return)/beta, the portfolio's excess return $= 0.080 \times 0.750 = 6.0\%$ and its Sharpe measure $= 6.0\%/30.0\% = 0.20$.
考察: Portfolio Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

86. Darlene is a risk analyst who evaluates the creditworthiness of loan applicants at her financial institution. Her department is testing a new logistic regression model. If the model performs well in testing, it will be deployed to assist in the underwriting decision-making process. The training data is a sub-sample ($n = 800$) from the same LendingClub database used in reading. In the logistic regression, the dependent variable is a 0/1 for the terminal state of the loan being either zero (fully paid off) or one (deemed irrecoverable or defaulted). In the actual code, this dependent variable is labeled 'outcome'.
The following are the features (aka, independent variables) as given by their textual labels: Amount, Term, Interest_rate, Installment, Employ_hist, Income, and Bankruptcies. In regard to units in the database, please note the following: Amount is thousands of dollars ($000s); Term is months; Interest_rate is effectively multiplied by one hundred such that 7 equates to 7% or 0.070; Installment is dollars; Employment_hist is years; Income is thousand of dollars ($000); and Bankruptcies is a whole number $\{0, 1, 2, ...\}$.
The table below displays the logistic regression results:

Coeff_labels	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	−2.329	0.841	−2.769	0.006
Amount	0.000	0.000	1.339	0.181
Term	−0.041	0.027	−1.519	0.129
Interest rate	0.140	0.034	4.108	0.000
Installment	−0.003	0.003	−1.033	0.302
Employ Hist	−0.032	0.031	−1.025	0.305
Income	0.000	0.000	−0.937	0.349
Bankruptcies	0.457	0.230	1.991	0.046

In regard to this logistic regression, each of the following statements is true EXCEPT which is false?

Darlene 是一名风险分析师，她在其金融机构负责评估贷款申请人的信用状况。她所在的部门正在测试一个新的逻辑回归模型。如果该模型在测试中表现良好，将被部署用来辅助承保决策过程。训练数据是来自同一 LendingClub 数据库的一个子样本（$n = 800$），该数据库也在阅读材料中被引用。在逻辑回归模型中，因变量为 0/1 变量，代表贷款的最终状态为零（已全部还清）或一（被视为无法收回或违约）。在实际代码中，该因变量被标记为'outcome'。

以下是特征（即自变量）的文本标签：Amount, Term, Interest_rate, Installment, Employ_hist, Income, 以及 Bankruptcies。关于数据库中的单位，请注意：Amount 单位为千美元（$000）；Term 单位为月；Interest_rate 实际上乘以一百，因此 7 表示 7% 或 0.070；Installment 单位为美元；Employment_hist 单位为年；Income 单位为千美元（$000）；Bankruptcies 为整数 $\{0, 1, 2, ...\}$。下表展示了逻辑回归的结果：

Coeff_labels	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	−2.329	0.841	−2.769	0.006
Amount	0.000	0.000	1.339	0.181
Term	−0.041	0.027	−1.519	0.129
Interest rate	0.140	0.034	4.108	0.000
Installment	−0.003	0.003	−1.033	0.302
Employ Hist	−0.032	0.031	−1.025	0.305
Income	0.000	0.000	−0.937	0.349
Bankruptcies	0.457	0.230	1.991	0.046

关于此次逻辑回归，以下每个表述都是正确的，除了哪一项是错误的?

A. A single additional bankruptcy increases the expected odds of default by almost 58 percent
   一次额外的破产将违约的预期几率几乎增加了 58 个百分点

B. If she requires significance at the 5% level or better, then two of the coefficients (in addition to the intercept) are significant
   如果她要求在 5% 水平或更好，那么两个系数（除了截距）是显著的

C. Each +100 basis points increase in the interest rate (e.g., from 8.0% to 9.0%), the expected default odds increase by about 15 percent
   每次利率增加 100 个基点（例如，从 8.0% 到 9.0%），违约的预期几率大约增加 15 个百分点

D. The key weakness of this regression is that Darlene cannot override the binary (yes/no) classification based on the standard $Z = 0.50$ threshold in order to avoid accepting too many applications
   这个回归的关键弱点是，Darlene 无法覆盖基于标准 $Z = 0.50$ 阈值的二元（是/否）分类，以避免接受太多申请

答案:D
解析:
D 选项正确。 In regard to (A), (B), and (C), each is TRUE.
考察: Quantitative Methods: Logistic Regression　　**重要程度:** [■■□]
科目: Quantitative Analysis

87. Consider the following four mutual funds and their respective behaviors:

- Fund A is an open-end fund that engages in "late trading" by accepting orders after 4 pm
- Fund B engages in "front running" by allowing its traders to make profitable trades in their own accounts ahead of the fund's large block trades
- Fund C engages in "directed brokerage" because it has an informal, reciprocal arrangement that guarantees the fund will use the brokerage for trades but only if the brokerage recommends its clients to the mutual fund
- Fund D engages in "timing the market" because, among its watchlist of 100 stocks, it buys (sells) whenever the price moves two standard deviations below (above) its 200-day moving average

Each of these behaviors is undesirable (or illegal) EXCEPT which is acceptable; i.e., which of the following statements is TRUE?
考虑以下四个共同基金及其各自的行为:

- 基金 A 是一个开放式基金，通过在下午 4 点后接受订单进行"晚交易"
- 基金 B 通过允许其交易员在基金的大额交易之前在其自己的账户中进行盈利交易进行"抢先交易"
- 基金 C 通过"定向经纪"进行交易，因为它有一个非正式的、互惠的安排，保证基金只会使用经纪商进行交易，但只有在经纪商推荐其客户给共同基金时才会这样做
- 基金 D 通过"市场时机"进行交易，因为它在其 100 只股票的观察名单中，每当价格移动两个标准差低于（高于）其 200 天移动平均线时买入（卖出）

每个行为都是不受欢迎的（或非法的），除了哪一个是可接受的；即，以下哪个陈述是正确的?

A. Fund A's behavior is permitted by the SEC
   基金 A 的行为被 SEC 允许

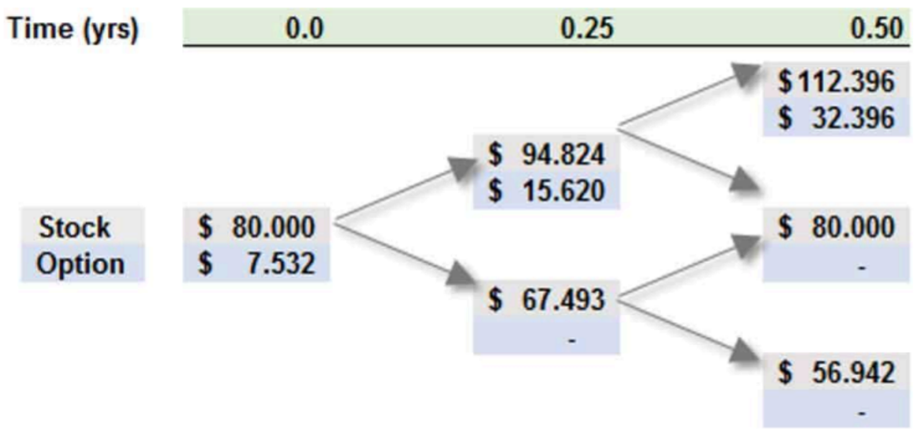
B. Fund B's behavior is legal
   基金 B 的行为是合法的

C. Fund C's behavior is encouraged by regulators
   基金 C 的行为被监管机构鼓励

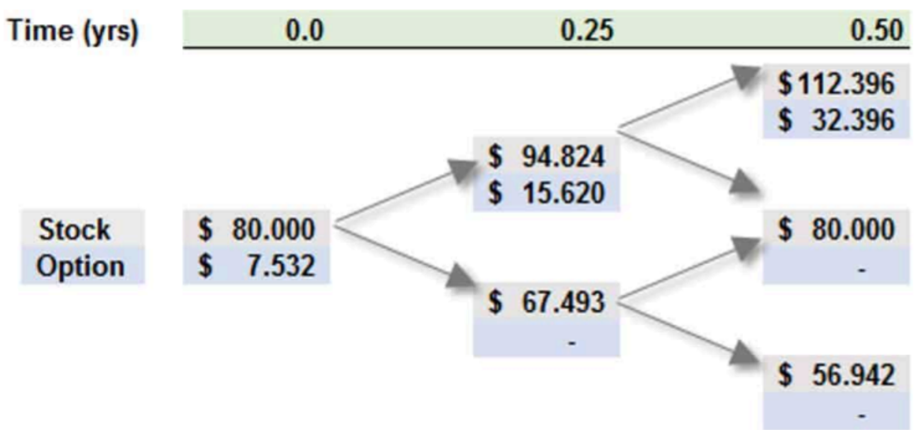
D. Fund D's behavior is permitted by regulators
   基金 D 的行为被监管机构允许

答案:D
解析:
Option D is correct. Fund D's behavior (timing the market) is permitted by regulators, and is practiced by many investors. This timing is not to be confused with the "market timing" associated with buying/selling open-ended mutual fund shares at 4 pm when the actual values may be higher/lower than the slightly stale net asset value (NAV).
In regard to (A), (B), and (C), each is an undesirable trading behavior
考察: Finance: Mutual Funds　　重要程度: [■■□]
科目: Quantitative Analysis

88. Consider a six-month at-the-money (ATM) European call option on a non-dividend-paying stock with a current price of \$80.00. Peter the Risk Analyst has employed a two-step (i.e., three months per step) binomial model to price the option, as displayed below:



Peter's model matches the up and down movements to his estimate of the stock's prospective volatility, which he assumes is 34.0% per annum. The risk-free rate is 4.0%. Which of the following is nearest to the risk-neutral probability of the stock price going up in a single step?
考虑一个到期期限为六个月的平值（ATM）欧式看涨期权，该期权标的是一只不支付股息的股票，当前价格为 \$80.00。风险分析师 Peter 使用了两步二项树模型（即每步为三个月）对该期权定价，并如下面的图所示：



Peter 的模型将价格的上下变动与他对该股票预期波动率的估计相匹配，他假设年化波动率为 34.0%，无风险利率为 4.0%。下列哪个数值最接近股票价格在一步内向上变动的风险中性概率?

A. 45.61%

B. 46.44%

C. 48.70%

D. 52.52%

答案:C

解析:
Option C is correct. 48.70%.
$u = \exp[\sigma\sqrt{\Delta t}] = \exp[0.340\sqrt{0.250}] = 1.18530$,
$d = \exp[-\sigma\sqrt{\Delta t}]$ or $1/d = 0.84366$, and
$p = [\exp(r\Delta t) - d]/(u - d) = [\exp(0.040 \times 0.25) - 0.84366]/(1.18530 - 0.84366) = 0.4870195$.
考察: Finance: Derivatives **重要程度:** [■■□]
科目: Quantitative Analysis

89. About risk reporting practices, the Committee says "[51.] Accurate, complete and timely data is a foundation for effective risk management. However, data alone does not guarantee that the board and senior management will receive appropriate information to make effective decisions about risk. To manage risk effectively, the right information needs to be presented to the right people at the right time. Risk reports based on risk data should be accurate, clear and complete. They should contain the correct content and be presented to the appropriate decision-makers in a time that allows for an appropriate response. To effectively achieve their objectives, risk reports should comply with the following principles. Compliance with these principles should not be at the expense of each other ..."[1]
Each of the following is a risk reporting principle EXCEPT which is not a risk reporting principle?
[1] "Principles for Effective Data Aggregation and Risk Reporting,"(Basel Committee on Banking Supervision Publication, January 2013)
关于风险报告实践，委员会表示:"[51.] 准确、完整和及时的数据是有效风险管理的基础。然而，仅有数据并不能保证董事会和高级管理层会收到足以做出有效风险决策的适当信息。为了有效管理风险，必须在正确的时间将正确的信息传递给正确的人。基于风险数据的报告应该准确、清晰且完整。它们应包含正确信息，并以允许及时响应的时间点呈现给适当的决策者。为了有效实现其目标，风险报告应遵循以下原则。这些原则的遵循不应以牺牲彼此为代价……"[1]
下列各项都是风险报告原则，除了哪一项不是风险报告原则?
[1] "有效数据聚合与风险报告原则"(巴塞尔银行监管委员会出版物，2013 年 1 月)

A. Accuracy
准确性

B. Comprehensiveness
全面性

C. Clarity and usefulness
清晰和有用性

D. Manual workarounds
手动工作

答案:D
解析:
D 选项正确。 Manual workarounds are "Employing human-based processes and tools to transfer, manipulate or alter data used to be aggregated or reported"and they are mentioned in risk data aggregation (39. Supervisors expect banks to document and explain all of their risk data aggregation processes whether automated or manual (judgment based or otherwise). Documentation should include an explanation of the appropriateness of any manual workarounds, a description of their criticality to the accuracy of risk data aggregation and proposed actions to reduce the impact."[1]
In regard to (A), (B) and (C) each is Risk Reporting practice. Basel Committee's Principles for effective risk data aggregation and risk reporting: "III. Risk reporting practices

- Principle 7: Accuracy –Risk management reports should accurately and precisely convey aggregated risk data and reflect risk in an exact manner. Reports should be reconciled and validated.
- Principle 8: Comprehensiveness –Risk management reports should cover all material risk areas within the organisation. The depth and scope of these reports should be consistent with the size and complexity of the bank's operations and risk profile, as well as the requirements of the recipients.

- Principle 9: Clarity and usefulness –Risk management reports should communicate information in a clear and concise manner. Reports should be easy to understand yet comprehensive enough to facilitate informed decision-making. Reports should include an appropriate balance between risk data, analysis and interpretation, and qualitative explanations. Reports should include meaningful information tailored to the needs of the recipients.

- Principle 10: Frequency –The board and senior management (or other recipients as appropriate) should set the frequency of risk management report production and distribution. Frequency requirements should reflect the needs of the recipients, the nature of the risk reported, and the speed at which the risk can change, as well as the importance of reports in contributing to sound risk management and effective and efficient decision-making across the bank. The frequency of reports should be increased during times of stress/crisis.

- Principle 11: Distribution –Risk management reports should be distributed to the relevant parties and while ensuring confidentiality is maintained."[1]

[1] "Principles for Effective Data Aggregation and Risk Reporting,"(Basel Committee on Banking Supervision Publication, January 2013)

考察: Risk Management　**重要程度:** [■■□]

科目: Quantitative Analysis

90. Peter wants to model the following AR(2) time series: $Y(t) = 0.750 * Y(t-1) - 0.1250 * T(t-2) + e(t)$. He wonders if this AR(2) is stationary. He realizes that he can write this as a lag polynomial:

- $Y(t) = 0.750 * L - 0.1250 * L^2 + \varepsilon(t)$; first he expresses the model with lag operators
- $(1 - 0.750 * L + 0.1250 * L^2) * Y(t) = \varepsilon(t)$; then he isolates the lag polynomial
- $(1 - 0.50 * L)(1 - 0.250 * L) * Y(t) = \varepsilon(t)$; finally, he factors the polynomial

Which of the following statements about this AR(2) time series is true?

彼得想对以下 AR(2) 时间序列进行建模: $Y(t) = 0.750 * Y(t-1) - 0.1250 * T(t-2) + e(t)$。他想知道这个 AR(2) 是否是平稳的。他意识到他可以将其写成滞后多项式:

- $Y(t) = 0.750 * L - 0.1250 * L^2 + \varepsilon(t)$; 首先他用滞后算子表示模型
- $(1 - 0.750 * L + 0.1250 * L^2) * Y(t) = \varepsilon(t)$; 然后他用滞后多项式隔离
- $(1 - 0.50 * L)(1 - 0.250 * L) * Y(t) = \varepsilon(t)$; 最后他用多项式因式分解

下列关于这个 AR(2) 时间序列的陈述中哪一个是正确的?

A. This AR(2) is non-stationary and Peter incorrectly factored the polynomial
这个 AR(2) 是非平稳的，彼得错误地因式分解了多项式

B. This AR(2) is non-stationary because the sum of the coefficients is less than one
这个 AR(2) 是非平稳的，因为系数之和小于一

C. This AR(2) is stationary because the sum of the coefficients is less than one
这个 AR(2) 是平稳的，因为系数之和小于一

D. This AR(2) is stationary because both roots of the characteristic equation are greater than one
这个 AR(2) 是平稳的，因为特征方程的两个根都大于一

答案:D

解析:

Option D is correct. This AR(2) is stationary because both roots of the characteristic equation are greater than one

Given this factored polynomial, $(1 - 0.50 * L)(1 - 0.250 * L) * Y(t)$, the roots solve for: $(1 - 0.50 * L) = 0$ such that $L = 2.0$; and $(1 - 0.250 * L) = 0$ such that $L = 4.0$.

If the absolute value of both roots are greater than unity (1.0), then the process is stationary.

考察: Quantitative Analysis　**重要程度:** [■■□]

科目: Quantitative Analysis

91. The daily standard deviation of Bitcoin (BTC) is \$200.00 while the standard deviation of the futures contract price (on the same BTC) is \$340.00. The correlation between the two changes is 0.850. If the size of one futures contract is five Bitcoins (5 BTC), and the trader seeks to hedge the purchase of 100 Bitcoins (100 BTC), then how many contracts should be shorted to hedge?
比特币（BTC）的每日标准差为 \$200.00，而期货合约价格（与 BTC 相关）的标准差为 \$340.00。这两个变化的关联性为 0.850。如果一个期货合约的价值为五个比特币（5 BTC），而交易者希望对购买 100 比特币（100 BTC）进行对冲，那么应该做空多少个合约?

A. One (1)
一（1）

B. Three (3)
三（3）

C. Ten (10)
十（10）

D. Twenty five (25)
二十五（25）

答案:C
解析:
C 选项正确。 The optimal hedge ratio, h^*, is given by $h^* = \beta(\Delta S/\Delta F) = \rho \cdot \sigma(S)/\sigma(F)$; in this case, $h^* = 0.850 \cdot \$200.00/\$340.00 = 0.50$.
The number of contracts to short is given by, $N^* = h^* \times Q_A/Q_F$; in this case, $N^* = 0.50 \cdot 100/5 = 10.0$.
考察: Risk Management: Hedging with Futures **重要程度:** [■■□]
科目: Quantitative Analysis

92. After Barbara wrote (i.e., sold) 100 put option contracts, the trade's position delta was +7,200. The put options were in-the-money as the stock price was \$70.00 while the options' strike price was \$100.00. Each contract was for 100 put options. The percentage gamma of each put option was +0.0120. If the stock price subsequently, immediately decreased by \$10.00 to \$60.00, which is **nearest** to an estimate of the trade's new position delta?
巴勃拉卖出（即出售）了 100 份看跌期权合约，该交易的头寸 delta 为 +7,200。看跌期权处于价内状态，因为股票价格为 \$70.00，而期权的行权价为 \$100.00。每个合约包含 100 份看跌期权。每个看跌期权的百分比 gamma 为 +0.0120。如果股票价格随后立即下跌 \$10.00 至 \$60.00，哪一项最接近对该交易新头寸 delta 的估计?

A. -3,500

B. +7,200 (unchanged)

C. +8,400

D. +11,900

答案:C
解析:
C 选项正确。 The percentage delta of the short put position was $7,200/-10,000 = -0.720$. Given a percentage gamma of $+0.0120$, a drop of \$10.00 in the stock price implies a change in delta given by $-\$10.00 \times 0.0120 = -0.120$, such that the new delta would be $-0.720 - 0.120 = -0.840$. Notice that, with the stock price drop, the put options become even more "in the money" and we do expect the percentage delta to drop (toward its limit at -1.0). Consequently, the new position delta is $-10,000 \times -0.840 = +8,400$.
考察: Valuation and Risk Models: Option Greeks **重要程度:** [■■□]

科目: Quantitative Analysis

93. The riskfree rate is 3.0% and the market portfolio's expected return is 10.0% (put another way, the market's excess expected return is 7.0%). Consider two portfolios:

- Portfolio A has a high volatility, $\sigma(A) = 50.0\%$ per annum, but its correlation to the market portfolio is only, $\rho(A, M) = 0.30$
- Portfolio B has a moderate volatility, $\sigma(B) = 30.0\%$ per annum, and its correlation to the market portfolio is, $\rho(B, M) = 0.70$

If both portfolios lie on the security market line (SML), and if Portfolio A has an expected return, $ER(A) = 7.20\%$, then what is the expected return of Portfolio B? 无风险利率为 3.0%，市场投资组合的预期收益率为 10.0%（换句话说，市场的超额预期收益为 7.0%）。考虑两个投资组合:

- 投资组合 A 具有高波动性，$\sigma(A) = 50.0\%$ 每年，但其与市场投资组合的相关系数仅为 $\rho(A, M) = 0.30$
- 投资组合 B 具有中等波动性，$\sigma(B) = 30.0\%$ 每年，其与市场投资组合的相关系数为 $\rho(B, M) = 0.70$

如果两个投资组合都位于证券市场线（SML）上，且投资组合 A 的预期收益为 $ER(A) = 7.20\%$，那么投资组合 B 的预期收益是多少?

A. 5.100%

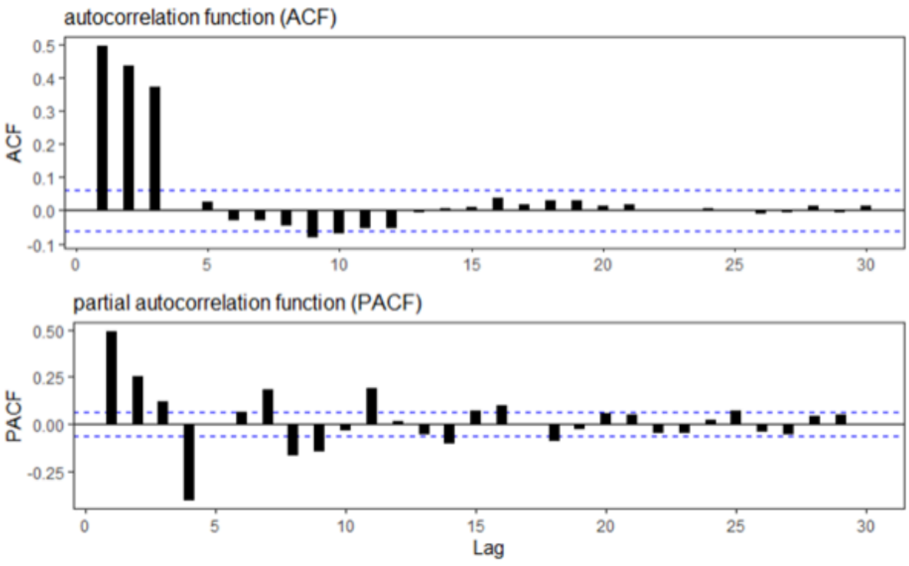
B. 7.900%

C. 8.880%

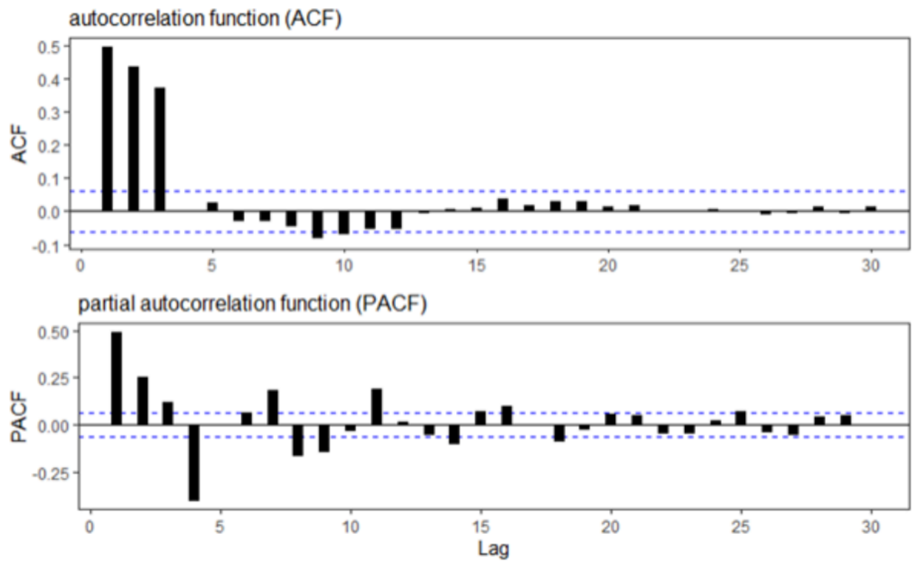
D. 9.540%

答案:C
解析:
C 选项正确。 True: 8.880%. If the both portfolios lie on the SML, then $ER(A) = Rf + \beta(A, M) * [ER(M) - Rf]$ and $ER(B) = Rf + \beta(B, M) * [ER(M) - Rf]$. We can infer Portfolio A's beta: since $7.20\% = 3.0\% + \beta(A, M) * 7.0\%$, $\beta(A, M) = 0.60$. And since $\beta(A, M) = \rho(A, M) * \sigma(A)/\sigma(M)$, we can infer that $\sigma(M) = [\rho(A, M) * \sigma(A)]/\beta(A, M) = [0.30 * 0.50]/0.60 = 0.25$. Then we can compute the beta of portfolio B: $\beta(B, M) = \rho(B, M) * \sigma(B)/\sigma(M) = 0.70 * 0.30/0.25 = 0.840$. The expected return of of portfolio B is given by $ER(B) = 3.0\% + 0.840 * 7.0\% = 8.880\%$.
考察: Finance: CAPM **重要程度:** [■■□]
科目: Financial Markets and Products

94. Below are plotted the autocorrelation function (ACF) and partial autocorrelation function (PACF) for a simulated time-series process. Please note that the horizontal dashed blue lines represent the 95.0% confidence interval.



Which of the following time-series models is **most likely** described by these ACF and PACF plots? 下图展示了模拟时间序列过程的自相关函数（ACF）和偏自相关函数（PACF）。请注意，水平虚蓝线代表 95.0% 的置信区间。



下列哪个时间序列模型最有可能被这些 ACF 和 PACF 图表描述?

A. MA(1) with a coefficient of 1.3
一阶移动平均模型，系数为 1.3

B. MA(3) with coefficients of 0.4, 0.6, and 0.8
三阶移动平均模型，系数为 0.4, 0.6, 和 0.8

C. AR(2) with coefficients of 1.5 and 0.8
二阶自回归模型，系数为 1.5 和 0.8

D. AR(3) with coefficients of 0.3, -1.4 and 0.7
三阶自回归模型，系数为 0.3, -1.4 和 0.7

答案:B
解析:
B 选项正确。 True: MA(3) with coefficients of 0.4, 0.6, and 0.8
The ACF cuts off (i.e., drops off sharply after) 3 lags. The PACF decays gradually. This suggests an MA(3) process. Explains GARP (emphasis ours): "The ACF is always zero for lags larger than q and the PACF is also complex. In general, the PACF of an MA(q) is non-zero at all lags ... The PACF oscillates and decays slowly towards zero ... These both [Figure 10.6] show the same pattern—one that is key to distinguishing MA models from AR models: The ACF cuts off sharply at the order of the MA, whereas the PACF oscillates and decays towards zero over many lags. This is the opposite pattern from what is seen in Figure 10.2 and Figure 10.4, where the ACFs decayed slowly while the PACF cut off sharply." — 2020 FRM Part I: Quantitative Analysis, 10th Edition.
考察: Quantitative Analysis: Time Series　**重要程度:** [■■□]
科目: Quantitative Analysis

95. Peter is a new analyst who has been asked to perform a valuation of a mortgage-backed security (MBS) using one of the firm's models. The pool is likely to be stripped into interest-only (IO) and principal-only (PO) strips. Each of the following statements is true EXCEPT which is false?
彼得是一名新分析师，被要求使用公司的一个模型对抵押贷款支持证券（MBS）进行估值。该池可能被剥离为利息支付（IO）和本金支付（PO）。下列每个陈述都是正确的，除了哪一个是错误的?

A. Interest rate uncertainty tends to reduce the value of IOs but increase the value of POs
利率不确定性倾向于降低 IO 的价值，但增加 PO 的价值

B. His prepayment model will include an annualized prepayment rate that is likely to predict refinancings according to an incentive function
他的提前还款模型将包括一个年化提前还款率，该率可能根据激励函数预测再融资

C. The prepayment rate in his model can be affected (in addition to interest rate levels) by average loan size, average loan age, season (aka, seasonality), and geographical location
他的提前还款率可以受到（除了利率水平之外）平均贷款规模、平均贷款年龄、季节（即季节性）和地理位置的影响

D. His preferred model is a recombining tree because the cash flows are path independent because the permutation of up/down jumps prior to arrival at a future interest rate not is irrelevant
他的首选模型是一个重构树，因为现金流是路径独立的，因为提前到达未来利率的上升/下降跳跃的排列是不相关的

答案:D
解析:
D 选项正确。 False. Instead, he will want to model path dependence and, therefore, will prefer Monte Carlo Simulation. In regard to (A), (B), and (C), each is TRUE.
考察: Fixed Income: Mortgage-Backed Securities **重要程度:** [■■□]
科目: Quantitative Analysis

96. Trader Joe (who is unrelated to the awesome grocery store!) takes a long position in 100 out-of-the-money (OTM) put option contracts when the non-dividend-paying stock price is \$88.00 and its volatility is 28.0%; each contract is for 100 options. The strike price is \$80.00, and the option term is six months. The risk-free rate is 5.0%. The value of each option is \$4.38, and the position's value is $\$43,800.00$. With respect to these puts, the immediate percentage delta, $\Delta = -0.2550$ and gamma, $\Gamma = 0.0130$. If the stock price jumps by +\$7.00 to \$95.00, which is nearest to the position's value as approximated by delta and gamma; i.e., without a full re-pricing of the position?
交易员乔（与著名的杂货店无关！）在非分红股票价格为 \$88.00 且波动率为 28.0% 时买入 100 份价外（OTM）看跌期权合约；每个合约包含 100 份期权。行权价为 \$80.00，期权期限为六个月。无风险利率为 5.0%。每个期权的价值为 \$4.38，该头寸的价值为 $\$43,800.00$。关于这些看跌期权，立即的百分比 delta 为 $\Delta = -0.2550$，gamma 为 $\Gamma = 0.0130$。如果股票价格跳涨 \$7.00 至 \$95.00，哪一项最接近该头寸的价值，即基于 delta 和 gamma 的近似值，不进行完整的价格重新计算?

A. $\$18,280$

B. $\$25,950$

C. $\$29,135$

D. $\$33,640$

答案:C
解析:
C 选项正确。 True: $\$29,135$
The delta-gamma approximated change is given by $\Delta S \cdot \Delta + 0.5 \cdot \Gamma \cdot (\Delta S)^2$; in this case of $\Delta S = +7.00$, estimated change $= 7.0 \cdot (-0.2550) + 0.5 \cdot (0.0130) \cdot 7.0^2 = -1.4665$. Each option is estimated to be worth $\$4.38 - \$1.4665 = \$2.9135$ and the position is estimated to be worth $\$29,135.00$.
In regard to false (B), $\$25,950$ is the estimated value based on only a delta approximation; i.e., if we exclude the gamma adjustment term.
The exact (re-priced with BSM) value is $\$28,841$, so the approximation is off by about 1.0%.
考察: Market Risk: The Greeks **重要程度:** [■■□]
科目: Quantitative Analysis

97. Which of the following is TRUE about the bank's audit function?
关于银行审计功能，下列哪个陈述是正确的?

A. Internal audit is necessary because it is virtually impossible to rate (i.e., assign ratings to) the risk management function
内部审计是必要的，因为几乎不可能对风险管理功能进行评级（即分配评级）

B. The firm's operational risk management should report directly to the internal audit function (who will, therefore, oversee risk management)
公司的运营风险管理应直接向内部审计功能报告（因此将监督风险管理）

C. Internal auditors are responsible for reviewing monitoring procedures, tracking the progress of risk management system upgrades, and affirming the efficacy of vetting processes
内部审计师负责审查监控程序，跟踪风险管理系统的升级进度，并确认筛选过程的有效性

D. The assistance of internal audit should not be required to the risk governance function: a properly designed risk governance function should be able to ascertain compliance alone
风险治理功能不应需要内部审计的协助：一个合理设计的风险治理功能应能够独立确认合规性

答案:C
解析:
Option C is correct. Internal auditors are responsible for reviewing monitoring procedures, tracking the progress of risk management system upgrades, and affirming the efficacy of vetting processes.
In regard to (A), (B) and (D), each is FALSE.

- GARP recommends rating risk management practices: "A risk management function can be rated. This rating may be used internally or by third parties (e.g., rating agencies) that undertake comparative analyses of multiple enterprises. There is no one formula for excellence in risk management. Despite this, the rating of risk management practices would be instrumental in facilitating comparisons across an organization so that both the internal and external parties can benefit from such objective critiques," explains GARP (2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions)

- Risk management must be independent, even from internal audit: "Within the industry, there has been an active debate as to whether the audit function should have effective oversight of the firm's operational risk management. Note that the audit has a natural interest in the quality of internal controls. While subject to auditor review, however, the implementation of risk management must remain separate from the auditing function. As a basic principle, auditor independence from the underlying activity is essential to ensure confidence in any assurances or opinions rendered by the auditors to the board, and this applies equally to the risk management function and its associated processes. Unless this independence is maintained, conflicts of interest could compromise the quality of both risk management and audit activity and seriously jeopardize risk governance.", explains GARP.

- Internal audit is an essential complement to risk management: "Adherence to [the risk management] process can prevent the assumption of unbridled excessive risk. However, the risk governance function alone cannot ascertain compliance to the policies established by the board and external regulations. This is where the audit function comes in. It is incumbent upon the internal audit function to ensure the set-up, implementation, and efficacy of risk management/governance." explains GARP [1]

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions
考察: Foundations of Risk Management　　**重要程度:** [■□□□□]
科目: Foundations of Risk Management

98. For each of three banks (Azure Bank, Breeze Bank, and Coastal Bank), Stella collected the following data on three features:

	Banks			Statistics		
	Azure	Breeze	Coastal	Mean	STDEV.S	STDEV.P
No. of customers (MM)	**1.0**	**2.5**	**6.5**	3.33	2.84	2.32
Size of loan book (BB)	**$5.0**	**$12.5**	**$38.0**	18.50	17.30	14.12
Number of branches	**40**	**90**	**180**	103.33	70.95	57.93

	Standardized (STDEV.S)		
	Azure	Breeze	Coastal
No. of customers (MM)	-0.821	-0.293	1.114
Size of loan book (BB)	-0.780	-0.347	1.127
Number of branches	-0.893	-0.188	1.081

Statistics displayed (top-right panel) include the mean, sample standard deviation (STDEV.S), and population standard deviation (STDEV.P). The standardized features (lower-left panel) are based on the sample standard deviation.
With respect to the standardized features (not the raw values), which of the following is **nearest** to the Euclidean distance between Breeze Bank and Coastal Bank?
对于三家银行（Azure Bank、Breeze Bank 和 Coastal Bank），Stella 收集了以下三个特征的数据：

	Banks			Statistics		
	Azure	Breeze	Coastal	Mean	STDEV.S	STDEV.P
No. of customers (MM)	1.0	2.5	6.5	3.33	2.84	2.32
Size of loan book (BB)	$5.0	$12.5	$38.0	18.50	17.30	14.12
Number of branches	40	90	180	103.33	70.95	57.93

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显示的数据（右上角面板）包括均值、样本标准差（STDEV.S）和总体标准差（STDEV.P）。标准化特征（左下角面板）是基于样本标准差计算的。
针对标准化特征（而非原始值），下列哪个数值最接近 Breeze Bank 与 Coastal Bank 之间的欧几里得距离?

A. 2.40

B. 4.15

C. 93.63

D. 119.50

答案:A
解析:
A 选项正确。 True: 2.40 is the Euclidean distance between the standardized features of Breeze Bank and Coastal Bank. The Euclidean distance between Breeze and Coastal is given by:

$$\sqrt{(-0.2930-1.1140)^2+(-0.3470-1.1270)^2+(-0.1880-1.0810)^2}=2.40056$$

In regard to (B), (C), and (D), each is false. Instead, they are alternative measures between Breeze Bank and Coastal Bank. Specifically:

- 4.15 is the Manhattan distance between their standardized features
- 93.63 is the Euclidean distance between their raw features
- 119.50 is the Manhattan distance between their raw features

考察: Quantitative Methods: Distance Metrics **重要程度:** [■■□]
科目: Quantitative Analysis

99. Among over-the-counter (OTC) interest rate derivatives, interest rate swap contracts are the most actively traded, according to BIS (source: https://www.bis.org/publ/qtrpdf/r_qt2212e.htm). Further, as Libor phases out, swaps will increasingly reference overnight rates such as SOFR and SONIA. About these modern overnight indexed swap (OIS) contracts, each of the following statements is true EXCEPT which is false?
根据 BIS（来源：https://www.bis.org/publ/qtrpdf/r_qt2212e.htm），场外（OTC）利率衍生品中，利率互换合约是最活跃交易的，而随着 Libor 的逐步退出，互换将越来越多地参考隔夜利率，如 SOFR 和 SONIA。关于这些现代隔夜指数互换（OIS）合约，下列每个陈述都是正确的，除了哪一个是错误的?

A. In contrast to LIBOR, overnight rates such as SOFR and SONIA are nearly (or effectively) risk-free rates
与 LIBOR 相比，隔夜利率（如 SOFR 和 SONIA）几乎是（或实际上是无风险的）无风险利率

B. The N-year swap rate is set equal to the average yield to maturity of bonds trading with N-year maturities
N 年互换利率等于 N 年到期债券的平均到期收益率

C. While LIBOR rates are known at the beginning of a period, overnight rates are known only at the end of each period
虽然 LIBOR 利率在期初是已知的，但隔夜利率只有在每个期末是已知的

D. The confirmation specifies the swap's day count convention, payment exchange dates, payment calculation methodology, and the applicable holiday calendar
确认规定了互换的日计数惯例、支付交换日期、支付计算方法以及适用的节假日日历

答案:B
解析:
B 选项正确。 False. The N-year swap rate is the market's N-year Par Yield because it's the N-year coupon rate that prices currently initiated swaps to par. Put another way, the swap rate is a par yield. On the other hand, the yield-to-maturity (aka yield) of current N-year bonds will vary as a function of their coupon rates.
In regard to (A), (C), and (D), each is TRUE.
考察: Financial Markets and Products: Interest Rate Swaps **重要程度:** [■■□]
科目: Quantitative Analysis

100. Analyst Patricia is analyzing the following four bonds:

- Bond A is a \$100.00 face value bond with 7.0 years to maturity that pays a monthly coupon at a rate of 6.0% per annum and offers a yield of 5.0% per annum (with monthly compound frequency)
- Bond B is a \$100.00 face value bond with 10.0 years to maturity that pays a semi-annual coupon at a rate of 4.0% per annum and offers a yield of 5.0% per annum (with semi-annual compound frequency)
- Bond C is a \$100.00 face value bond with 10.0 years to maturity that pays an annual coupon at a rate of 7.0% per annum and offers a yield of 6.0% per annum (with annual compound frequency)
- Bond D is a \$1,000.00 face value zero-coupon bond with 30.0 years to maturity that offers a yield (aka, yield to maturity) of 8.0% per annum with semi-annual compound frequency

In terms of their current theoretical prices, which bonds are, respectively, the cheapest and most expensive (among the four)?
分析师帕特丽夏正在分析以下四种债券:

- 债券 A 是一张面值为 \$100.00、期限为 7 年的债券，每月支付 6% 的年利率，年化收益率为 5%（每月复利频率）
- 债券 B 是一张面值为 \$100.00、期限为 10 年的债券，每半年支付 4% 的年利率，年化收益率为 5%（半年复利频率）
- 债券 C 是一张面值为 \$100.00、期限为 10 年的债券，每年支付 7% 的年利率，年化收益率为 6%（年复利频率）
- 债券 D 是一张面值为 \$1,000.00、期限为 30 年的零息债券，年化收益率为 8%（半年复利频率）

按照当前的理论价格，哪两种债券分别是四种债券中最便宜和最昂贵的?

A. Bond A is the cheapest and Bond D is the most expensive
债券 A 是最便宜的，债券 D 是最昂贵的

B. Bond B is the cheapest and Bond C is the most expensive
债券 B 是最便宜的，债券 C 是最昂贵的

C. Bond C is the cheapest and Bond A is the most expensive
债券 C 是最便宜的，债券 A 是最昂贵的

D. Bond D is the cheapest and Bond B is the most expensive
债券 D 是最便宜的，债券 B 是最昂贵的

答案:B
解析:
Option B is correct. Bond B is the cheapest (at $92.21) and Bond C is the most expensive (at $107.36). The following are the theoretical prices:

- Bond A: $7 * 12 = 84$ N, $5/12 = 0.4167$ I/Y, $0.06 * 100/12 = 0.50$ PMT, 100 FV and CPT PV [+/-] = $105.90
- Bond B: $10 * 2 = 20$ N, $5/2 = 2.5$ I/Y, $0.04 * 100/2 = 2.0$ PMT, 100 FV and CPT PV [+/-] = $92.21
- Bond C: $10 * 1 = 10$ N, $6/1 = 6.0$ I/Y, $0.07 * 100/1 = 7.0$ PMT, 100 FV and CPT PV [+/-] = $107.36
- Bond D: $30 * 2 = 60$ N, $8/2 = 4.0$ I/Y, 0 PMT, 100 FV and CPT PV [+/-] = $95.06

考察: Fixed Income　　**重要程度:** [■■□]
科目: Quantitative Analysis

101. Alex is a junior analyst in the operational risk department at Global Trust Bank. His manager Sarah asks him to classify recent operational risk events according to Basel Committee's seven event categories. Which operational risk is correctly categorized?
Alex 是 Global Trust Bank 运营风险部门的一名初级分析师。他的经理 Sarah 要求他根据巴塞尔委员会的七个事件类别对最近的运营风险事件进行分类。哪一种运营风险被正确分类?

A. Reuters reports that a senior trader at Global Trust Bank manipulated the LIBOR submission process for three years, resulting in $5 million illicit gains. Alex categorizes this as business disruption and system failures.
Reuters 报道，Global Trust Bank 的一名高级交易员连续三年操纵 LIBOR 提交过程，非法获利 500 万美元。Alex 将此事归类为业务中断和系统故障。

B. The bank's core trading platform experienced a four-hour outage during peak market hours, causing significant trading delays. Alex classifies this as execution, delivery, and process management.
银行的核心交易平台在高峰市场时段发生四小时宕机，导致重大交易延误。Alex 将此事归类为执行、交付和流程管理。

C. A severe earthquake damaged several branch offices in the coastal region, leading to $20 million in property losses. Alex categorizes this as internal fraud.
一场严重的地震毁坏了沿海地区的一些分支机构，造成 2000 万美元的财产损失。Alex 将此事归类为内部欺诈。

D. Local media reveals that external hackers breached the bank's security system and stole sensitive client data, demanding $3 million in ransom. Alex classifies this as external fraud.
当地媒体透露，外部黑客入侵了银行的安保系统，窃取了敏感的客户数据，并索要 300 万美元的赎金。Alex 将此事归类为外部欺诈。

答案:D
解析:
A 选项错误。

- LIBOR 操纵属于内部欺诈范畴，因为涉及员工故意违规获取个人利益。该事件包含了欺诈、故意违反规定等典型的内部欺诈特征，而不是业务中断和系统故障。

B 选项错误。

- 交易平台故障应归类为业务中断和系统故障，因为这涉及技术系统的运行中断。而执行、交付和流程管理通常指操作流程和交易处理中的人为错误。

C 选项错误。

- 地震造成的财产损失应归类为实物资产损坏，这类风险源于自然灾害或其他外部事件对银行实物资产的破坏，而不是内部欺诈行为。

D 选项正确。

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考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

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解析:
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考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
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解析:
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考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
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考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
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C. A severe earthquake damaged several branch offices in the coastal region, leading to $20 million in property losses. Alex categorizes this as internal fraud.
一场严重的地震毁坏了沿海地区的一些分支机构，造成 2000 万美元的财产损失。Alex 将此事归类为内部欺诈。

D. Local media reveals that external hackers breached the bank's security system and stole sensitive client data, demanding $3 million in ransom. Alex classifies this as external fraud.
当地媒体透露，外部黑客入侵了银行的安保系统，窃取了敏感的客户数据，并索要 300 万美元的赎金。Alex 将此事归类为外部欺诈。

答案:D

解析:

A 选项错误。

- LIBOR 操纵属于内部欺诈范畴，因为涉及员工故意违规获取个人利益。该事件包含了欺诈、故意违反规定等典型的内部欺诈特征，而不是业务中断和系统故障。

B 选项错误。

- 交易平台故障应归类为业务中断和系统故障，因为这涉及技术系统的运行中断。而执行、交付和流程管理通常指操作流程和交易处理中的人为错误。

C 选项错误。

- 地震造成的财产损失应归类为实物资产损坏，这类风险源于自然灾害或其他外部事件对银行实物资产的破坏，而不是内部欺诈行为。

D 选项正确。

- 外部黑客入侵和勒索属于典型的外部欺诈案例。这类事件由外部人员实施，目的是通过非法手段获取经济利益，符合外部欺诈的定义特征。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]

科目: Financial Markets and Products